Chapter 1 Problems

SS Student solution available in interactive e-text.

SS **1.1** Suppose that you want to design an experiment to study the proportion of unpopped kernels of popcorn. Complete steps 1–3 of the guidelines for designing experiments in Section 1.4. Are there any major sources of variation that would be difficult to control?

1.2 Suppose that you want to investigate the factors that potentially affect cooking rice.

(a) What would you use as a response variable in this experiment? How would you measure the response?

(b) List all of the potential sources of variability that could impact the response.

(c) Complete the first three steps of the guidelines for designing experiments in Section 1.4.

1.3 Suppose that you want to compare the growth of garden flowers with different conditions of sunlight, water, fertilizer, and soil conditions. Complete steps 1–3 of the guidelines for designing experiments in Section 1.4.

1.4 Select an experiment of interest to you. Complete steps 1–3 of the guidelines for designing experiments in Section 1.4.

1.5 Search the World Wide Web for information about Sir Ronald A. Fisher and his work on experimental design in agricultural science at the Rothamsted Experimental Station.

1.6 Find a website for a business that you are interested in. Develop a list of factors that you would use in an experiment to improve the effectiveness of this website.

1.7 Almost everyone is concerned about the price of gasoline. Construct a cause-and-effect diagram identifying the factors that potentially influence the gasoline mileage that you get in your car. How would you go about conducting an experiment to determine any of these factors actually affect your gasoline mileage?

1.8 What is replication? Why do we need replication in an experiment? Present an example that illustrates the difference between replication and repeated measurements.

1.9 Why is randomization important in an experiment? SS

1.10 What are the potential risks of a single, large, comprehensive experiment in contrast to a sequential approach? SS

Chapter 2 Problems

2.1 Computer output for a random sample of data is shown below. Some of the quantities are missing. Compute the values of the missing quantities.

Variable	N	Mean	SE Mean	Std. Dev.	Variance	Minimum	Maximum
Y	9	19.96	?	3.12	?	15.94	27.16

2.2 Computer output for a random sample of data is shown below. Some of the quantities are missing. Compute the values of the missing quantities.

Variable	N	Mean	SE Mean	Std. Dev.	Sum
Y	16	?	0.159	?	399.851

2.3 Suppose that we are testing $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$. Calculate the P-value for the following observed values of the test statistic:

(a) $Z_0 = 2.25$ **(b)** $Z_0 = 1.55$ **(c)** $Z_0 = 2.10$

(d) $Z_0 = 1.95$ **(e)** $Z_0 = -0.10$

2.4 Suppose that we are testing $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$. Calculate the P-value for the following observed values of the test statistic:

(a) $Z_0 = 2.45$ **(b)** $Z_0 = -1.53$ **(c)** $Z_0 = 2.15$

(d) $Z_0 = 1.95$ **(e)** $Z_0 = -0.25$

2.5 Suppose that we are testing $H_0 : \mu_1 = \mu_2$ versus $H_0 : \mu_1 \neq \mu_2$ where the two sample sizes are $n_1 = n_2 = 12$. Both sample variances are unknown but assumed equal. Find bounds on the P-value for the following observed values of the test statistic.

(a) $t_0 = 2.30$ **(b)** $t_0 = 3.41$ **(c)** $t_0 = 1.95$ **(d)** $t_0 = -2.45$

2.6 Suppose that we are testing $H_0 : \mu_1 = \mu_2$ versus $H_0 : \mu_1 > \mu_2$ where the two sample sizes are $n_1 = n_2 = 10$. Both sample variances are unknown but assumed equal. Find bounds on the P-value for the following observed values of the test statistic.

(a) $t_0 = 2.31$ **(b)** $t_0 = 3.60$ **(c)** $t_0 = 1.95$ **(d)** $t_0 = 2.19$

2.7 Consider the following sample data: 9.37, 13.04, 11.69, 8.21, 11.18, 10.41, 13.15, 11.51, 13.21, and 7.75. Is it reasonable to assume that this data is a sample from a normal distribution? Is there evidence to support a claim that the mean of the population is 10?

2.8 A computer program has produced the following output for a hypothesis-testing problem:

```
Difference in sample means: 2.35
Degrees of freedom: 18
Standard error of the difference in sample means: ?
Test statistic: t0 = 2.01
P-value: 0.0298
```

(a) What is the missing value for the standard error?

(b) Is this a two-sided or a one-sided test?

(c) If $\alpha = 0.05$, what are your conclusions?

(d) Find a 90 percent two-sided CI on the difference in means.

2.9 A computer program has produced the following output for a hypothesis-testing problem:

```
Difference in sample means: 11.5
Degrees of freedom: 24
Standard error of the difference in sample means: ?
Test statistic: t0 = -1.88
P-value: 0.0723
```

(a) What is the missing value for the standard error?

(b) Is this a two-sided or a one-sided test?

(c) If $\alpha = 0.05$, what are your conclusions?

(d) Find a 95 percent two-sided CI on the difference in means.

2.10 A two-sample t-test has been conducted and the sample sizes are $n_1 = n_2 = 10$. The computed value of the test statistic is $t_0 = 2.15$. If the null hypothesis is two-sided, an upper bound on the P-value is

(a) 0.10 **(b)** 0.05 **(c)** 0.025

(d) 0.01 **(e)** none of the above.

2.11 A two-sample t-test has been conducted and the sample sizes are $n_1 = n_2 = 12$. The computed value of the test statistic is $t_0 = 2.27$. If the null hypothesis is two-sided, an upper bound on the P-value is

(a) 0.10 **(b)** 0.05 **(c)** 0.025

(d) 0.01 **(e)** none of the above.

2.12 Suppose that we are testing $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$ with a sample size of $n = 15$. Calculate bounds on the P-value for the following observed values of the test statistic:

(a) $t_0 = 2.35$ **(b)** $t_0 = 3.55$ **(c)** $t_0 = 2.00$ **(d)** $t_0 = 1.55$

2.13 Suppose that we are testing $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ with a sample size of $n = 10$. Calculate bounds on the P-value for the following observed values of the test statistic:

(a) $t_0 = 2.48$ **(b)** $t_0 = -3.95$ **(c)** $t_0 = 2.69$

(d) $t_0 = 1.88$ **(e)** $t_0 = -1.25$

2.14 Consider the computer output shown below.

One-Sample T: Y

Test of mu = 25 vs. > 25

Variable	N	Mean	Std. Dev.	SE Mean	95% Lower Bound	T	P
Y	12	25.6818	?	0.3360	?	?	0.034

(a) How many degrees of freedom are there on the t-test statistic?

(b) Fill in the missing information.

2.15 Consider the computer output shown below.

Two-Sample t-Test and CI: Y1, Y2

Two-sample T for Y1 vs Y2

	N	Mean	Std. Dev.	SE Mean
Y1	20	50.19	1.71	0.38
Y2	20	52.52	2.48	0.55

Difference = mu (X1) - mu (X2)
Estimate for difference: -2.33341
95% CI for difference: (-3.69547, -0.97135)
T-Test of difference = 0 (vs not =): T-Value = -3.47
P-Value = 0.001 DF = 38
Both use Pooled Std. Dev. = 2.1277

(a) Can the null hypothesis be rejected at the 0.05 level? Why?

(b) Is this a one-sided or a two-sided test?

(c) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$, would you reject the null hypothesis at the 0.05 level?,

(d) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 < 2$, would you reject the null hypothesis at the 0.05 level? Can you answer this question without doing any additional calculations? Why?

(e) Use the output and the t table to find a 95 percent upper confidence bound on the difference in means.

(f) What is the P-value if the hypotheses are $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$?

2.16 The breaking strength of a fiber is required to be at least 150 psi. Past experience has indicated that the standard deviation of breaking strength is $\sigma = 3$ psi. A random sample of four specimens is tested, and the results are $y_1 = 145$, $y_2 = 153, y_3 = 150$, and $y_4 = 147$.

(a) State the hypotheses that you think should be tested in this experiment.

(b) Test these hypotheses using $\alpha = 0.05$. What are your conclusions?

(c) Find the P-value for the test in part (b).

(d) Construct a 95 percent confidence interval on the mean breaking strength.

2.17 The viscosity of a liquid detergent is supposed to average 800 centistokes at 25°C. A random sample of 16 batches of detergent is collected, and the average viscosity is 812. Suppose that we know that the standard deviation of viscosity is $\sigma = 25$ centistokes.

(a) State the hypotheses that should be tested.

(b) Test these hypotheses using $\alpha = 0.05$. What are your conclusions?

(c) What is the P-value for the test?

(d) Find a 95 percent confidence interval on the mean.

2.18 A normally distributed random variable has an unknown mean μ and a known variance $\sigma^2 = 9$. Find the sample size required to construct a 95 percent confidence interval on the mean that has a total length of 1.0.

2.19 The shelf life of a carbonated beverage is of interest. Ten bottles are randomly selected and tested, and the following results are obtained:

Days	
108	138
124	163
124	159
106	134
115	139

(a) We would like to demonstrate that the mean shelf life exceeds 120 days. Set up appropriate hypotheses for investigating this claim.

(b) Test these hypotheses using $\alpha = 0.01$. What are your conclusions?

(c) Find the P-value for the test in part (b).

(d) Construct a 99 percent confidence interval on the mean shelf life.

2.20 Consider the shelf life data in Problem 2.19. Can shelf life be described or modeled adequately by a normal distribution? What effect would the violation of this assumption have on the test procedure you used in solving Problem 2.15?

2.21 The time to repair an electronic instrument is a normally distributed random variable measured in hours. The repair times for 16 such instruments chosen at random are as follows:

Hours			
159	280	101	212
224	379	179	264
222	362	168	250
149	260	485	170

(a) You wish to know if the mean repair time exceeds 225 hours. Set up appropriate hypotheses for investigating this issue.

(b) Test the hypotheses you formulated in part (a). What are your conclusions? Use $\alpha = 0.05$.

(c) Find the P-value for the test.

(d) Construct a 95 percent confidence interval on mean repair time.

2.22 Reconsider the repair time data in Problem 2.21. Can repair time, in your opinion, be adequately modeled by a normal distribution?

2.23 Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The filling processes can be assumed to be normal, with standard deviations of $\sigma_1 = 0.015$ and $\sigma_2 = 0.018$. The quality engineering department suspects that both machines fill to the same net volume, whether or not this volume is 16.0 ounces. An experiment is performed by taking a random sample from the output of each machine.

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

(a) State the hypotheses that should be tested in this experiment.

(b) Test these hypotheses using $\alpha = 0.05$. What are your conclusions?

(c) Find the P-value for this test.

(d) Find a 95 percent confidence interval on the difference in mean fill volume for the two machines.

2.24 Two types of plastic are suitable for use by an electronic calculator manufacturer. The breaking strength of this plastic is important. It is known that $\sigma_1 = \sigma_2 = 1.0$ psi. From random samples of $n_1 = 10$ and $n_2 = 12$, we obtain $\bar{y}_1 = 162.5$ and $\bar{y}_2 = 155.0$. The company will not adopt plastic 1 unless its breaking strength exceeds that of plastic 2 by at least 10 psi. Based on the sample information, should they use plastic 1? In answering this question, set up and test appropriate hypotheses using $\alpha = 0.01$. Construct a 99 percent confidence interval on the true mean difference in breaking strength.

2.25 The following are the burning times (in minutes) of chemical flares of two different formulations. The design engineers are interested in both the mean and variance of the burning times.

Type 1		Type 2	
65	82	64	56
81	67	71	69
57	59	83	74
66	75	59	82
82	70	65	79

(a) Test the hypothesis that the two variances are equal. Use $\alpha = 0.05$.

(b) Using the results of (a), test the hypothesis that the mean burning times are equal. Use $\alpha = 0.05$. What is the P-value for this test?

(c) Discuss the role of the normality assumption in this problem. Check the assumption of normality for both types of flares.

2.26 An article in *Solid State Technology*, "Orthogonal Design for Process Optimization and Its Application to Plasma Etching" by G. Z. Yin and D. W. Jillie (May 1987) describes an experiment to determine the effect of the C_2F_6 flow rate on the uniformity of the etch on a silicon wafer used in integrated circuit manufacturing. All of the runs were made in random order. Data for two flow rates are as follows:

C_2F_6 Flow (SCCM)	Uniformity Observation					
	1	2	3	4	5	6
125	2.7	4.6	2.6	3.0	3.2	3.8
200	4.6	3.4	2.9	3.5	4.1	5.1

(a) Does the C_2F_6 flow rate affect average etch uniformity? Use $\alpha = 0.05$.

(b) What is the P-value for the test in part (a)?

(c) Does the C_2F_6 flow rate affect the wafer-to-wafer variability in etch uniformity? Use $\alpha = 0.05$.

(d) Draw box plots to assist in the interpretation of the data from this experiment.

2.27 Photoresist is a light-sensitive material applied to semiconductor wafers so that the circuit pattern can be imaged on to the wafer. After application, the coated wafers are baked to remove the solvent in the photoresist mixture and to harden the resist. Here are measurements of photoresist thickness (in kA) for eight wafers baked at two different temperatures. Assume that all of the runs were made in random order.

95°C	100°C
11.176	5.263
7.089	6.748
8.097	7.461
11.739	7.015
11.291	8.133
10.759	7.418
6.467	3.772
8.315	8.963

(a) Is there evidence to support the claim that the higher baking temperature results in wafers with a lower mean photoresist thickness? Use $\alpha = 0.05$.

(b) What is the P-value for the test conducted in part (a)?

(c) Find a 95 percent confidence interval on the difference in means. Provide a practical interpretation of this interval.

(d) Draw dot diagrams to assist in interpreting the results from this experiment.

(e) Check the assumption of normality of the photoresist thickness.

(f) Find the power of this test for detecting an actual difference in means of 2.5 kA.

(g) What sample size would be necessary to detect an actual difference in means of 1.5 kA with a power of at least 0.9?

2.28 Front housings for cell phones are manufactured in an injection molding process. The time the part is allowed to cool in the mold before removal is thought to influence the occurrence of a particularly troublesome cosmetic defect, flow lines, in the finished housing. After manufacturing, the housings are inspected visually and assigned a score between 1 and 10 based on their appearance, with 10 corresponding to a perfect part and 1 corresponding to a completely defective part. An experiment was conducted using two cool-down times, 10 and 20 seconds, and 20 housings were evaluated at each level of cool-down time. All 40 observations in this experiment were run in random order. The data are as follows:

10 seconds		20 seconds	
1	3	7	6
2	6	8	9
1	5	5	5
3	3	9	7
5	2	5	4
1	1	8	6
5	6	6	8
2	8	4	5
3	2	6	8
5	3	7	7

(a) Is there evidence to support the claim that the longer cool-down time results in fewer appearance defects? Use $\alpha = 0.05$.

(b) What is the P-value for the test conducted in part (a)?

(c) Find a 95 percent confidence interval on the difference in means. Provide a practical interpretation of this interval.

(d) Draw dot diagrams to assist in interpreting the results from this experiment.

(e) Check the assumption of normality for the data from this experiment.

2.29 The diameter of a ball bearing was measured by 12 inspectors, each using two different kinds of calipers. The results are as follows:

Inspector	Caliper 1	Caliper 2
1	0.265	0.264
2	0.265	0.265
3	0.266	0.264
4	0.267	0.266
5	0.267	0.267
6	0.265	0.268
7	0.267	0.264
8	0.267	0.265
9	0.265	0.265
10	0.268	0.267
11	0.268	0.268
12	0.265	0.269

(a) Is there a significant difference between the means of the population of measurements from which the two samples were selected? Use $\alpha = 0.05$.

(b) Find the P-value for the test in part (a).

(c) Construct a 95 percent confidence interval on the difference in mean diameter measurements for the two types of calipers.

2.30 An article in the journal *Neurology* (1998, Vol. 50, pp. 1246–1252) observed that monozygotic twins share numerous physical, psychological, and pathological traits. The investigators measured an intelligence score of 10 pairs of twins. The data obtained are as follows:

Pair	Birth Order: 1	Birth Order: 2
1	6.08	5.73
2	6.22	5.80
3	7.99	8.42
4	7.44	6.84
5	6.48	6.43
6	7.99	8.76
7	6.32	6.32
8	7.60	7.62
9	6.03	6.59
10	7.52	7.67

(a) Is the assumption that the difference in score is normally distributed reasonable?

(b) Find a 95 percent confidence interval on the difference in mean score. Is there any evidence that mean score depends on birth order?

(c) Test an appropriate set of hypotheses indicating that the mean score does not depend on birth order.

2.31 An article in the *Journal of Strain Analysis* (Vol. 18, no. 2, 1983) compares several procedures for predicting the shear strength for steel plate girders. Data for nine girders in the form of the ratio of predicted to observed load for two of these procedures, the Karlsruhe and Lehigh methods, are as follows:

Girder	Karlsruhe Method	Lehigh Method
S1/1	1.186	1.061
S2/1	1.151	0.992
S3/1	1.322	1.063
S4/1	1.339	1.062
S5/1	1.200	1.065
S2/1	1.402	1.178
S2/2	1.365	1.037
S2/3	1.537	1.086
S2/4	1.559	1.052

(a) Is there any evidence to support a claim that there is a difference in mean performance between the two methods? Use $\alpha = 0.05$.

(b) What is the P-value for the test in part (a)?

(c) Construct a 95 percent confidence interval for the difference in mean predicted to observed load.

(d) Investigate the normality assumption for both samples.

(e) Investigate the normality assumption for the difference in ratios for the two methods.

(f) Discuss the role of the normality assumption in the paired t-test.

2.32 The deflection temperature under load for two different formulations of ABS plastic pipe is being studied. Two samples of 12 observations each are prepared using each formulation and the deflection temperatures (in °F) are reported below:

Formulation 1			Formulation 2		
206	193	192	177	176	198
188	207	210	197	185	188
205	185	194	206	200	189
187	189	178	201	197	203

(a) Construct normal probability plots for both samples. Do these plots support assumptions of normality and equal variance for both samples?

(b) Do the data support the claim that the mean deflection temperature under load for formulation 1 exceeds that of formulation 2? Use $\alpha = 0.05$.

(c) What is the P-value for the test in part (a)?

2.33 Refer to the data in Problem 2.32. Do the data support a claim that the mean deflection temperature under load for formulation 1 exceeds that of formulation 2 by at least 3°F?

2.34 In semiconductor manufacturing, wet chemical etching is often used to remove silicon from the backs of wafers prior to metallization. The etch rate is an important characteristic of this process. Two different etching solutions are being evaluated. Eight randomly selected wafers have been etched in each solution, and the observed etch rates (in mils/min) are as follows:

Solution 1		Solution 2	
9.9	10.6	10.2	10.6
9.4	10.3	10.0	10.2
10.0	9.3	10.7	10.4
10.3	9.8	10.5	10.3

(a) Do the data indicate that the claim that both solutions have the same mean etch rate is valid? Use $\alpha = 0.05$ and assume equal variances.

(b) Find a 95 percent confidence interval on the difference in mean etch rates.

(c) Use normal probability plots to investigate the adequacy of the assumptions of normality and equal variances.

2.35 Two popular pain medications are being compared on the basis of the speed of absorption by the body. Specifically, tablet 1 is claimed to be absorbed twice as fast as tablet 2. Assume that σ_1^2 and σ_2^2 are known. Develop a test statistic for

$$H_0 : 2\mu_1 = \mu_2$$
$$H_1 : 2\mu_1 \neq \mu_2$$

2.36 Continuation of Problem 2.35. An article in *Nature* (1972, pp. 225–226) reported on the levels of monoamine oxidase in blood platelets for a sample of 43 schizophrenic patients resulting in $\bar{y}_1 = 2.69$ and $s_1 = 2.30$ while for a sample of 45 normal patients the results were $\bar{y}_2 = 6.35$ and $s_2 = 4.03$. The units are nm/mg protein/h. Use the results of the previous problem to test the claim that the mean monoamine oxidase level for normal patients is at least twice the mean level for schizophrenic patients. Assume that the sample sizes are large enough to use the sample standard deviations as the true parameter values.

2.37 Suppose that we are testing

$$H_0 : \mu_1 = \mu_2$$
$$H_1 : \mu_1 \neq \mu_2$$

where $\sigma_1^2 > \sigma_2^2$ are known. Our sampling resources are constrained such that $n_1 + n_2 = N$. Show that an allocation of the observations n_1 and n_2 to the two samples leads to the most powerful test is in the ratio $n_1/n_2 = \sigma_1/\sigma_2$.

2.38 Continuation of Problem 2.37. Suppose that we want to construct a 95 percent two-sided confidence interval on the difference in two means where the two sample standard deviations are known to be $\sigma_1 = 4$ and $\sigma_2 = 8$. The total sample size is restricted to $N = 30$. What is the length of the 95 percent CI if the sample sizes used by the experimenter are $n_1 = n_2 = 15$? How much shorter would the 95 percent CI have been if the experimenter had used an optimal sample size allocation?

2.39 Develop Equation 2.46 for a $100(1 - \alpha)$ percent confidence interval for the variance of a normal distribution.

2.40 Develop Equation 2.50 for a $100(1 - \alpha)$ percent confidence interval for the ratio σ_1^2/σ_2^2, where σ_1^2 and σ_1^2 are the variances of two normal distributions.

2.41 Develop an equation for finding a $100(1 - \alpha)$ percent confidence interval on the difference in the means of two normal distributions where $\sigma_1^2 \neq \sigma_2^2$. Apply your equation to the Portland cement experiment data, and find a 95 percent confidence interval.

2.42 Construct a data set for which the paired t-test statistic is very large, but for which the usual two-sample or pooled t-test statistic is small. In general, describe how you created the data. Does this give you any insight regarding how the paired t-test works?

2.43 Consider the experiment described in Problem 2.25. If the mean burning times of the two flares differ by as much as 2 minutes, find the power of the test. What sample size would be required to detect an actual difference in mean burning time of 1 minute with a power of at least 0.90?

2.44 Reconsider the bottle filling experiment described in Problem 2.23. Rework this problem assuming that the two population variances are unknown but equal.

2.45 Consider the data from Problem 2.23. If the mean fill volume of the two machines differ by as much as 0.25 ounces, what is the power of the test used in Problem 2.18? What sample size would result in a power of at least 0.9 if the actual difference in mean fill volume is 0.25 ounces?

2.46 An experiment has been performed with a factor that has only two levels. Samples of size $n_1 = n_2 = 12$ have been taken, and the resulting sample data are as follows:

$$\bar{y}_1 = 12.5, \bar{y}_2 = 13.1, S_1 = 1.8, S_2 = 2.1.$$

Can you conclude that there is no difference in means using $\alpha = 0.05$? What are bounds on the P-value for this test? Find a 95 percent confidence interval on the difference in the two means. Does the confidence interval provide any information that is useful in interpreting the test of the hypothesis on the difference in the two means?

2.47 Reconsider the situation in Problem 2.46. Suppose that the two sample sizes were $n_1 = n_2 = 5$. What difference in conclusions (if any) would you have obtained from the hypothesis test? From the confidence interval?

2.48 Suppose that you are testing the hypothesis $H_0 : \mu = 50$ against the usual two-sided alternative. The data are normally distributed with known standard deviation $\sigma = 1$. The sample average obtained in the experiment is 50.5, and it is known that if the true population mean is actually 50.5, then this has no practical significance on the problem that motivated the experiment. Find the P-value for the t-test for the following sample sizes:

(a) $n = 5$ **(b)** $n = 10$ **(c)** $n = 25$

(d) $n = 50$ **(e)** $n = 100$ **(f)** $n = 1000$

Discuss your findings. What does this tell you about relying on P-values in hypothesis testing situations when sample sizes are large?

2.49 Consider the situation in Problem 2.48. Calculate the 95 percent confidence interval on the mean for each of the sample sizes given. How does the length of the confidence interval change with sample size?

2.50 Is the assumption of sampling from a normal distribution critical in the application of the t-test? Justify your answer.

2.51 An experiment has been performed with a factor that has only two levels. Samples of size $n_1 = n_2 = 10$ have been taken, and the resulting sample data are as follows:

$$\bar{y}_1 = 10.7, \bar{y}_2 = 15.1, S_1 = 1.5, S_2 = 4.1.$$

It seems likely that the two population variances are not the same. Can you conclude that there is no difference in means using $\alpha = 0.05$? What are bounds on the P-value for this test? Find a 95 percent confidence interval on the difference in the two means. Does the confidence interval provide any information that is useful in interpreting the test of the hypothesis on the difference in the two means?

2.52 Do you think that using a significance level of $\alpha = 0.05$ is appropriate for all experiments? In the early stages of research and development work, is there a lot of harm in identifying a factor as important when it really isn't? Would that seem to justify higher levels of significance such as $\alpha = 0.10$ or perhaps even $\alpha = 0.15$ in some situations?

2.53 In the early stages of research and development experimentation, which type of error do you think is most important, type I or type II? Justify your answer.

Chapter 3 Problems

3.1 An experimenter has conducted a single-factor experiment with four levels of the factor, and each factor level has been replicated six times. The computed value of the F-statistic is $F_0 = 3.26$. Find bounds on the P-value.

3.2 An experimenter has conducted a single-factor experiment with six levels of the factor, and each factor level has been replicated three times. The computed value of the F-statistic is $F_0 = 5.81$. Find bounds on the P-value.

SS **3.3** An experimenter has conducted a single-factor completely randomized design with five levels of the factor and three replicates. The computed value of the F-statistic is 4.87. Find bounds on the P-value.

3.4 An experimenter has conducted a single-factor completely randomized design with three levels of the factor and five replicates. The computed value of the F-statistic is 2.91. Find bounds on the P-value.

3.5 The mean square for error in the ANOVA provides an estimate of

(a) The variance of the random error

(b) The variance of an individual treatment average

(c) The standard deviation of an individual observation

(d) None of the above

SS **3.6** It is always a good idea to check the normality assumption in the ANOVA by applying a test for normality such as the Anderson–Darling test to the residuals.

(a) True

(b) False

3.7 A computer ANOVA output is shown below. Fill in the blanks. You may give bounds on the P-value.

One-way ANOVA

Source	DF	SS	MS	F	P
Factor	?	?	246.93	?	?
Error	25	186.53	?		
Total	29	1174.24			

3.8 An article appeared in *The Wall Street Journal* on Tuesday, April 27, 2010, with the title "Eating Chocolate Is Linked to Depression." The article reported on a study funded by the National Heart, Lung and Blood Institute (part of the National Institutes of Health) and conducted by faculty at the University of California, San Diego, and the University of California, Davis. The research was also published in the *Archives of Internal Medicine* (2010, pp. 699–703). The study examined 931 adults who were not taking antidepressants and did not have known cardiovascular disease or diabetes. The group was about 70% men and the average age of the group was reported to be about 58. The participants were asked about chocolate consumption and then screened for depression using a questionnaire. People who score less than 16 on the questionnaire were not considered depressed, while those with scores above 16 and less than or equal to 22 were considered possibly depressed, while those with scores above 22 were considered likely to be depressed. The survey found that people who were not depressed ate an average 5.4 servings of chocolate per month, possibly depressed individuals ate an average of 8.4 servings of chocolate per month, while those individuals who scored above 22 and were likely to be depressed ate the most chocolate, an average of 11.8 servings per month. No differentiation was made between dark and milk chocolate. Other foods were also examined, but no pattern emerged between other foods and depression. Is this study really a designed experiment? Does it establish a cause-and-effect link between chocolate consumption and depression? How would the study have to be conducted to establish such a cause-and-effect link?

3.9 The tensile strength of Portland cement is being studied. Four different mixing techniques can be used economically. A completely randomized experiment was conducted and the following data were collected:

Mixing Technique	Tensile Strength (lb/in^2)			
1	3129	3000	2865	2890
2	3200	3300	2975	3150
3	2800	2900	2985	3050
4	2600	2700	2600	2765

(a) Test the hypothesis that mixing techniques affect the strength of the cement. Use $\alpha = 0.05$.

(b) Construct a graphical display as described in Section 3.5.3 to compare the mean tensile strengths for the four mixing techniques. What are your conclusions?

(c) Use the Fisher LSD method with $\alpha = 0.05$ to make comparisons between pairs of means.

(d) Construct a normal probability plot of the residuals. What conclusion would you draw about the validity of the normality assumption?

(e) Plot the residuals versus the predicted tensile strength. Comment on the plot.

(f) Prepare a scatter plot of the results to aid the interpretation of the results of this experiment.

3.10 (a) Rework part (c) of Problem 3.9 using Tukey's test with $\alpha = 0.05$. Do you get the same conclusions from Tukey's test that you did from the graphical procedure and/or the Fisher LSD method?

(b) Explain the difference between the Tukey and Fisher procedures.

SS **3.11** Reconsider the experiment in Problem 3.9. Find a 95 percent confidence interval on the mean tensile strength of the Portland cement produced by each of the four mixing techniques. Also find a 95 percent confidence interval on the difference in means for techniques 1 and 3. Does this aid you in interpreting the results of the experiment?

3.12 A product developer is investigating the tensile strength of a new synthetic fiber that will be used to make cloth for men's shirts. Strength is usually affected by the percentage of cotton used in the blend of materials for the fiber. The engineer conducts a completely randomized experiment with five levels of cotton content and replicates the experiment five times. The data are shown in the following table.

Cotton Weight Percent	Observations				
15	7	7	15	11	9
20	12	17	12	18	18
25	14	19	19	18	18
30	19	25	22	19	23
35	7	10	11	15	11

(a) Is there evidence to support the claim that cotton content affects the mean tensile strength? Use $\alpha = 0.05$.

(b) Use the Fisher LSD method to make comparisons between the pairs of means. What conclusions can you draw?

(c) Analyze the residuals from this experiment and comment on model adequacy.

3.13 Reconsider the experiment described in Problem 3.12. Suppose that 30 percent cotton content is a control. Use Dunnett's test with $\alpha = 0.05$ to compare all of the other means with the control.

3.14 A pharmaceutical manufacturer wants to investigate S the bioactivity of a new drug. A completely randomized single-factor experiment was conducted with three dosage levels, and the following results were obtained.

Dosage	Observations			
20 g	24	28	37	30
30 g	37	44	31	35
40 g	42	47	52	38

(a) Is there evidence to indicate that dosage level affects bioactivity? Use $\alpha = 0.05$.

(b) If it is appropriate to do so, make comparisons between the pairs of means. What conclusions can you draw?

(c) Analyze the residuals from this experiment and comment on model adequacy.

3.15 A rental car company wants to investigate whether the type of car rented affects the length of the rental period. An experiment is run for one week at a particular location, and 10 rental contracts are selected at random for each car type. The results are shown in the following table.

Type of Car	Observations									
Subcompact	3	5	3	7	6	5	3	2	1	6
Compact	1	3	4	7	5	6	3	2	1	7
Midsize	4	1	3	5	7	1	2	4	2	7
Full size	3	5	7	5	10	3	4	7	2	7

(a) Is there evidence to support a claim that the type of car rented affects the length of the rental contract? Use $\alpha = 0.05$. If so, which types of cars are responsible for the difference?

(b) Analyze the residuals from this experiment and comment on model adequacy.

(c) Notice that the response variable in this experiment is a count. Should this cause any potential concerns about the validity of the analysis of variance?

3.16 I belong to a golf club in my neighborhood. I divide S the year into three golf seasons: summer (June–September), winter (November–March), and shoulder (October, April, and May). I believe that I play my best golf during the summer (because I have more time and the course isn't crowded)

and shoulder (because the course isn't crowded) seasons, and my worst golf is during the winter (because when all of the part-year residents show up, the course is crowded, play is slow, and I get frustrated). Data from the last year are shown in the following table.

Season	Observations									
Summer	83	85	85	87	90	88	88	84	91	90
Shoulder	91	87	84	87	85	86	83			
Winter	94	91	87	85	87	91	92	86		

(a) Do the data indicate that my opinion is correct? Use $\alpha = 0.05$.

(b) Analyze the residuals from this experiment and comment on model adequacy.

3.17 A regional opera company has tried three approaches to solicit donations from 24 potential sponsors. The 24 potential sponsors were randomly divided into three groups of eight, and one approach was used for each group. The dollar amounts of the resulting contributions are shown in the following table.

Approach	Contributions (in $)							
1	1000	1500	1200	1800	1600	1100	1000	1250
2	1500	1800	2000	1200	2000	1700	1800	1900
3	900	1000	1200	1500	1200	1550	1000	1100

(a) Do the data indicate that there is a difference in results obtained from the three different approaches? Use $\alpha = 0.05$.

(b) Analyze the residuals from this experiment and comment on model adequacy.

3.18 An experiment was run to determine whether four specific firing temperatures affect the density of a certain type of brick. A completely randomized experiment led to the following data:

Temperature	Density				
100	21.8	21.9	21.7	21.6	21.7
125	21.7	21.4	21.5	21.4	
150	21.9	21.8	21.8	21.6	21.5
175	21.9	21.7	21.8	21.4	

(a) Does the firing temperature affect the density of the bricks? Use $\alpha = 0.05$.

(b) Is it appropriate to compare the means using the Fisher LSD method (for example) in this experiment?

(c) Analyze the residuals from this experiment. Are the analysis of variance assumptions satisfied?

(d) Construct a graphical display of the treatment as described in Section 3.5.3. Does this graph adequately summarize the results of the analysis of variance in part (a)?

3.19 Rework part (d) of Problem 3.18 using the Tukey method. What conclusions can you draw? Explain carefully how you modified the technique to account for unequal sample sizes.

3.20 A manufacturer of television sets is interested in the effect on tube conductivity of four different types of coating for color picture tubes. A completely randomized experiment is conducted and the following conductivity data are obtained:

Coating Type	Conductivity			
1	143	141	150	146
2	152	149	137	143
3	134	136	132	127
4	129	127	132	129

(a) Is there a difference in conductivity due to coating type? Use $\alpha = 0.05$.

(b) Estimate the overall mean and the treatment effects.

(c) Compute a 95 percent confidence interval estimate of the mean of coating type 4. Compute a 99 percent confidence interval estimate of the mean difference between coating types 1 and 4.

(d) Test all pairs of means using the Fisher LSD method with $\alpha = 0.05$.

(e) Use the graphical method discussed in Section 3.5.3 to compare the means. Which coating type produces the highest conductivity?

(f) Assuming that coating type 4 is currently in use, what are your recommendations to the manufacturer? We wish to minimize conductivity.

3.21 Reconsider the experiment from Problem 3.20. Analyze the residuals and draw conclusions about model adequacy.

SS **3.22** An article in *Environment International* (Vol. 18, No. 4, 1992) describes an experiment in which the amount of radon released in showers was investigated. Radon-enriched water was used in the experiment, and six different orifice diameters were tested in shower heads. The data from the experiment are shown in the following table:

Orifice Diameter	Radon Released (%)			
0.37	80	83	83	85
0.51	75	75	79	79
0.71	74	73	76	77
1.02	67	72	74	74
1.40	62	62	67	69
1.99	60	61	64	66

(a) Does the size of the orifice affect the mean percentage of radon released? Use $\alpha = 0.05$.

(b) Find the P-value for the F-statistic in part (a).

(c) Analyze the residuals from this experiment.

(d) Find a 95 percent confidence interval on the mean percent of radon released when the orifice diameter is 1.40.

(e) Construct a graphical display to compare the treatment means as described in Section 3.5.3. What conclusions can you draw?

3.23 The response time in milliseconds was determined for three different types of circuits that could be used in an automatic valve shutoff mechanism. The results from a completely randomized experiment are shown in the following table:

Circuit Type	Response Time				
1	9	12	10	8	15
2	20	21	23	17	30
3	6	5	8	16	7

(a) Test the hypothesis that the three circuit types have the same response time. Use $\alpha = 0.01$.

(b) Use Tukey's test to compare pairs of treatment means. Use $\alpha = 0.01$.

(c) Use the graphical procedure in Section 3.5.3 to compare the treatment means. What conclusions can you draw? How do they compare with the conclusions from part (b)?

(d) Construct a set of orthogonal contrasts, assuming that at the outset of the experiment you suspected the response time of circuit type 2 to be different from the other two.

(e) If you were the design engineer and you wished to minimize the response time, which circuit type would you select?

(f) Analyze the residuals from this experiment. Are the basic analysis of variance assumptions satisfied?

3.24 The effective life of insulating fluids at an accelerated load of 35 kV is being studied. Test data have been obtained for four types of fluids. The results from a completely randomized experiment are as follows:

Fluid Type	Life (in h) at 35 kV Load					
1	17.6	18.9	16.3	17.4	20.1	21.6
2	16.9	15.3	18.6	17.1	19.5	20.3
3	21.4	23.6	19.4	18.5	20.5	22.3
4	19.3	21.1	16.9	17.5	18.3	19.8

(a) Is there any indication that the fluids differ? Use $\alpha = 0.05$.

(b) Which fluid would you select, given that the objective is long life?

(c) Analyze the residuals from this experiment. Are the basic analysis of variance assumptions satisfied?

3.25 Four different designs for a digital computer circuit are being studied to compare the amount of noise present. The following data have been obtained:

Circuit Design	Noise Observed				
1	19	20	19	30	8
2	80	61	73	56	80
3	47	26	25	35	50
4	95	46	83	78	97

(a) Is the same amount of noise present for all four designs? Use $\alpha = 0.05$.

(b) Analyze the residuals from this experiment. Are the analysis of variance assumptions satisfied?

(c) Which circuit design would you select for use? Low noise is best.

3.26 Four chemists are asked to determine the percentage of methyl alcohol in a certain chemical compound. Each chemist makes three determinations, and the results are the following:

Chemist	Percentage of Methyl Alcohol		
1	84.99	84.04	84.38
2	85.15	85.13	84.88
3	84.72	84.48	85.16
4	84.20	84.10	84.55

(a) Do chemists differ significantly? Use $\alpha = 0.05$.

(b) Analyze the residuals from this experiment.

(c) If chemist 2 is a new employee, construct a meaningful set of orthogonal contrasts that might have been useful at the start of the experiment.

S **3.27** Three brands of batteries are under study. It is suspected that the lives (in weeks) of the three brands are different. Five randomly selected batteries of each brand are tested with the following results:

Weeks of Life		
Brand 1	Brand 2	Brand 3
100	76	108
96	80	100
92	75	96
96	84	98
92	82	100

(a) Are the lives of these brands of batteries different?

(b) Analyze the residuals from this experiment.

(c) Construct a 95 percent confidence interval estimate on the mean life of battery brand 2. Construct a 99 percent confidence interval estimate on the mean difference between the lives of battery brands 2 and 3.

(d) Which brand would you select for use? If the manufacturer will replace without charge any battery that fails in less than 85 weeks, what percentage would the company expect to replace?

3.28 Four catalysts that may affect the concentration of one component in a three-component liquid mixture are being investigated. The following concentrations are obtained from a completely randomized experiment:

Catalyst			
1	2	3	4
58.2	56.3	50.1	52.9
57.2	54.5	54.2	49.9
58.4	57.0	55.4	50.0
55.8	55.3		51.7
54.9			

(a) Do the four catalysts have the same effect on the concentration?

(b) Analyze the residuals from this experiment.

(c) Construct a 99 percent confidence interval estimate of the mean response for catalyst 1.

3.29 A semiconductor manufacturer has developed three different methods for reducing particle counts on wafers. All three methods are tested on five different wafers and the after treatment particle count obtained. The data are shown below:

Method	Count				
1	31	10	21	4	1
2	62	40	24	30	35
3	53	27	120	97	68

(a) Do all methods have the same effect on mean particle count?

(b) Plot the residuals versus the predicted response. Construct a normal probability plot of the residuals. Are there potential concerns about the validity of the assumptions?

(c) Based on your answer to part (b), conduct another analysis of the particle count data and draw appropriate conclusions.

3.30 Several ovens in a metal working shop are used to SS heat metal specimens. All the ovens are supposed to operate at the same temperature, although it is suspected that this may not be true. Three ovens are selected at random, and their

temperatures on successive heats are noted. The data collected are as follows:

Oven	Temperature					
1	491.50	498.30	498.10	493.50	493.60	
2	488.50	484.65	479.90	477.35		
3	490.10	484.80	488.25	473.00	471.85	478.65

(a) Is there significant variation in temperature between ovens? Use $\alpha = 0.05$.

(b) Estimate the components of variance for this model.

(c) Analyze the residuals from this experiment and draw conclusions about model adequacy.

3.31 An article in the *Journal of the Electrochemical Society* (Vol. 139, No. 2, 1992, pp. 524–532) describes an experiment to investigate the low-pressure vapor deposition of polysilicon. The experiment was carried out in a large-capacity reactor at Sematech in Austin, Texas. The reactor has several wafer positions, and four of these positions are selected at random. The response variable is film thickness uniformity. Three replicates of the experiment were run, and the data are as follows:

Wafer Position	Uniformity		
1	2.76	5.67	4.49
2	1.43	1.70	2.19
3	2.34	1.97	1.47
4	0.94	1.36	1.65

(a) Is there a difference in the wafer positions? Use $\alpha = 0.05$.

(b) Estimate the variability due to wafer positions.

(c) Estimate the random error component.

(d) Analyze the residuals from this experiment and comment on model adequacy.

3.32 Consider the vapor-deposition experiment described in Problem 3.31.

(a) Estimate the total variability in the uniformity response.

(b) How much of the total variability in the uniformity response is due to the difference between positions in the reactor?

(c) To what level could the variability in the uniformity response be reduced if the position-to-position variability in the reactor could be eliminated? Do you believe this is a significant reduction?

3.33 A single-factor completely randomized design has four levels of the factor. There are three replicates and the total sum of squares is 330.56. The treatment sum of squares is 250.65.

(a) What is the estimate of the error variance σ^2?

(b) What proportion of the variability in the response variable is explained by the treatment effect?

3.34 A single-factor completely randomized design has six levels of the factor. There are five replicates and the total sum of squares is 900.25. The treatment sum of squares is 750.50.

(a) What is the estimate of the error variance σ^2?

(b) What proportion of the variability in the response variable is explained by the treatment effect?

3.35 Find a 95% confidence interval on the intraclass correlation coefficient for the experiment in Problem 3.33.

3.36 Find a 95% confidence interval on the intraclass correlation coefficient for the experiment in Problem 3.34.

3.37 Consider testing the equality of the means of two normal populations, where the variances are unknown but are assumed to be equal. The appropriate test procedure is the pooled t-test. Show that the pooled t-test is equivalent to the single-factor analysis of variance.

3.38 Show that the variance of the linear combination $\sum_{i=1}^{a} c_i y_{i.}$ is $\sigma^2 \sum_{i=1}^{a} n_i c_i^2$.

3.39 In a fixed effects experiment, suppose that there are n observations for each of the four treatments. Let Q_1^2, Q_2^2, Q_3^2 be single-degree-of-freedom components for the orthogonal contrasts. Prove that $SS_{\text{Treatments}} = Q_1^2 + Q_2^2 + Q_3^2$.

3.40 Use Bartlett's test to determine if the assumption of equal variances is satisfied in Problem 3.27. Use $\alpha = 0.05$. Did you reach the same conclusion regarding equality of variances by examining residual plots?

3.41 Use the modified Levene test to determine if the assumption of equal variances is satisfied in Problem 3.27. Use $\alpha = 0.05$. Did you reach the same conclusion regarding the equality of variances by examining residual plots?

3.42 Refer to Problem 3.23. If we wish to detect a maximum difference in mean response times of 10 milliseconds with a probability of at least 0.90, what sample size should be used? How would you obtain a preliminary estimate of σ^2?

3.43 Refer to Problem 3.27.

(a) If we wish to detect a maximum difference in battery life of 10 hours with a probability of at least 0.90, what

sample size should be used? Discuss how you would obtain a preliminary estimate of σ^2 for answering this question.

(b) If the maximum difference between brands is 8 hours, what sample size should be used if we wish to detect this with a probability of at least 0.90?

3.44 Consider the experiment in Problem 3.27. If we wish to construct a 95 percent confidence interval on the difference in two mean battery lives that has an accuracy of ± 2 weeks, how many batteries of each brand must be tested?

3.45 Suppose that four normal populations have means of $\mu_1 = 50$, $\mu_2 = 60$, $\mu_3 = 50$, and $\mu_4 = 60$. How many observations should be taken from each population so that the probability of rejecting the null hypothesis of equal population means is at least 0.90? Assume that $\alpha = 0.05$ and that a reasonable estimate of the error variance is $\sigma^2 = 25$.

3.46 Refer to Problem 3.45.

(a) How would your answer change if a reasonable estimate of the experimental error variance were $\sigma^2 = 36$?

(b) How would your answer change if a reasonable estimate of the experimental error variance were $\sigma^2 = 49$?

(c) Can you draw any conclusions about the sensitivity of your answer in this particular situation about how your estimate of σ affects the decision about sample size?

(d) Can you make any recommendations about how we should use this general approach to choosing n in practice?

3.47 Refer to the aluminum smelting experiment described in Section 3.8.3. Verify that ratio control methods do not affect average cell voltage. Construct a normal probability plot of the residuals. Plot the residuals versus the predicted values. Is there an indication that any underlying assumptions are violated?

3.48 Refer to the aluminum smelting experiment in Section 3.8.3. Verify the ANOVA for pot noise summarized in Table 3.17. Examine the usual residual plots and comment on the experimental validity.

3.49 Four different feed rates were investigated in an experiment on a CNC machine producing a component part used in an aircraft auxiliary power unit. The manufacturing engineer in charge of the experiment knows that a critical part dimension of interest may be affected by the feed rate. However, prior experience has indicated that only dispersion effects are likely to be present. That is, changing the feed rate does not affect the *average* dimension, but it could affect dimensional variability. The engineer makes five production runs at each feed rate and obtains the standard deviation of the critical dimension (in 10^{-3} mm). The data are shown below. Assume that all runs were made in random order.

Feed Rate (in/min)	Production Run				
	1	2	3	4	5
10	0.09	0.10	0.13	0.08	0.07
12	0.06	0.09	0.12	0.07	0.12
14	0.11	0.08	0.08	0.05	0.06
16	0.19	0.13	0.15	0.20	0.11

(a) Does feed rate have any effect on the standard deviation of this critical dimension?

(b) Use the residuals from this experiment to investigate model adequacy. Are there any problems with experimental validity?

3.50 Consider the data shown in Problem 3.23.

(a) Write out the least squares normal equations for this problem and solve them for $\hat{\mu}$ and $\hat{\tau}_i$, using the usual constraint $\left(\sum_{i=1}^{3} \hat{\tau}_i = 0\right)$. Estimate $\tau_1 - \tau_2$.

(b) Solve the equations in (a) using the constraint $\hat{\tau}_3 = 0$. Are the estimators $\hat{\tau}_i$ and $\hat{\mu}$ the same as you found in (a)? Why? Now estimate $\tau_1 - \tau_2$ and compare your answer with that for (a). What statement can you make about estimating contrasts in the τ_i?

(c) Estimate $\mu + \tau_1$, $2\tau_1 - \tau_2 - \tau_3$, and $\mu + \tau_1 + \tau_2$ using the two solutions to the normal equations. Compare the results obtained in each case.

3.51 Apply the general regression significance test to the experiment in Example 3.6. Show that the procedure yields the same results as the usual analysis of variance.

3.52 Use the Kruskal–Wallis test for the experiment in Problem 3.24. Compare the conclusions obtained with those from the usual analysis of variance.

3.53 Use the Kruskal–Wallis test for the experiment in Problem 3.25/3.4R. Are the results comparable to those found by the usual analysis of variance?

3.54 Consider the experiment in Example 3.6S. Suppose that the largest observation on etch rate is incorrectly recorded as 250 Å/min. What effect does this have on the usual analysis of variance? What effect does it have on the Kruskal–Wallis test?

3.55 The normality assumption is extremely important in the analysis of variance.

(a) **True**

(b) **False**

3.56 The analysis of variance treats both quantitative and qualitative factors alike so far as the basic computations for sums of squares are concerned.

(a) True

(b) False

3.57 If a single-factor experiment has a levels of the factor and a polynomial of degree $a - 1$ is fit to the experimental data, the error sum of squares for the polynomial model will be exactly the same as the error sum of squares for the standard ANOVA.

(a) True

(b) False

3.58 Fisher's LSD procedure is an extremely conservative method for comparing pairs of treatment means following an ANOVA.

(a) True

(b) False

3.59 The REML method of estimating variance components is a technique based on maximum likelihood, while the ANOVA method is a method-of-moments procedure.

(a) True

(b) False

3.60 One advantage of the REML method of estimating variance components is that it automatically produces confidence intervals on the variance components.

(a) True

(b) False

3.61 The Tukey method is used to compare all treatment means to a control.

(a) True

(b) False

3.62 The estimate of the standard deviation of any observation in the experiment in Problem 3.10R is

(a) 7.03

(b) 2.65

(c) 5.91

(d) 1.95

(e) None of the above

Chapter 4 Problems

4.1 Suppose that a single-factor experiment with four levels of the factor has been conducted. There are six replicates and the experiment has been conducted in blocks. The error sum of squares is 500 and the block sum of squares is 250. If the experiment had been conducted as a completely randomized design the estimate of the error variance σ^2 would be.

(a) 25.0 **(b)** 25.5 **(c)** 35.0

(d) 37.5 **(e)** None of the above

SS **4.2** Suppose that a single-factor experiment with five levels of the factor has been conducted. There are three replicates and the experiment has been conducted as a complete randomized design. If the experiment had been conducted in blocks, the pure error degrees of freedom would be reduced by

(a) 3 **(b)** 5 **(c)** 2

(d) 4 **(e)** None of the above

4.3 Blocking is a technique that can be used to control the variability transmitted by uncontrolled nuisance factors in an experiment.

(a) True

(b) False

4.4 The number of blocks in the RCBD must always equal the number of treatments or factor levels.

(a) True

(b) False

4.5 The key concept of the phrase "Block if you can, randomize if you can't" is that:

(a) It is usually better to not randomize within blocks.

(b) Blocking violates the assumption of constant variance.

(c) Create blocks by using each level of the nuisance factor as a block and randomize within blocks.

(d) Randomizing the runs is preferable to randomizing blocks.

4.6 Consider the single-factor completely randomized experiment shown in Problem 3.7. Suppose that this experiment had been conducted in a randomized complete block design and that the sum of squares for blocks was 80.00. Modify the ANOVA for this experiment to show the correct analysis for the randomized complete block experiment.

4.7 A chemist wishes to test the effect of four chemical agents on the strength of a particular type of cloth. Because there might be variability from one bolt to another, the chemist decides to use a randomized block design, with the bolts of cloth considered as blocks. She selects five bolts and applies all four chemicals in random order to each bolt. The resulting tensile strengths follow. Analyze the data from this experiment (use $\alpha = 0.05$) and draw appropriate conclusions.

	Bolt				
Chemical	**1**	**2**	**3**	**4**	**5**
1	73	68	74	71	67
2	73	67	75	72	70
3	75	68	78	73	68
4	73	71	75	75	69

4.8 Three different washing solutions are being compared to study their effectiveness in retarding bacteria growth in 5-gallon milk containers. The analysis is done in a laboratory, and only three trials can be run on any day. Because days could represent a potential source of variability, the experimenter decides to use a randomized block design. Observations are taken for four days, and the data are shown here. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

	Days			
Solution	**1**	**2**	**3**	**4**
1	13	22	18	39
2	16	24	17	44
3	5	4	1	22

4.9 Plot the mean tensile strengths observed for each chemical type in Problem 4.7 and compare them to an appropriately scaled t distribution. What conclusions would you draw from this display?

4.10 Plot the average bacteria counts for each solution in Problem 4.8 and compare them to a scaled t distribution. What conclusions can you draw?

4.11 Consider the hardness testing experiment described in Section 4.1. Suppose that the experiment was conducted as described and that the following Rockwell C-scale data (coded by subtracting 40 units) obtained:

	Coupon			
Tip	1	2	3	4
1	9.3	9.4	9.6	10.0
2	9.4	9.3	9.8	9.9
3	9.2	9.4	9.5	9.7
4	9.7	9.6	10.0	10.2

(a) Analyze the data from this experiment.

(b) Use the Fisher LSD method to make comparisons among the four tips to determine specifically which tips differ in mean hardness readings.

(c) Analyze the residuals from this experiment.

4.12 A consumer products company relies on direct mail marketing pieces as a major component of its advertising campaigns. The company has three different designs for a new brochure and wants to evaluate their effectiveness, as there are substantial differences in costs between the three designs. The company decides to test the three designs by mailing 5000 samples of each to potential customers in four different regions of the country. Since there are known regional differences in the customer base, regions are considered as blocks. The number of responses to each mailing is as follows:

	Region			
Design	NE	NW	SE	SW
1	250	350	219	375
2	400	525	390	580
3	275	340	200	310

(a) Analyze the data from this experiment.

(b) Use the Fisher LSD method to make comparisons among the three designs to determine specifically which designs differ in the mean response rate.

(c) Analyze the residuals from this experiment.

4.13 The effect of three different lubricating oils on fuel economy in diesel truck engines is being studied. Fuel economy is measured using brake-specific fuel consumption after the engine has been running for 15 minutes. Five different truck engines are available for the study, and the experimenters conduct the following RCBD.

	Truck				
Oil	1	2	3	4	5
1	0.500	0.634	0.487	0.329	0.512
2	0.535	0.675	0.520	0.435	0.540
3	0.513	0.595	0.488	0.400	0.510

(a) Analyze the data from this experiment.

(b) Use the Fisher LSD method to make comparisons among the three lubricating oils to determine specifically which oils differ in brake-specific fuel consumption.

(c) Analyze the residuals from this experiment.

4.14 An article in *Communications of the ACM* (Vol. 30, No. 5, 1987) studied different algorithms for estimating software development costs. Six algorithms were applied to several different software development projects and the percent error in estimating the development cost was observed. Some of the data from this experiment is shown in the table below.

(a) Do the algorithms differ in their mean cost estimation accuracy?

(b) Analyze the residuals from this experiment.

(c) Which algorithm would you recommend for use in practice?

	Project					
Algorithm	1	2	3	4	5	6
1(SLIM)	1244	21	82	2221	905	839
2(COCOMO-A)	281	129	396	1306	336	910
3(COCOMO-R)	220	84	458	543	300	794
4(COCONO-C)	225	83	425	552	291	826
5(FUNCTION POINTS)	19	11	−34	121	15	103
6(ESTIMALS)	−20	35	−53	170	104	199

4.15 An article in *Nature Genetics* (2003, Vol. 34, pp. 85–90) "Treatment-Specific Changes in Gene Expression Discriminate in vivo Drug Response in Human Leukemia Cells"

studied gene expression as a function of different treatments for leukemia. Three treatment groups are as follows: mercaptopurine (MP) only; low-dose methotrexate (LDMTX) and MP; and high-dose methotrexate (HDMTX) and MP. Each group contained ten subjects. The responses from a specific gene are shown in the table below.

(a) Is there evidence to support the claim that the treatment means differ?

(b) Check the normality assumption. Can we assume these samples are from normal populations?

(c) Take the logarithm of the raw data. Is there evidence to support the claim that the treatment means differ for the transformed data?

(d) Analyze the residuals from the transformed data and comment on model adequacy.

Treatments	Observations									
MP ONLY	334.5	31.6	701	41.2	61.2	69.6	67.5	66.6	120.7	881.9
MP + HDMTX	919.4	404.2	1024.8	54.1	62.8	671.6	882.1	354.2	321.9	91.1
MP + LDMTX	108.4	26.1	240.8	191.1	69.7	242.8	62.7	396.9	23.6	290.4

4.16 Consider the ratio control algorithm experiment described in Section 3.8. The experiment was actually conducted as a randomized block design, where six time periods were selected as the blocks, and all four ratio control algorithms were tested in each time period. The average cell voltage and the standard deviation of voltage (shown in parentheses) for each cell are as follows:

Ratio Control Algorithm	Time Period		
	1	2	3
1	4.93 (0.05)	4.86 (0.04)	4.75 (0.05)
2	4.85 (0.04)	4.91 (0.02)	4.79 (0.03)
3	4.83 (0.09)	4.88 (0.13)	4.90 (0.11)
4	4.89 (0.03)	4.77 (0.04)	4.94 (0.05)

Ratio Control Algorithm	Time Period		
	4	5	6
1	4.95 (0.06)	4.79 (0.03)	4.88 (0.05)
2	4.85 (0.05)	4.75 (0.03)	4.85 (0.02)
3	4.75 (0.15)	4.82 (0.08)	4.90 (0.12)
4	4.86 (0.05)	4.79 (0.03)	4.76 (0.02)

(a) Analyze the average cell voltage data. (Use $\alpha = 0.05$.) Does the choice of ratio control algorithm affect the average cell voltage?

(b) Perform an appropriate analysis on the standard deviation of voltage. (Recall that this is called "pot noise.") Does the choice of ratio control algorithm affect the pot noise?

(c) Conduct any residual analyses that seem appropriate.

(d) Which ratio control algorithm would you select if your objective is to reduce both the average cell voltage *and* the pot noise?

4.17 An aluminum master alloy manufacturer produces grain refiners in ingot form. The company produces the product in four furnaces. Each furnace is known to have its own unique operating characteristics, so any experiment run in the foundry that involves more than one furnace will consider furnaces as a nuisance variable. The process engineers suspect that stirring rate affects the grain size of the product. Each furnace can be run at four different stirring rates. A randomized block design is run for a particular refiner, and the resulting grain size data are as follows.

Stirring Rate (rpm)	Furnace			
	1	2	3	4
5	8	4	5	6
10	14	5	6	9
15	14	6	9	2
20	17	9	3	6

(a) Is there any evidence that stirring rate affects grain size?

(b) Graph the residuals from this experiment on a normal probability plot. Interpret this plot.

(c) Plot the residuals versus furnace and stirring rate. Does this plot convey any useful information?

(d) What should the process engineers recommend concerning the choice of stirring rate and furnace for this particular grain refiner if small grain size is desirable?

4.18 Analyze the data in Problem 4.8 using the general regression significance test.

4.19 Assuming that chemical types and bolts are fixed, estimate the model parameters τ_i and β_j in Problem 4.7.

4.20 Draw an operating characteristic curve for the design in Problem 4.8. Does the test seem to be sensitive to small differences in the treatment effects?

4.21 Suppose that the observation for chemical type 2 and bolt 3 is missing in Problem 4.7. Analyze the problem by estimating the missing value. Perform the exact analysis and compare the results.

4.22 Consider the hardness testing experiment in Problem 4.11. Suppose that the observation for tip 2 in coupon 3 is missing. Analyze the problem by estimating the missing value.

4.23 An industrial engineer is conducting an experiment on eye focus time. He is interested in the effect of the distance of the object from the eye on the focus time. Four different distances are of interest. He has five subjects available for the experiment. Because there may be differences among individuals, he decides to conduct the experiment in a randomized block design. The data obtained follow. Analyze the data from this experiment (use $\alpha = 0.05$) and draw appropriate conclusions.

	Subject				
Distance (ft)	**1**	**2**	**3**	**4**	**5**
4	10	6	6	6	6
6	7	6	6	1	6
8	5	3	3	2	5
10	6	4	4	2	3

4.24 The effect of five different ingredients (A, B, C, D, E) on the reaction time of a chemical process is being studied. Each batch of new material is only large enough to permit five runs to be made. Furthermore, each run requires approximately $1\frac{1}{2}$ hours, so only five runs can be made in one day. The experimenter decides to run the experiment as a Latin square so that day and batch effects may be systematically controlled. She obtains the data that follow. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

	Day				
Batch	**1**	**2**	**3**	**4**	**5**
1	$A = 8$	$B = 7$	$D = 1$	$C = 7$	$E = 3$
2	$C = 11$	$E = 2$	$A = 7$	$D = 3$	$B = 8$
3	$B = 4$	$A = 9$	$C = 10$	$E = 1$	$D = 5$
4	$D = 6$	$C = 8$	$E = 6$	$B = 6$	$A = 10$
5	$E = 4$	$D = 2$	$B = 3$	$A = 8$	$C = 8$

4.25 An industrial engineer is investigating the effect of four assembly methods (A, B, C, D) on the assembly time for a color television component. Four operators are selected for the study. Furthermore, the engineer knows that each assembly method produces such fatigue that the time required for the last assembly may be greater than the time required for the first, regardless of the method. That is, a trend develops in the required assembly time. To account for this source of variability, the engineer uses the Latin square design that follows. Analyze the data from this experiment ($\alpha = 0.05$) and draw appropriate conclusions.

	Operator			
Order of Assembly	**1**	**2**	**3**	**4**
1	$C = 10$	$D = 14$	$A = 7$	$B = 8$
2	$B = 7$	$C = 18$	$D = 11$	$A = 8$
3	$A = 5$	$B = 10$	$C = 11$	$D = 9$
4	$D = 10$	$A = 10$	$B = 12$	$C = 14$

4.26 Consider the randomized complete block design in Problem 4.8. Assume that the days are random. Estimate the block variance component.

4.27 Consider the randomized complete block design in Problem 4.11. Assume that the coupons are random. Estimate the block variance component.

4.28 Consider the randomized complete block design in Problem 4.13. Assume that the trucks are random. Estimate the block variance component.

4.29 Consider the randomized complete block design in Problem 4.14. Assume that the software projects that were used as blocks are random. Estimate the block variance component.

4.30 Consider the gene expression experiment in Problem 4.15. Assume that the subjects used in this experiment are random. Estimate the block variance component.

4.31 Suppose that in Problem 4.24 the observation from batch 3 on day 4 is missing. Estimate the missing value and perform the analysis using the value.

4.32 Consider a $p \times p$ Latin square with rows (α_i), columns (β_k), and treatments (τ_j) fixed. Obtain least squares estimates of the model parameters α_i, β_k, and τ_j.

4.33 Derive the missing value formula (Equation 4.28) for the Latin square design.

4.34 ***Designs involving several Latin squares.*** [See Cochran and Cox (1957), John (1971).] The $p \times p$ Latin square contains only p observations for each treatment. To obtain more replications, the experimenter may use several squares, say n. It is immaterial whether the squares used are the same or different. The appropriate model is

$$y_{ijkh} = \begin{matrix} \mu + \rho_h + \alpha_{i(h)} \\ +\tau_j + \beta_{k(h)} \\ +(\tau\rho)_{jh} + \varepsilon_{ijkh} \end{matrix} \begin{cases} i = 1,2,\ldots,p \\ j = 1,2,\ldots,p \\ k = 1,2,\ldots,p \\ h = 1,2,\ldots,n \end{cases}$$

where y_{ijkh} is the observation on treatment j in row i and column k of the hth square. Note that $\alpha_{i(h)}$ and $\beta_{k(h)}$ are the row and column effects in the hth square, ρ_h is the effect of the hth square, and $(\tau\rho)_{jh}$ is the interaction between treatments and squares.

(a) Set up the normal equations for this model and solve for estimates of the model parameters. Assume that appropriate side conditions on the parameters are $\sum_h \hat{\rho}_h = 0$, $\sum_i \hat{\alpha}_{i(h)} = 0$, and $\sum_k \beta_{k(h)} = 0$ for each h, $\sum_j \hat{\tau}_j = 0$, $\sum_j (\hat{\tau}\rho)_{jh} = 0$ for each h, and $\sum_h (\hat{\tau}\rho)_{jh} = 0$ for each j.

(b) Write down the analysis of variance table for this design.

4.35 Discuss how you would determine the sample size for use with the Latin square design.

4.36 Suppose that in Problem 4.24 the data taken on day 5 were incorrectly analyzed and had to be discarded. Develop an appropriate analysis for the remaining data.

4.37 The yield of a chemical process was measured using five batches of raw material, five acid concentrations, five standing times (A, B, C, D, E), and five catalyst concentrations ($\alpha, \beta, \gamma, \delta, \varepsilon$). The Graeco-Latin square that follows was used. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

	Acid Concentration		
Batch	**1**	**2**	**3**
1	$A\alpha = 26$	$B\beta = 16$	$C\gamma = 19$
2	$B\gamma = 18$	$C\delta = 21$	$D\varepsilon = 18$
3	$C\varepsilon = 20$	$D\alpha = 12$	$E\beta = 16$
4	$D\beta = 15$	$E\gamma = 15$	$A\delta = 22$
5	$E\delta = 10$	$A\varepsilon = 24$	$B\alpha = 17$

	Acid Concentration	
Batch	**4**	**5**
1	$D\delta = 16$	$E\varepsilon = 13$
2	$E\alpha = 11$	$A\beta = 21$
3	$A\gamma = 25$	$B\delta = 13$
4	$B\varepsilon = 14$	$C\alpha = 17$
5	$C\beta = 17$	$D\gamma = 14$

4.38 Suppose that in Problem 4.25 the engineer suspects that the workplaces used by the four operators may represent an additional source of variation. A fourth factor, workplace ($\alpha, \beta, \gamma, \delta$) may be introduced and another experiment conducted, yielding the Graeco-Latin square that follows. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

	Operator			
Order of Assembly	**1**	**2**	**3**	**4**
1	$C\beta = 11$	$B\gamma = 10$	$D\delta = 14$	$A\alpha = 8$
2	$B\alpha = 8$	$C\delta = 12$	$A\gamma = 10$	$D\beta = 12$
3	$A\delta = 9$	$D\alpha = 11$	$B\beta = 7$	$C\gamma = 15$
4	$D\gamma = 9$	$A\beta = 8$	$C\alpha = 18$	$B\delta = 6$

4.39 Construct a 5×5 hypersquare for studying the effects of five factors. Exhibit the analysis of variance table for this design.

4.40 Consider the data in Problems 4.25 and 4.38. Suppressing the Greek letters in problem 4.38, analyze the data using the method developed in Problem 4.34.

4.41 Consider the randomized block design with one missing value in Problem 4.22. Analyze this data by using the exact analysis of the missing value problem discussed in Section 4.1.4. Compare your results to the approximate analysis of these data given from Problem 4.22.

4.42 An engineer is studying the mileage performance characteristics of five types of gasoline additives. In the road test, he wishes to use cars as blocks; however, because of a time constraint, he must use an incomplete block design. He runs the balanced design with the five blocks that follow. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions. SS

	Car				
Additive	**1**	**2**	**3**	**4**	**5**
1		17	14	13	12
2	14	14		13	10
3	12		13	12	9
4	13	11	11	12	
5	11	12	10		8

4.43 Construct a set of orthogonal contrasts for the data in Problem 4.42. Compute the sum of squares for each contrast. SS

4.44 Seven different hardwood concentrations are being studied to determine their effect on the strength of the paper produced. However, the pilot plant can only produce three runs each day. As days may differ, the analyst uses the BIBD that SS

follows. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

Hardwood Concentration (%)	Days 1	2	3	4
2	114			
4	126	120		
6		137	117	
8	141		129	149
10		145		150
12			120	
14				136

Hardwood Concentration (%)	Days 5	6	7
2	120		117
4		119	
6			134
8			
10	143		
12	118	123	
14		130	127

4.45 Analyze the data in Example 4.4 using the general regression significance test.

4.46 Prove that $k\sum_{i=1}^{a} Q_i^2/(\lambda a)$ is the adjusted sum of squares for treatments in a BIBD.

4.47 An experimenter wishes to compare four treatments in blocks of two runs. Find a BIBD for this experiment with six blocks.

4.48 An experimenter wishes to compare eight treatments in blocks of four runs. Find a BIBD with 14 blocks and $\lambda = 3$.

4.49 Perform the interblock analysis for the design in Problem 4.42.

4.50 Perform the interblock analysis for the design in Problem 4.44.

4.51 Verify that a BIBD with the parameters $a = 8$, $r = 8, k = 4$, and $b = 16$ does not exist.

4.52 Show that the variance of the intrablock estimators $\{\hat{\tau}_i\}$ is $k(a-1)\sigma^2/(\lambda a^2)$.

4.53 Suppose that a single-factor experiment with five levels of the factor has been conducted. There are three replicates and the experiment has been conducted as a complete randomized design. If the experiment had been conducted in blocks, the pure error degrees of freedom would be reduced by (choose the correct answer):

(a) 3 **(b)** 5 **(c)** 2

(d) 4 **(e)** none of the above

4.54 Physics graduate student Laura Van Ertia has conducted a complete randomized design with a single factor, hoping to solve the mystery of the unified theory and complete her dissertation. The results of this experiment are summarized in the following ANOVA display:

Source	DF	SS	MS	F
Factor	?	?	14.18	?
Error	?	37.75	?	
Total	23	108.63		

Answer the following questions about this experiment.

(a) The sum of squares for the factor is _____.

(b) The number of degrees of freedom for the single factor in the experiment is _____.

(c) The number of degrees of freedom for error is _____.

(d) The mean square for error is _____.

(e) The value of the test statistic is _____.

(f) If the significance level is 0.05, your conclusions are not to reject the null hypothesis.

Yes

No

(g) An upper bound on the P-value for the test statistic is _____.

(h) A lower bound on the P-value for the test statistic is _____.

(i) Laura used _____ levels of the factor in this experiment.

(j) Laura replicated this experiment _____ times.

(k) Suppose that Laura had actually conducted this experiment as a randomized complete block design and the sum of squares for blocks was 12. Reconstruct the ANOVA display above to reflect this new situation. How much has blocking reduced the estimate of experimental error?

Chapter 5 Problems

5.1 An interaction effect in the model from a factorial experiment involving quantitative factors is a way of incorporating curvature into the response surface model representation of the results.

(a) True

(b) False

5.2 A factorial experiment may be conducted as a RCBD by running each replicate of the experiment in a unique block.

(a) True

(b) False

5.3 If an interaction effect in a factorial experiment is significant, the main effects of the factors involved in that interaction are difficult to interpret individually.

(a) True

(b) False

5.4 A biomedical researcher has conducted a two-factor factorial experiment as part of the research to develop a new product. She performed the statistical analysis using a computer software package. A portion of the output is shown below:

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	*F* Value	Prob > *F*
Model	874.00	5	174.80	3.28	0.0904
A	*776.00*	*?*	*388.00*	*7.27*	*0.0249*
B	*5.33*	*1*	*5.33*	*0.10*	*0.7625*
AB	*92.67*	*2*	*46.33*	*0.87*	*0.4663*
Pure Error	320.00	?	53.33		
Cor Total	1194.00	11			

(a) Interpret the *F*-statistic in the "Model" row of the ANOVA. Specifically, what hypothesies are being tested?

(b) What conclusions should be drawn regarding the individual model effects?

(c) How many levels of factor *A* were used in this experiment?

(d) How many replicates were run?

5.5 Consider the following incomplete ANOVA table:

Source	SS	DF	MS	*F*
A	?	1	50.00	?
B	80.00	?	40.00	?
AB	30.00	2	15.00	?
Error	?	12	?	
Total	172.00	17		

(a) Complete the ANOVA calculations.

(b) Provide an interpretation of this experiment.

(c) The pure error estimate of the standard deviation of the sample observations is 1.

True

False

5.6 The following output was obtained from a computer program that performed a two-factor ANOVA on a factorial experiment.

SS

Two-way ANOVA: y versus A, B

Source	DF	SS	MS	F	P
A	1	?	0.0002	?	?
B	?	180.378	?	?	?
Interaction	3	8.479	?	?	0.932
Error	8	158.797	?		
Total	15	347.653			

(a) Fill in the blanks in the ANOVA table. You can use bounds on the *P*-values.

(b) How many levels were used for factor *B*?

(c) How many replicates of the experiment were performed?

(d) What conclusions would you draw about this experiment?

5.7 The yield of a chemical process is being studied. The two most important variables are thought to be pressure and temperature. Three levels of each factor are selected, and a factorial experiment with two replicates is performed. The yield data are as follows:

	Pressure (psig)		
Temperature (°C)	**200**	**215**	**230**
150	90.4	90.7	90.2
	90.2	90.6	90.4
160	90.1	90.5	89.9
	90.3	90.6	90.1
170	90.5	90.8	90.4
	90.7	90.9	90.1

(a) Analyze the data and draw conclusions. Use $\alpha = 0.05$.

(b) Prepare appropriate residual plots and comment on the model's adequacy.

(c) Under what conditions would you operate this process?

5.8 An engineer suspects that the surface finish of a metal part is influenced by the feed rate and the depth of cut. He selects three feed rates and four depths of cut. He then conducts a factorial experiment and obtains the following data:

Feed Rate	Depth of Cut (in)			
(in/min)	**0.15**	**0.18**	**0.20**	**0.25**
	74	79	82	99
0.20	64	68	88	104
	60	73	92	96
	92	98	99	104
0.25	86	104	108	110
	88	88	95	99
	99	104	108	114
0.30	98	99	110	111
	102	95	99	107

(a) Analyze the data and draw conclusions. Use $\alpha = 0.05$.

(b) Prepare appropriate residual plots and comment on the model's adequacy.

(c) Obtain point estimates of the mean surface finish at each feed rate.

(d) Find the P-values for the tests in part (a).

5.9 For the data in Problem 5.8, compute a 95 percent confidence interval estimate of the mean difference in response for feed rates of 0.20 and 0.25 in/min.

5.10 An article in *Industrial Quality Control* (1956, pp. 5–8) SS describes an experiment to investigate the effect of the type of glass and the type of phosphor on the brightness of a television tube. The response variable is the current necessary (in microamps) to obtain a specified brightness level. The data are as follows:

	Phosphor Type		
Glass Type	**1**	**2**	**3**
	280	300	290
1	290	310	285
	285	295	290
	230	260	220
2	235	240	225
	240	235	230

(a) Is there any indication that either factor influences brightness? Use $\alpha = 0.05$.

(b) Do the two factors interact? Use $\alpha = 0.05$.

(c) Analyze the residuals from this experiment.

5.11 The factors that influence the breaking strength of a SS synthetic fiber are being studied. Four production machines and three operators are chosen and a factorial experiment is run using fiber from the same production batch. The results are as follows:

	Machine			
Operator	**1**	**2**	**3**	**4**
1	109	110	108	110
	110	115	109	108
2	110	110	111	114
	112	111	109	112
3	116	112	114	120
	114	115	119	117

(a) Analyze the data and draw conclusions. Use $\alpha = 0.05$.

(b) Prepare appropriate residual plots and comment on the model's adequacy.

5.12 An experiment is conducted to study the influence of operating temperature and three types of faceplate glass in the light output of an oscilloscope tube. The following data are collected:

	Temperature		
Glass Type	**100**	**125**	**150**
	580	1090	1392
1	568	1087	1380
	570	1085	1386
	550	1070	1328
2	530	1035	1312
	579	1000	1299
	546	1045	867
3	575	1053	904
	599	1066	889

(a) Use $\alpha = 0.05$ in the analysis. Is there a significant interaction effect? Does glass type or temperature affect the response? What conclusions can you draw?

(b) Fit an appropriate model relating light output to glass type and temperature.

(c) Analyze the residuals from this experiment. Comment on the adequacy of the models you have considered.

5.13 Consider the experiment in Problem 5.7. Fit an appropriate model to the response data. Use this model to provide guidance concerning operating conditions for the process.

5.14 Use Tukey's test to determine which levels of the pressure factor are significantly different for the data in Problem 5.7.

5.15 An experiment was conducted to determine whether either firing temperature or furnace position affects the baked density of a carbon anode. The data are shown below:

	Temperature (°C)		
Position	**800**	**825**	**850**
	570	1063	565
1	565	1080	510
	583	1043	590
	528	988	526
2	547	1026	538
	521	1004	532

Suppose we assume that no interaction exists. Write down the statistical model. Conduct the ANOVA and test hypotheses on the main effects. What conclusions can be drawn? Comment on the model's adequacy.

5.16 Derive the expected mean squares for a two-factor analysis of variance with one observation per cell, assuming that both factors are fixed.

5.17 Consider the following data from a two-factor factorial experiment. Analyze the data and draw conclusions. Perform a test for nonadditivity. Use $\alpha = 0.05$.

	Column Factor			
Row Factor	**1**	**2**	**3**	**4**
1	36	39	36	32
2	18	20	22	20
3	30	37	33	34

5.18 The shear strength of an adhesive is thought to be affected by the application pressure and temperature. A factorial experiment is performed in which both factors are assumed to be fixed. Analyze the data and draw conclusions. Perform a test for nonadditivity.

	Temperature (°F)		
Pressure (lb/in²)	**250**	**260**	**270**
120	9.60	11.28	9.00
130	9.69	10.10	9.57
140	8.43	11.01	9.03
150	9.98	10.44	9.80

5.19 Consider the three-factor model

SS

$$y_{ijk} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\beta\gamma)_{jk} + \epsilon_{ijk} \quad \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, b \\ k = 1, 2, \ldots, c \end{cases}$$

Notice that there is only one replicate. Assuming all the factors are fixed, write down the ANOVA table, including the expected mean squares. What would you use as the "experimental error" to test hypotheses?

5.20 The percentage of hardwood concentration in raw pulp, the vat pressure, and the cooking time of the pulp are being investigated for their effects on the strength of paper. Three levels of hardwood concentration, three levels of pressure, and two cooking times are selected. A factorial experiment with two replicates is conducted, and the following data are obtained:

Percentage of Hardwood Concentration	Cooking Time 3.0 Hours		
	Pressure		
	400	500	650
2	196.6	197.7	199.8
	196.0	196.0	199.4
4	198.5	196.0	198.4
	197.2	196.9	197.6
8	197.5	195.6	197.4
	196.6	196.2	198.1

Percentage of Hardwood Concentration	Cooking Time 4.0 Hours		
	Pressure		
	400	500	650
2	198.4	199.6	200.6
	198.6	200.4	200.9
4	197.5	198.7	199.6
	198.1	198.0	199.0
8	197.6	197.0	198.5
	198.4	197.8	199.8

(a) Analyze the data and draw conclusions. Use $\alpha = 0.05$.

(b) Prepare appropriate residual plots and comment on the model's adequacy.

(c) Under what set of conditions would you operate this process? Why?

5.21 The quality control department of a fabric finishing plant is studying the effect of several factors on the dyeing of cotton–synthetic cloth used to manufacture men's shirts. Three operators, three cycle times, and two temperatures were selected, and three small specimens of cloth were dyed under each set of conditions. The finished cloth was compared to a standard, and a numerical score was assigned. The results are as follows. Analyze the data and draw conclusions. Comment on the model's adequacy.

	Temperature					
	300°C			350°C		
	Operator			Operator		
Cycle Time	1	2	3	1	2	3
	23	27	31	24	38	34
40	24	28	32	23	36	36
	25	26	29	28	35	39
	36	34	33	37	34	34
50	35	38	34	39	38	36
	36	39	35	35	36	31
	28	35	26	26	36	28
60	24	35	27	29	37	26
	27	34	25	25	34	24

5.22 In Problem 5.7, suppose that we wish to reject the null hypothesis with a high probability if the difference in the true mean yield at any two pressures is as great as 0.5. If a reasonable prior estimate of the standard deviation of yield is 0.1, how many replicates should be run?

5.23 Consider the data in Problem 5.10. Analyze the data, assuming that replicates are blocks.

5.24 Consider the data in Problem 5.11. Analyze the data, assuming that replicates are blocks. SS

5.25 An article in the *Journal of Testing and Evaluation* (Vol. 16, no. 2, pp. 508–515) investigated the effects of cyclic loading and environmental conditions on fatigue crack growth at a constant 22 MPa stress for a particular material. The data from this experiment are shown below (the response is crack growth rate): SS

Frequency	Environment		
	Air	H_2O	Salt H_2O
	2.29	2.06	1.90
10	2.47	2.05	1.93
	2.48	2.23	1.75
	2.12	2.03	2.06
	2.65	3.20	3.10
1	2.68	3.18	3.24
	2.06	3.96	3.98
	2.38	3.64	3.24
	2.24	11.00	9.96
0.1	2.71	11.00	10.01
	2.81	9.06	9.36
	2.08	11.30	10.40

(a) Analyze the data from this experiment (use $\alpha = 0.05$).

(b) Analyze the residuals.

(c) Repeat the analyses from parts (a) and (b) using ln (y) as the response. Comment on the results.

5.26 An article in the *IEEE Transactions on Electron Devices* (Nov. 1986, pp. 1754) describes a study on polysilicon doping. The experiment shown below is a variation of their study. The response variable is base current.

Polysilicon Doping (ions)	Anneal Temperature (°C)		
	900	950	1000
1×10^{20}	4.60	10.15	11.01
	4.40	10.20	10.58
2×10^{20}	3.20	9.38	10.81
	3.50	10.02	10.60

(a) Is there evidence (with $\alpha = 0.05$) indicating that either polysilicon doping level or anneal temperature affects base current?

(b) Prepare graphical displays to assist in interpreting this experiment.

(c) Analyze the residuals and comment on model adequacy.

(d) Is the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \varepsilon$$

supported by this experiment (x_1 = doping level, x_2 = temperature)? Estimate the parameters in this model and plot the response surface.

5.27 An experiment was conducted to study the life (in hours) of two different brands of batteries in three different devices (radio, camera, and portable DVD player). A completely randomized two-factor factorial experiment was conducted and the following data resulted.

Brand of Battery	Device		
	Radio	Camera	DVD Player
A	8.6	7.9	5.4
	8.2	8.4	5.7
B	9.4	8.5	5.8
	8.8	8.9	5.9

(a) Analyze the data and draw conclusions, using $\alpha = 0.05$.

(b) Investigate model adequacy by plotting the residuals.

(c) Which brand of batteries would you recommend?

5.28 A manufacturer of laundry products is investigating the performance of a newly formulated stain remover. The new formulation is compared to the original formulation with respect to its ability to remove a standard tomato-like stain in a test article of cotton cloth using a factorial experiment. The other factors in the experiment are the number of times the test article is washed (1 or 2) and whether or not a detergent booster is used. The response variable is the stain shade after washing (12 is the darkest, 0 is the lightest). The data are shown in the following table.

Formulation	Number of Washings		Number of Washings	
	1		2	
	Booster		Booster	
	Yes	No	Yes	No
New	6, 5	6, 5	3, 2	4, 1
Original	10, 9	11, 11	10, 9	9, 10

(a) Conduct an analysis of variance. Using $\alpha = 0.05$, what conclusions can you draw?

(b) Investigate model adequacy by plotting the residuals.

5.29 Bone anchors are used by orthopedic surgeons in repairing torn rotator cuffs (a common shoulder tendon injury among baseball players). The bone anchor is a threaded insert that is screwed into a hole that has been drilled into the shoulder bone near the site of the torn tendon. The torn tendon is then sutured to the anchor. In a successful operation, the tendon is stabilized and reattaches itself to the bone. However, bone anchors can pull out if they are subjected to high loads. An experiment was performed to study the force required to pull out the anchor for three anchor types and two different foam densities (the foam simulates the natural variability found in real bone). Two replicates of the experiment were performed. The experimental design and the pullout force response data are as follows.

Anchor Type	Foam Density	
	Low	High
A	190, 200	241, 255
B	185, 190	230, 237
C	210, 205	256, 260

(a) Analyze the data from this experiment.

(b) Investigate model adequacy by constructing appropriate residual plots.

(c) What conclusions can you draw?

5.30 An experiment was performed to investigate the keyboard feel on a computer (crisp or mushy) and the size of the keys (small, medium, or large). The response variable is typing speed. Three replicates of the experiment were performed. The experimental design and the data are as follow.

	Keyboard Feel	
Key Size	**Mushy**	**Crisp**
Small	31, 33, 35	36, 40, 41
Medium	36, 35, 33	40, 41, 42
Large	37, 34, 33	38, 36, 39

(a) Analyze the data from this experiment.

(b) Investigate model adequacy by constructing appropriate residual plots.

(c) What conclusions can you draw?

5.31 An article in *Quality Progress* (May 2011, pp. 42–48) describes the use of factorial experiments to improve a silver powder production process. This product is used in conductive pastes to manufacture a wide variety of products ranging from silicon wafers to elastic membrane switches. Powder density (g/cm^2) and surface area (cm^2/g) are the two critical characteristics of this product. The experiments involved three factors—reaction temperature, ammonium percent, and stirring rate. Each of these factors had two levels and the design was replicated twice. The design is shown below.

Ammonium (%)	Stir Rate (RPM)	Temperature (°C)	Density	Surface Area
2	100	8	14.68	0.40
2	100	8	15.18	0.43
30	100	8	15.12	0.42
30	100	8	17.48	0.41
2	150	8	7.54	0.69
2	150	8	6.66	0.67
30	150	8	12.46	0.52
30	150	8	12.62	0.36
2	100	40	10.95	0.58
2	100	40	17.68	0.43
30	100	40	12.65	0.57
30	100	40	15.96	0.54
2	150	40	8.03	0.68
2	150	40	8.84	0.75
30	150	40	14.96	0.41
30	150	40	14.96	0.41

(a) Analyze the density response. Are any interactions significant? Draw appropriate conclusions about the effects of the significant factors on the response.

(b) Prepare appropriate residual plots and comment on model adequacy.

(c) Construct contour plots to aid in practical interpretation of the density response.

(d) Analyze the surface area response. Are any interactions significant? Draw appropriate conclusions about the effects of the significant factors on the response.

(e) Prepare appropriate residual plots and comment on model adequacy.

(f) Construct contour plots to aid in practical interpretation of the surface area response.

5.32 Continuation of Problem 5.31. Suppose that the specifications require that surface area must be between 0.3 and 0.6 cm^2/g and that density must be less than 14 g/cm^3. Find a set of operating conditions that will result in a product that meets these requirements.

5.33 An article in *Biotechnology Progress* (2001, Vol. 17, pp. 366–368) described an experiment to investigate nisin extraction in aqueous two-phase solutions. A two-factor factorial experiment was conducted using factors A = concentration of PEG and B = concentration of Na_2SO_4. Data similar to that reported in the paper are shown below.

A	*B*	Extraction (%)
13	11	62.9
13	11	65.4
15	11	76.1
15	11	72.3
13	13	87.5
13	13	84.2
15	13	102.3
15	13	105.6

(a) Analyze the extraction response. Draw appropriate conclusions about the effects of the significant factors on the response.

(b) Prepare appropriate residual plots and comment on model adequacy.

(c) Construct contour plots to aid in practical interpretation of the density response.

5.34 Reconsider the experiment in Problem 5.2R. Suppose that this experiment had been conducted in three blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell

comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.35 Reconsider the experiment in Problem 5.9. Suppose that this experiment had been conducted in three blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.36 Reconsider the experiment in Problem 5.11. Suppose that this experiment had been conducted in two blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.37 Reconsider the three-factor factorial experiment in Problem 5.20. Suppose that this experiment had been conducted in two blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.38 Reconsider the three-factor factorial experiment in Problem 5.21. Suppose that this experiment had been conducted in three blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.39 Reconsider the bone anchor experiment in Problem 5.29. Suppose that this experiment had been conducted in two blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.40 Reconsider the keyboard experiment in Problem 5.30. Suppose that this experiment had been conducted in three blocks, with each replicate a block. Assume that the observations in the data table are given in order, that is, the first observation in each cell comes from the first replicate, and so on. Reanalyze the data as a factorial experiment in blocks and estimate the variance component for blocks. Does it appear that blocking was useful in this experiment?

5.41 The C. F. Eye Care company manufactures lenses for transplantation into the eye following cataract surgery. An engineering group has conducted an experiment involving two factors to determine their effect on the lens polishing process. The results of this experiment are summarized in the following ANOVA display:

Source	DF	SS	MS	F	P-Value
Factor A	?	?	0.0833	0.05	0.952
Factor B	?	96.333	96.3333	57.80	<0.001
Interaction	2	12.167	6.0833	3.65	?
Error	6	10.000	?	?	
Total	11	118.667			

Answer the following questions about this experiment.

(a) The sum of squares for factor A is ____.

(b) The number of degrees of freedom for factor A in the experiment is ____.

(c) The number of degrees of freedom for factor B is ____.

(d) The mean square for error is ____.

(e) An upper bound for the P-value for the interaction test statistic is ____.

(f) The engineers used ____ levels of the factor A in this experiment.

(g) The engineers used ____ levels of the factor B in this experiment.

(h) There are ____ replicates of this experiment.

(i) Would you conclude that the effect of factor B depends on the level of factor A?

Yes

No

(j) An estimate of the standard deviation of the response variable is ____.

5.42 Reconsider the lens polishing experiment in Problem 5.41. Suppose that this experiment had been conducted as a randomized complete block design. The sum of squares for blocks is 4.00. Reconstruct the ANOVA given this new information. What impact does the blocking have on the conclusions from the original experiment?

5.43 In Problem 4.54 you met physics PhD student Laura Van Ertia who had conducted a single-factor experiment in her pursuit of the unified theory. She is at it again, and this time she has moved on to a two-factor factorial conducted as a completely randomized design. From her experiment, Laura has constructed the following incomplete ANOVA display:

Source	SS	DF	MS	*F*
A	350.00	2	?	?
B	300.00	?	150	?
AB	200.00	?	50	?
Error	150.00	18		
Total	1000.00			

(a) How many levels of factor *B* did she use in the experiment?

(b) How many degrees of freedom are associated with interaction?

(c) The error mean square is ____.

(d) The mean square for factor *A* is ____.

(e) How many replicates of the experiment were conducted? ____

(f) What are your conclusions about interaction and the two main effects?

(g) An estimate of the standard deviation of the response variable is ____.

(h) If this experiment had been run in blocks there would have been ____ degrees of freedom for blocks.

5.44 Continuation of Problem 5.43. Suppose that Laura did actually conduct the experiment in Problem 5.43 as a randomized complete block design. Assume that the block sum of squares is 60.00. Reconstruct the ANOVA display under this new set of assumptions.

5.45 Consider the following incomplete ANOVA table:

Source	SS	DF	MS	*F*
A	50.00	1	50.00	?
B	80.00	2	40.00	?
AB	30.00	2	15.00	?
Error	?	12	?	
Total	172.00	17		

In addition to the ANOVA table, you know that the experiment has been replicated three times and that the totals of the three replicates are 10, 12, and 14, respectively. The original experiment was run as a completely randomized design. Answer the following questions:

(a) The pure error estimate of the standard deviation of the sample observations is 1.

Yes

No

(b) Suppose that the experiment had been run in blocks, so that it is an randomized complete block design. The number of degrees of freedom for blocks would be ____.

(c) The block sum of squares is ____.

(d) The error sum of squares in the randomized complete block design is now ____.

(e) For the randomized complete block design, what is the estimate of the standard deviation of the sample observations?

5.46 Consider the following incomplete ANOVA table:

Source	SS	DF	MS	*F*
A	50.00	1	50.00	?
B	80.00	2	40.00	?
AB	30.00	2	15.00	?
Blocks	10.00	1	?	
Error	?	?	?	
Total	185.00	11		

(a) The pure error estimate of the standard deviation of the sample observations is 1.73.

True

False

(b) Suppose that the experiment had not been run in blocks; that is, it is now a CRD. The number of degrees of freedom for error would now be ____.

(c) The error mean square in the CRD would be ____.

(d) The F-test statistic for interaction in the CRD is significant at $\alpha = 0.05$.

True

False

Chapter 6 Problems

6.1 In a 2^4 factorial design, the number of degrees of freedom for the model, assuming the complete factorial model, is

(a) 7 **(b)** 5 **(c)** 6

(d) 11 **(e)** 12 **(f)** none of the above

6.2 A 2^3 factorial is replicated twice. The number of pure error or residual degrees of freedom is

(a) 4 **(b)** 12 **(c)** 15

(d) 2 **(e)** 8 **(f)** none of the above

6.3 A 2^3 factorial is replicated twice. The ANOVA indicates that all main effects are significant but the interactions are not significant. The interaction terms are dropped from the model. The number of residual degrees of freedom for the reduced model is

(a) 12 **(b)** 8 **(c)** 16

(d) 14 **(e)** 10 **(f)** none of the above

6.4 A 2^3 factorial is replicated three times. The ANOVA indicates that all main effects are significant but two of the interactions are not significant. The interaction terms are dropped from the model. The number of residual degrees of freedom for the reduced model is

(a) 12 **(b)** 14 **(c)** 6

(d) 10 **(e)** 8 **(f)** none of the above

SS **6.5** An engineer is interested in the effects of cutting speed (A), tool geometry (B), and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a 2^3 factorial design are run. The results are as follows:

			Treatment	Replicate		
A	*B*	*C*	Combination	I	II	III
−	−	−	(1)	22	31	25
+	−	−	*a*	32	43	29
−	+	−	*b*	35	34	50
+	+	−	*ab*	55	47	46
−	−	+	*c*	44	45	38
+	−	+	*ac*	40	37	36
−	+	+	*bc*	60	50	54
+	+	+	*abc*	39	41	47

(a) Estimate the factor effects. Which effects appear to be large?

(b) Use the analysis of variance to confirm your conclusions for part (a).

(c) Write down a regression model for predicting tool life (in hours) based on the results of this experiment.

(d) Analyze the residuals. Are there any obvious problems?

(e) On the basis of an analysis of main effect and interaction plots, what coded factor levels of A, B, and C would you recommend using?

6.6 Reconsider part (c) of Problem 6.5. Use the regression model to generate response surface and contour plots of the tool life response. Interpret these plots. Do they provide insight regarding the desirable operating conditions for this process? SS

6.7 Find the standard error of the factor effects and approximate 95 percent confidence limits for the factor effects in Problem 6.5. Do the results of this analysis agree with the conclusions from the analysis of variance?

6.8 Plot the factor effects from Problem 6.5 on a graph relative to an appropriately scaled t distribution. Does this graphical display adequately identify the important factors? Compare the conclusions from this plot with the results from the analysis of variance.

SS **6.9** A router is used to cut locating notches on a printed circuit board. The vibration level at the surface of the board as it is cut is considered to be a major source of dimensional variation in the notches. Two factors are thought to influence vibration: bit size (A) and cutting speed (B). Two bit sizes ($\frac{1}{16}$ and $\frac{1}{8}$in.) and two speeds (40 and 90 rpm) are selected, and four boards are cut at each set of conditions shown below. The response variable is vibration measured as the resultant vector of three accelerometers (x, y, and z) on each test circuit board.

		Treatment	Replicate			
A	B	Combination	I	II	III	IV
−	−	(1)	18.2	18.9	12.9	14.4
+	−	a	27.2	24.0	22.4	22.5
−	+	b	15.9	14.5	15.1	14.2
+	+	ab	41.0	43.9	36.3	39.9

(a) Analyze the data from this experiment.

(b) Construct a normal probability plot of the residuals, and plot the residuals versus the predicted vibration level. Interpret these plots.

(c) Draw the AB interaction plot. Interpret this plot. What levels of bit size and speed would you recommend for routine operation?

SS **6.10** Reconsider the experiment described in Problem 6.5. Suppose that the experimenter only performed the eight trials from replicate I. In addition, he ran four center points and obtained the following response values: 36, 40, 43, 45.

(a) Estimate the factor effects. Which effects are large?

(b) Perform an analysis of variance, including a check for pure quadratic curvature. What are your conclusions?

(c) Write down an appropriate model for predicting tool life, based on the results of this experiment. Does this model differ in any substantial way from the model in Problem 6.5, part (c)?

(d) Analyze the residuals.

(e) What conclusions would you draw about the appropriate operating conditions for this process?

6.11 An experiment was performed to improve the yield of a chemical process. Four factors were selected, and two replicates of a completely randomized experiment were run. The results are shown in the following table:

Treatment	Replicate		Treatment	Replicate	
Combination	I	II	Combination	I	II
(1)	90	93	d	98	95
a	74	78	ad	72	76
b	81	85	bd	87	83
ab	83	80	abd	85	86
c	77	78	cd	99	90
ac	81	80	acd	79	75
bc	88	82	bcd	87	84
abc	73	70	$abcd$	80	80

(a) Estimate the factor effects.

(b) Prepare an analysis of variance table and determine which factors are important in explaining yield.

(c) Write down a regression model for predicting yield, assuming that all four factors were varied over the range from −1 to +1 (in coded units).

(d) Plot the residuals versus the predicted yield and on a normal probability scale. Does the residual analysis appear satisfactory?

(e) Two three-factor interactions, ABC and ABD, apparently have large effects. Draw a cube plot in the factors A, B, and C with the average yields shown at each corner. Repeat using the factors A, B, and D. Do these two plots aid in data interpretation? Where would you recommend that the process be run with respect to the four variables?

6.12 A bacteriologist is interested in the effects of two different culture media and two different times on the growth of a particular virus. He or she performs six replicates of a 2^2 design, making the runs in random order. Analyze the bacterial growth data that follow and draw appropriate conclusions. Analyze the residuals and comment on the model's adequacy.

	Culture Medium			
Time (h)	1		2	
	21	22	25	26
12	23	28	24	25
	20	26	29	27
	37	39	31	34
18	38	38	29	33
	35	36	30	35

6.13 An industrial engineer employed by a beverage bottler is interested in the effects of two different types of 32-ounce bottles on the time to deliver 12-bottle cases of the product. The two bottle types are glass and plastic. Two workers are used to perform a task consisting of moving 40 cases of the product 50 feet on a standard type of hand truck and stacking the cases in a display. Four replicates of a 2^2 factorial design are performed, and the times observed are listed in the following table. Analyze the data and draw appropriate conclusions. Analyze the residuals and comment on the model's adequacy.

	Worker			
Bottle Type	**1**		**2**	
Glass	5.12	4.89	6.65	6.24
	4.98	5.00	5.49	5.55
Plastic	4.95	4.43	5.28	4.91
	4.27	4.25	4.75	4.71

6.14 An article in the *AT&T Technical Journal* (March/April 1986, Vol. 65, pp. 39–50) describes the application of two-level factorial designs to integrated circuit manufacturing. A basic processing step is to grow an epitaxial layer on polished silicon wafers. The wafers mounted on a susceptor are positioned inside a bell jar, and chemical vapors are introduced. The susceptor is rotated, and heat is applied until the epitaxial layer is thick enough. An experiment was run using two factors: arsenic flow rate (A) and deposition time (B). Four replicates were run, and the epitaxial layer thickness was measured (μm). The data are shown in Table P6.1.

(a) Estimate the factor effects.

(b) Conduct an analysis of variance. Which factors are important?

(c) Write down a regression equation that could be used to predict epitaxial layer thickness over the region of arsenic flow rate and deposition time used in this experiment.

(d) Analyze the residuals. Are there any residuals that should cause concern?

(e) Discuss how you might deal with the potential outlier found in part (d).

6.15 ***Continuation of Problem 6.14.*** Use the regression model in part (c) of Problem 6.14 to generate a response surface contour plot for epitaxial layer thickness. Suppose it is critically important to obtain layer thickness of 14.5 μm. What settings of arsenic flow rate and decomposition time would you recommend?

6.16 ***Continuation of Problem 6.15.*** How would your answer to Problem 6.15 change if arsenic flow rate was more difficult to control in the process than the deposition time?

6.17 An experimenter has run a single replicate of a 2^4 design. The following effect estimates have been calculated:

$A = 76.95$	$AB = -51.32$	$ABC = -2.82$
$B = -67.52$	$AC = 11.69$	$ABD = -6.50$
$C = -7.84$	$AD = 9.78$	$ACD = 10.20$
$D = -18.73$	$BC = 20.78$	$BCD = -7.98$
	$BD = 14.74$	$ABCD = -6.25$
	$CD = 1.27$	

(a) Construct a normal probability plot of these effects.

(b) Identify a tentative model, based on the plot of the effects in part (a).

6.18 The effect estimates from a 2^4 factorial design are as follows: ABCD = −1.5138, ABC = −1.2661, ABD = −0.9852, ACD = −0.7566, BCD = −0.4842, CD = −0.0795, BD = −0.0793, AD = 0.5988, BC = 0.9216, AC = 1.1616, AB = 1.3266, D = 4.6744, C = 5.1458, B = 8.2469, and A = 12.7151. Are you comfortable with the conclusions that all main effects are active?

6.19 The effect estimates from a 2^4 factorial experiment are listed here. Are any of the effects significant? ABCD = −2.5251, BCD = 4.4054, ACD = −0.4932, ABD = −5.0842, ABC = −5.7696, CD = 4.6707, BD = −4.6620, BC = −0.7982, AD = −1.6564, AC = 1.1109, AB = −10.5229, D = −6.0275, C = −8.2045, B = −6.5304, and A = −0.7914.

■ **TABLE P6.1**
The 2^2 Design for Problem 6.14

		Replicate					Factor Levels	
A	B	I	II	III	IV		Low (−)	High (+)
−	−	14.037	16.165	13.972	13.907	A	55%	59%
+	−	13.880	13.860	14.032	13.914			
−	+	14.821	14.757	14.843	14.878	B	Short	Long
+	+	14.888	14.921	14.415	14.932		(10 min)	(15 min)

■ **TABLE P6.2**
Fill Height Experiment from Problem 6.20

	Coded Factors			Fill Height Deviation			Factor Levels	
Run	*A*	*B*	*C*	Replicate 1	Replicate 2		Low (−1)	High (+1)
1	−	−	−	−3	−1	*A* (%)	10	12
2	+	−	−	0	1	*B* (psi)	25	30
3	−	+	−	−1	0	*C* (b/m)	200	250
4	+	+	−	2	3			
5	−	−	+	−1	0			
6	+	−	+	2	1			
7	−	+	+	1	1			
8	+	+	+	6	5			

SS **6.20** Consider a variation of the bottle filling experiment from Example 5.3. Suppose that only two levels of carbonation are used so that the experiment is a 2^3 factorial design with two replicates. The data are shown in Table P6.2.

(a) Analyze the data from this experiment. Which factors significantly affect fill height deviation?

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy?

(c) Obtain a model for predicting fill height deviation in terms of the important process variables. Use this model to construct contour plots to assist in interpreting the results of the experiment.

(d) In part (a), you probably noticed that there was an interaction term that was borderline significant. If you did not include the interaction term in your model, include it now and repeat the analysis. What difference did this make? If you elected to include the interaction term in part (a), remove it and repeat the analysis. What difference does the interaction term make?

SS **6.21** I am always interested in improving my golf scores. Since a typical golfer uses the putter for about 35–45 percent of his or her strokes, it seems reasonable that improving one's putting is a logical and perhaps simple way to improve a golf score ("The man who can putt is a match for any man."—Willie Parks, 1864–1925, two-time winner of the British Open). An experiment was conducted to study the effects of four factors on putting accuracy. The design factors are length of putt, type of putter, breaking putt versus straight putt, and level versus downhill putt. The response variable is distance from the ball to the center of the cup after the ball comes to rest. One golfer performs the experiment, a 2^4 factorial design with seven replicates was used, and all putts are made in random order. The results are shown in Table P6.3.

(a) Analyze the data from this experiment. Which factors significantly affect putting performance?

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy?

6.22 A company markets its products by direct mail. An experiment was conducted to study the effects of three factors on the customer response rate for a particular product. The three factors are A = type of mail used (3rd class, 1st class), B = type of descriptive brochure (color, black-and-white), and C = offered price ($19.95, $24.95). The mailings are made to two groups of 8000 randomly selected customers, with 1000 customers in each group receiving each treatment combination. Each group of customers is considered as a replicate. The response variable is the number of orders placed. The experimental data are shown in Table P6.4.

(a) Analyze the data from this experiment. Which factors significantly affect the customer response rate?

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy?

(c) What would you recommend to the company?

6.23 Consider the single replicate of the 2^4 design in Example 6.2. Suppose that we had arbitrarily decided to analyze the data assuming that all three- and four-factor interactions were negligible. Conduct this analysis and compare your results with those obtained in the example. Do you think that it is a good idea to arbitrarily assume interactions to be negligible even if they are relatively high-order ones?

SS **6.24** An experiment was run in a semiconductor fabrication plant in an effort to increase yield. Five factors, each at two levels, were studied. The factors (and levels) were A = aperture setting (small, large), B = exposure time (20% below nominal, 20% above nominal), C = development time (30 and 45 s), D = mask dimension (small, large), and E = etch time

■ **TABLE P6.3**
The Putting Experiment from Problem 6.21

Design Factors				Distance from Cup (replicates)						
Length of Putt (ft)	Type of Putter	Break of Putt	Slope of Putt	1	2	3	4	5	6	7
10	Mallet	Straight	Level	10.0	18.0	14.0	12.5	19.0	16.0	18.5
30	Mallet	Straight	Level	0.0	16.5	4.5	17.5	20.5	17.5	33.0
10	Cavity back	Straight	Level	4.0	6.0	1.0	14.5	12.0	14.0	5.0
30	Cavity back	Straight	Level	0.0	10.0	34.0	11.0	25.5	21.5	0.0
10	Mallet	Breaking	Level	0.0	0.0	18.5	19.5	16.0	15.0	11.0
30	Mallet	Breaking	Level	5.0	20.5	18.0	20.0	29.5	19.0	10.0
10	Cavity back	Breaking	Level	6.5	18.5	7.5	6.0	0.0	10.0	0.0
30	Cavity back	Breaking	Level	16.5	4.5	0.0	23.5	8.0	8.0	8.0
10	Mallet	Straight	Downhill	4.5	18.0	14.5	10.0	0.0	17.5	6.0
30	Mallet	Straight	Downhill	19.5	18.0	16.0	5.5	10.0	7.0	36.0
10	Cavity back	Straight	Downhill	15.0	16.0	8.5	0.0	0.5	9.0	3.0
30	Cavity back	Straight	Downhill	41.5	39.0	6.5	3.5	7.0	8.5	36.0
10	Mallet	Breaking	Downhill	8.0	4.5	6.5	10.0	13.0	41.0	14.0
30	Mallet	Breaking	Downhill	21.5	10.5	6.5	0.0	15.5	24.0	16.0
10	Cavity back	Breaking	Downhill	0.0	0.0	0.0	4.5	1.0	4.0	6.5
30	Cavity back	Breaking	Downhill	18.0	5.0	7.0	10.0	32.5	18.5	8.0

■ **TABLE P6.4**
The Direct Mail Experiment from Problem 6.22

Coded Factors				Number of Orders		Factor Levels		
Run	*A*	*B*	*C*	Replicate 1	Replicate 2		Low (−1)	High (+1)
1	−	−	−	50	54	*A* (class)	3rd	1st
2	+	−	−	44	42	*B* (type)	BW	Color
3	−	+	−	46	48	*C* ($)	$19.95	$24.95
4	+	+	−	42	43			
5	−	−	+	49	46			
6	+	−	+	48	45			
7	−	+	+	47	48			
8	+	+	+	56	54			

(14.5 and 15.5 min). The unreplicated 2^5 design shown below was run.

$(1) = 7$	$d = 8$	$e = 8$	$de = 6$
$a = 9$	$ad = 10$	$ae = 12$	$ade = 10$
$b = 34$	$bd = 32$	$be = 35$	$bde = 30$
$ab = 55$	$abd = 50$	$abe = 52$	$abde = 53$
$c = 16$	$cd = 18$	$ce = 15$	$cde = 15$
$ac = 20$	$acd = 21$	$ace = 22$	$acde = 20$
$bc = 40$	$bcd = 44$	$bce = 45$	$bcde = 41$
$abc = 60$	$abcd = 61$	$abce = 65$	$abcde = 63$

(a) Construct a normal probability plot of the effect estimates. Which effects appear to be large?

(b) Conduct an analysis of variance to confirm your findings for part (a).

(c) Write down the regression model relating yield to the significant process variables.

(d) Plot the residuals on normal probability paper. Is the plot satisfactory?

(e) Plot the residuals versus the predicted yields and versus each of the five factors. Comment on the plots.

(f) Interpret any significant interactions.

(g) What are your recommendations regarding process operating conditions?

(h) Project the 2^5 design in this problem into a 2^k design in the important factors. Sketch the design and show the average and range of yields at each run. Does this sketch aid in interpreting the results of this experiment?

6.25 ***Continuation of Problem 6.24.*** Suppose that the experimenter had run four center points in addition to the 32 trials in the original experiment. The yields obtained at the center point runs were 68, 74, 76, and 70. SS

(a) Reanalyze the experiment, including a test for pure quadratic curvature.

(b) Discuss what your next step would be.

6.26 In a process development study on yield, four factors were studied, each at two levels: time (A), concentration (B), pressure (C), and temperature (D). A single replicate of a 2^4 design was run, and the resulting data are shown in Table P6.5. SS

(a) Construct a normal probability plot of the effect estimates. Which factors appear to have large effects?

(b) Conduct an analysis of variance using the normal probability plot in part (a) for guidance in forming an error term. What are your conclusions?

■ **TABLE P6.5**
Process Development Experiment from Problem 6.26

Run Number	Actual Run Order	A	B	C	D	Yield (lbs)
1	5	−	−	−	−	12
2	9	+	−	−	−	18
3	8	−	+	−	−	13
4	13	+	+	−	−	16
5	3	−	−	+	−	17
6	7	+	−	+	−	15
7	14	−	+	+	−	20
8	1	+	+	+	−	15
9	6	−	−	−	+	10
10	11	+	−	−	+	25
11	2	−	+	−	+	13
12	15	+	+	−	+	24
13	4	−	−	+	+	19
14	16	+	−	+	+	21
15	10	−	+	+	+	17
16	12	+	+	+	+	23

Factor Levels		
	Low (−)	High (+)
A (h)	2.5	3
B (%)	14	18
C (psi)	60	80
D (°C)	225	250

(c) Write down a regression model relating yield to the important process variables.

(d) Analyze the residuals from this experiment. Does your analysis indicate any potential problems?

(e) Can this design be collapsed into a 2^3 design with two replicates? If so, sketch the design with the average and range of yield shown at each point in the cube. Interpret the results.

6.27 ***Continuation of Problem 6.26.*** Use the regression model in part (c) of Problem 6.26 to generate a response surface contour plot of yield. Discuss the practical value of this response surface plot.

6.28 ***The scrumptious brownie experiment.*** The author is an engineer by training and a firm believer in learning by doing. I have taught experimental design for many years to a wide variety of audiences and have always assigned the planning, conduct, and analysis of an actual experiment to the class participants. The participants seem to enjoy this practical experience and always learn a great deal from it. This problem uses the results of an experiment performed by Gretchen Krueger at Arizona State University.

There are many different ways to bake brownies. The purpose of this experiment was to determine how the pan material, the brand of brownie mix, and the stirring method affect the scrumptiousness of brownies. The factor levels were as follows:

Factor	Low (−)	High (+)
A = pan material	Glass	Aluminum
B = stirring method	Spoon	Mixer
C = brand of mix	Expensive	Cheap

The response variable was scrumptiousness, a subjective measure derived from a questionnaire given to the subjects who sampled each batch of brownies. (The questionnaire dealt with issues such as taste, appearance, consistency, aroma.) An eight-person test panel sampled each batch and filled out the questionnaire. The design matrix and the response data are as follows.

(a) Analyze the data from this experiment as if there were eight replicates of a 2^3 design. Comment on the results.

(b) Is the analysis in part (a) the correct approach? There are only eight batches; do we really have eight replicates of a 2^3 factorial design?

(c) Analyze the average and standard deviation of the scrumptiousness ratings. Comment on the results. Is this analysis more appropriate than the one in part (a)? Why or why not?

Brownie Batch	A	B	C	Test Panel Results 1	2	3	4	5	6	7	8
1	−	−	−	11	9	10	10	11	10	8	9
2	+	−	−	15	10	16	14	12	9	6	15
3	−	+	−	9	12	11	11	11	11	11	12
4	+	+	−	16	17	15	12	13	13	11	11
5	−	−	+	10	11	15	8	6	8	9	14
6	+	−	+	12	13	14	13	9	13	14	9
7	−	+	+	10	12	13	10	7	7	17	13
9	+	+	+	15	12	15	13	12	12	9	14

6.29 An experiment was conducted on a chemical process that produces a polymer. The four factors studied were temperature (A), catalyst concentration (B), time (C), and pressure (D). Two responses, molecular weight and viscosity, were observed. The design matrix and response data are shown in Table P6.6.

(a) Consider only the molecular weight response. Plot the effect estimates on a normal probability scale. What effects appear important?

(b) Use an analysis of variance to confirm the results from part (a). Is there indication of curvature?

(c) Write down a regression model to predict molecular weight as a function of the important variables.

(d) Analyze the residuals and comment on model adequacy.

(e) Repeat parts (a)–(d) using the viscosity response.

6.30 ***Continuation of Problem 6.29.*** Use the regression models for molecular weight and viscosity to answer the following questions.

(a) Construct a response surface contour plot for molecular weight. In what direction would you adjust the process variables to increase molecular weight?

(b) Construct a response surface contour plot for viscosity. In what direction would you adjust the process variables to decrease viscosity?

(c) What operating conditions would you recommend if it was necessary to produce a product with molecular weight between 2400 and 2500 and the lowest possible viscosity?

6.31 Consider the single replicate of the 2^4 design in Example 6.2. Suppose that we ran five points at the center (0, 0, 0, 0) and observed the responses 93, 95, 91, 89, and 96. Test for curvature in this experiment. Interpret the results.

6.32 An engineer has performed an experiment to study the effect of four factors on the surface roughness of a

■ **TABLE P6.6**
The 2^4 Experiment for Problem 6.29

Run Number	Actual Run Order	A	B	C	D	Molecular Weight	Viscosity	Factor Levels	Low (−)	High (+)
1	18	−	−	−	−	2400	1400	A (°C)	100	120
2	9	+	−	−	−	2410	1500	B (%)	4	8
3	13	−	+	−	−	2315	1520	C (min)	20	30
4	8	+	+	−	−	2510	1630	D (psi)	60	75
5	3	−	−	+	−	2615	1380			
6	11	+	−	+	−	2625	1525			
7	14	−	+	+	−	2400	1500			
8	17	+	+	+	−	2750	1620			
9	6	−	−	−	+	2400	1400			
10	7	+	−	−	+	2390	1525			
11	2	−	+	−	+	2300	1500			
12	10	+	+	−	+	2520	1500			
13	4	−	−	+	+	2625	1420			
14	19	+	−	+	+	2630	1490			
15	15	−	+	+	+	2500	1500			
16	20	+	+	+	+	2710	1600			
17	1	0	0	0	0	2515	1500			
18	5	0	0	0	0	2500	1460			
19	16	0	0	0	0	2400	1525			
20	12	0	0	0	0	2475	1500			

machined part. The factors (and their levels) are A = tool angle (12, 15°), B = cutting fluid viscosity (300, 400), C = feed rate (10 and 15 in./min), and D = cutting fluid cooler used (no, yes). The data from this experiment (with the factors coded to the usual −1, +1 levels) are shown in Table P6.7.

(a) Estimate the factor effects. Plot the effect estimates on a normal probability plot and select a tentative model.

(b) Fit the model identified in part (a) and analyze the residuals. Is there any indication of model inadequacy?

(c) Repeat the analysis from parts (a) and (b) using $1/y$ as the response variable. Is there an indication that the transformation has been useful?

(d) Fit a model in terms of the coded variables that can be used to predict the surface roughness. Convert this prediction equation into a model in the natural variables.

6.33 Resistivity on a silicon wafer is influenced by several factors. The results of a 2^4 factorial experiment performed during a critical processing step are shown in Table P6.8.

(a) Estimate the factor effects. Plot the effect estimates on a normal probability plot and select a tentative model.

(b) Fit the model identified in part (a) and analyze the residuals. Is there any indication of model inadequacy?

(c) Repeat the analysis from parts (a) and (b) using $\ln(y)$ as the response variable. Is there an indication that the transformation has been useful?

(d) Fit a model in terms of the coded variables that can be used to predict the resistivity.

6.34 ***Continuation of Problem 6.33***. Suppose that the experimenter had also run four center points along with the 16 runs in Problem 6.33. The resistivity measurements at the center points are 8.15, 7.63, 8.95, and 6.48. Analyze the experiment again incorporating the center points. What conclusions can you draw now?

6.35 An article in *Quality and Reliability Engineering International* (2010, Vol. 26, pp. 223–233) presents a 2^5 factorial design. The experiment is shown in Table P6.9.

(a) Analyze the data from this experiment. Identify the significant factors and interactions.

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy or violations of the assumptions?

■ **TABLE P6.7**
The Surface Roughness Experiment from Problem 6.32

Run	*A*	*B*	*C*	*D*	Surface Roughness
1	−	−	−	−	0.00340
2	+	−	−	−	0.00362
3	−	+	−	−	0.00301
4	+	+	−	−	0.00182
5	−	−	+	−	0.00280
6	+	−	+	−	0.00290
7	−	+	+	−	0.00252
8	+	+	+	−	0.00160
9	−	−	−	+	0.00336
10	+	−	−	+	0.00344
11	−	+	−	+	0.00308
12	+	+	−	+	0.00184
13	−	−	+	+	0.00269
14	+	−	+	+	0.00284
15	−	+	+	+	0.00253
16	+	+	+	+	0.00163

■ **TABLE P6.8**
The Resistivity Experiment from Problem 6.33

Run	*A*	*B*	*C*	*D*	Resistivity
1	−	−	−	−	1.92
2	+	−	−	−	11.28
3	−	+	−	−	1.09
4	+	+	−	−	5.75
5	−	−	+	−	2.13
6	+	−	+	−	9.53
7	−	+	+	−	1.03
8	+	+	+	−	5.35
9	−	−	−	+	1.60
10	+	−	−	+	11.73
11	−	+	−	+	1.16
12	+	+	−	+	4.68
13	−	−	+	+	2.16
14	+	−	+	+	9.11
15	−	+	+	+	1.07
16	+	+	+	+	5.30

(c) One of the factors from this experiment does not seem to be important. If you drop this factor, what type of design remains? Analyze the data using the full factorial model for only the four active factors. Compare your results with those obtained in part (a).

(d) Find settings of the active factors that maximize the predicted response.

6.36 A paper in the *Journal of Chemical Technology and Biotechnology* ("Response Surface Optimization of the Critical Media Components for the Production of Surfactin," 1997, Vol. 68, pp. 263–270) describes the use of a designed experiment to maximize surfactin production. A portion of the data from this experiment is shown in Table P6.10. Surfactin was assayed by an indirect method, which involves measurement of surface tensions of the diluted broth samples. Relative surfactin concentrations were determined by serially diluting the broth until the critical micelle concentration (CMC) was reached. The dilution at which the surface tension starts rising abruptly was denoted by CMC^{-1} and was considered proportional to the amount of surfactant present in the original sample.

(a) Analyze the data from this experiment. Identify the significant factors and interactions.

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy or violations of the assumptions?

(c) What conditions would optimize the surfactin production?

6.37 ***Continuation of Problem 6.36.*** The experiment in Problem 6.36 actually included six center points. The responses at these conditions were 35, 35, 35, 36, 36, and 34. Is there any indication of curvature in the response function? Are additional experiments necessary? What would you recommend doing now?

6.38 An article in the *Journal of Hazardous Materials* ("Feasibility of Using Natural Fishbone Apatite as a Substitute for Hydroxyapatite in Remediating Aqueous Heavy Metals," Vol. 69, Issue 2, 1999, pp. 187–196) describes an experiment to study the suitability of fishbone, a natural, apatite, rich substance, as a substitute for hydroxyapatite in the sequestering of aqueous divalent heavy metal ions. Direct comparison of hydroxyapatite and fishbone apatite was

■ TABLE P6.9
The 2^5 Design in Problem 6.35

A	B	C	D	E	y
−1.00	−1.00	−1.00	−1.00	−1.00	8.11
1.00	−1.00	−1.00	−1.00	−1.00	5.56
−1.00	1.00	−1.00	−1.00	−1.00	5.77
1.00	1.00	−1.00	−1.00	−1.00	5.82
−1.00	−1.00	1.00	−1.00	−1.00	9.17
1.00	−1.00	1.00	−1.00	−1.00	7.8
−1.00	1.00	1.00	−1.00	−1.00	3.23
1.00	1.00	1.00	−1.00	−1.00	5.69
−1.00	−1.00	−1.00	1.00	−1.00	8.82
1.00	−1.00	−1.00	1.00	−1.00	14.23
−1.00	1.00	−1.00	1.00	−1.00	9.2
1.00	1.00	−1.00	1.00	−1.00	8.94
−1.00	−1.00	1.00	1.00	−1.00	8.68
1.00	−1.00	1.00	1.00	−1.00	11.49
−1.00	1.00	1.00	1.00	−1.00	6.25
1.00	1.00	1.00	1.00	−1.00	9.12
−1.00	−1.00	−1.00	−1.00	1.00	7.93
1.00	−1.00	−1.00	−1.00	1.00	5
−1.00	1.00	−1.00	−1.00	1.00	7.47
1.00	1.00	−1.00	−1.00	1.00	12
−1.00	−1.00	1.00	−1.00	1.00	9.86
1.00	−1.00	1.00	−1.00	1.00	3.65
−1.00	1.00	1.00	−1.00	1.00	6.4
1.00	1.00	1.00	−1.00	1.00	11.61
−1.00	−1.00	−1.00	1.00	1.00	12.43
1.00	−1.00	−1.00	1.00	1.00	17.55
−1.00	1.00	−1.00	1.00	1.00	8.87
1.00	1.00	−1.00	1.00	1.00	25.38
−1.00	−1.00	1.00	1.00	1.00	13.06
1.00	−1.00	1.00	1.00	1.00	18.85
−1.00	1.00	1.00	1.00	1.00	11.78
1.00	1.00	1.00	1.00	1.00	26.05

performed using a three-factor two-level full factorial design. Apatite (30 or 60 mg) was added to 100 mL deionized water and gently agitated overnight in a shaker. The pH was then adjusted to 5 or 7 using nitric acid. Sufficient concentration of lead nitrate solution was added to each flask to result in a final volume of 200 mL and a lead concentration of 0.483 or 2.41 mM, respectively. The experiment was a 2^3 replicated twice and it was performed for both fishbone and synthetic apatite. Results are shown in Table P6.11.

■ TABLE P6.10
The Factorial Experiment in Problem 6.36

Run	Glucose ($g\ dm^{-3}$)	NH_4NO_3 ($g\ dm^{-3}$)	$FeSO_4$ ($g\ dm^{-3} \times 10^{-4}$)	$MnSO_4$ ($g\ dm^{-3} \times 10^{-2}$)	y $(CMC)^{-1}$
1	20.00	2.00	6.00	4.00	23
2	60.00	2.00	6.00	4.00	15
3	20.00	6.00	6.00	4.00	16
4	60.00	6.00	6.00	4.00	18
5	20.00	2.00	30.00	4.00	25
6	60.00	2.00	30.00	4.00	16
7	20.00	6.00	30.00	4.00	17
8	60.00	6.00	30.00	4.00	26
9	20.00	2.00	6.00	20.00	28
10	60.00	2.00	6.00	20.00	16
11	20.00	6.00	6.00	20.00	18
12	60.00	6.00	6.00	20.00	21
13	20.00	2.00	30.00	20.00	36
14	60.00	2.00	30.00	20.00	24
15	20.00	6.00	30.00	20.00	33
16	60.00	6.00	30.00	20.00	34

(a) Analyze the lead response for fishbone apatite. What factors are important?

(b) Analyze the residuals from this response and comment on model adequacy.

(c) Analyze the pH response for fishbone apatite. What factors are important?

(d) Analyze the residuals from this response and comment on model adequacy.

(e) Analyze the lead response for hydroxyapatite apatite. What factors are important?

(f) Analyze the residuals from this response and comment on model adequacy.

(g) Analyze the pH response for hydroxyapatite apatite. What factors are important?

(h) Analyze the residuals from this response and comment on model adequacy.

(i) What differences do you see between fishbone and hydroxyapatite apatite? The authors of this paper concluded that that fishbone apatite was comparable to hydroxyapatite apatite. Because the fishbone apatite is cheaper, it was recommended for adoption. Do you agree with these conclusions?

6.39 Often the fitted regression model from a 2^k factorial SS design is used to make predictions at points of interest in the

■ **TABLE P6.11**

The Experiment for Problem 6.38. For apatite, + is 60 mg and − is 30 mg per 200 mL metal solution. For initial pH, + is 7 and − is 4. For Pb + is 2.41 mM (500 ppm) and − is 0.483 mM (100 ppm)

			Fishbone		Hydroxyapatite	
Apatite	pH	Pb	Pb, mM	pH	Pb, mM	pH
+	+	+	1.82	5.22	0.11	3.49
+	+	+	1.81	5.12	0.12	3.46
+	+	−	0.01	6.84	0.00	5.84
+	+	−	0.00	6.61	0.00	5.90
+	−	+	1.11	3.35	0.80	2.70
+	−	+	1.04	3.34	0.76	2.74
+	−	−	0.00	5.77	0.03	3.36
+	−	−	0.01	6.25	0.05	3.24
−	+	+	2.11	5.29	1.03	3.22
−	+	+	2.18	5.06	1.05	3.22
−	+	−	0.03	5.93	0.00	5.53
−	+	−	0.05	6.02	0.00	5.43
−	−	+	1.70	3.39	1.34	2.82
−	−	+	1.69	3.34	1.26	2.79
−	−	−	0.05	4.50	0.06	3.28
−	−	−	0.05	4.74	0.07	3.28

design space. Assume that the model contains all main effects and two-factor interactions.

(a) Find the variance of the predicted response $\hat{y}$ at a point $x_1, x_2, \ldots, x_k$ in the design space. *Hint:* Remember that the x's are coded variables and assume a 2^k design with an equal number of replicates n at each design point so that the variance of a regression coefficient $\hat{\beta}$ is $\sigma^2/(n2^k)$ and that the covariance between any pair of regression coefficients is zero.

(b) Use the result in part (a) to find an equation for a $100(1-\alpha)$ percent confidence interval on the true mean response at the point $x_1, x_2, \ldots, x_k$ in design space.

6.40 Suppose that you want to run a 2^3 factorial design. The variance of an individual observation is expected to be about 4. Suppose that you want the length of a 95 percent confidence interval on any effect to be less than or equal to 1.5. How many replicates of the design do you need to run?

6.41 Suppose that a full 2^4 factorial uses the following factor levels:

Factor	Low (−)	High (+)
A: Acid strength (%)	85	95
B: Reaction time (min)	15	35
C: Amount of acid (mL)	35	45
D: Reaction temperature (°C)	60	80

The fitted model from this experiment is $\hat{y} = 24 + 16x_1 - 34x_2 + 12x_3 + 6x_4 - 10x_1x_2 + 16x_1x_3$. Predict the response at the following points:

(a) $A = 89, B = 20, C = 38, D = 66$

(b) $A = 90, B = 16, C = 40, D = 70$

(c) $A = 87, B = 28, C = 42, D = 61$

(d) $A = 90, B = 27, C = 37, D = 69$

6.42 An article in *Quality and Reliability Engineering International* (2010, Vol. 26, pp. 223–233) presents a 2^5 factorial design. The experiment is shown in Table P6.12.

■ **TABLE P6.12**

The Experiment for Problem 6.42

A	B	C	D	E	y
−1	−1	−1	−1	−1	8.11
1	−1	−1	−1	−1	5.56
−1	1	−1	−1	−1	5.77
1	1	−1	−1	−1	5.82
−1	−1	1	−1	−1	9.17
1	−1	1	−1	−1	7.8
−1	1	1	−1	−1	3.23
1	1	1	−1	−1	5.69
−1	−1	−1	1	−1	8.82
1	−1	−1	1	−1	14.23
−1	1	−1	1	−1	9.2
1	1	−1	1	−1	8.94
−1	−1	1	1	−1	8.68
1	−1	1	1	−1	11.49
−1	1	1	1	−1	6.25
1	1	1	1	−1	9.12
−1	−1	−1	−1	1	7.93
1	−1	−1	−1	1	5
−1	1	−1	−1	1	7.47
1	1	−1	−1	1	12
−1	−1	1	−1	1	9.86
1	−1	1	−1	1	3.65
−1	1	1	−1	1	6.4

■ **TABLE P6.12** *(Continued)*

A	B	C	D	E	y
1	1	1	−1	1	11.61
−1	−1	−1	1	1	12.43
1	−1	−1	1	1	17.55
−1	1	−1	1	1	8.87
1	1	−1	1	1	25.38
−1	−1	1	1	1	13.06
1	−1	1	1	1	18.85
−1	1	1	1	1	11.78
1	1	1	1	1	26.05

(a) Analyze the data from this experiment. Identify the significant factors and interactions.

(b) Analyze the residuals from this experiment. Are there any indications of model inadequacy or violations of the assumptions?

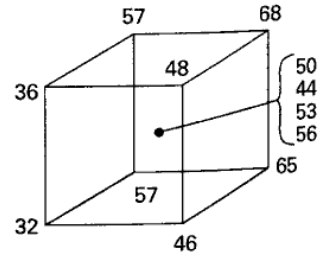
(c) One of the factors from this experiment does not seem to be important. If you drop this factor, what type of design remains? Analyze the data using the full factorial model for only the four active factors. Compare your results with those obtained in part (a).

(d) Find settings of the active factors that maximize the predicted response.

6.43 Consider the 2^3 shown below:

Run	Process Variables: Temp (°C)	Pressure (psig)	Conc (g/l)	Coded Variables: x_1	x_2	x_3	Yield, y
1	120	40	15	−1	−1	−1	32
2	160	40	15	1	−1	−1	46
3	120	80	15	−1	1	−1	57
4	160	80	15	1	1	−1	65
5	120	40	30	−1	−1	1	36
6	160	40	30	1	−1	1	48
7	120	80	30	−1	1	1	57
8	160	80	30	1	1	1	68
9	140	60	22.5	0	0	0	50
10	140	60	22.5	0	0	0	44
11	140	60	22.5	0	0	0	53
12	140	60	22.5	0	0	0	56

$$x_1 = \frac{\text{Temp} - 140}{20}, \quad x_2 = \frac{\text{Pressure} - 60}{20}, \quad x_3 = \frac{\text{Conc} - 22.5}{7.5}$$



When running a designed experiment, it is sometimes difficult to reach and hold the precise factor levels required by the design. Small discrepancies are not important, but large ones are potentially of more concern. To illustrate, the experiment presented in Table P6.13 shows a variation of the 2^3 design above, where many of the test combinations are not exactly the ones specified in the design. Most of the difficulty seems to have occurred with the temperature variable.

Fit a first-order model to both the original data and the data in Table P6.13. Compare the inference from the two models. What conclusions can you draw from this simple example?

6.44 In two-level design, the expected value of a nonsignificant factor effect is zero.

(a) True

(b) False

6.45 A half-normal plot of factor effects plots the expected normal percentile versus the effect estimate.

(a) True

(b) False

6.46 In an unreplicated design, the degrees of freedom associated with the "pure error" component of error are zero.

(a) True

(b) False

■ **TABLE P6.13**
Revised Experimental Data

Run	Process Variables: Temp (°C)	Pressure (psig)	Conc (g/l)	Coded Variables: x_1	x_2	x_3	Yield, y
1	125	41	14	−0.75	−0.95	−1.133	32
2	158	40	15	0.90	−1	−1	46
3	121	82	15	−0.95	1.1	−1	57
4	160	80	15	1	1	−1	65
5	118	39	33	−1.10	−1.05	1.14	36
6	163	40	30	1.15	−1	1	48
7	122	80	30	−0.90	1	1	57
8	165	83	30	1.25	1.15	1	68
9	140	60	22.5	0	0	0	50
10	140	60	22.5	0	0	0	44
11	140	60	22.5	0	0	0	53
12	140	60	22.5	0	0	0	56

6.47 In a replicated 2^3 design (16 runs), the estimate of the model intercept is equal to one-half of the total of all 16 runs.

(a) True

(b) False

6.48 Adding center runs to a 2^k design affects the estimate of the intercept term but not the estimates of any other factor effects.

(a) True

(b) False

6.49 The mean square for pure error in a replicated factorial design can get smaller if nonsignificant terms are added to a model.

(a) True

(b) False

6.50 A 2^k factorial design is a D-optimal design for fitting a first-order model.

(a) True

(b) False

6.51 If a D-optimal design algorithm is used to create a 12-run design for fitting a first-order model in three variables with all three two-factor interactions, the algorithm will construct a 2^3 factorial with four center runs.

(a) True

(b) False

6.52 Suppose that you want to replicate 2 of the 8 runs in a 2^3 factorial design. How many ways are there to choose the 2 runs to replicate? Suppose that you decide to replicate the run with all three factors at the high level and the run with all three factors at the low level.

(a) Is the resulting design orthogonal?

(b) What are the relative variances of the model coefficients if the main effects plus two-factor interaction model are fit to the data from this design?

(c) What is the power for detecting effects of two standard deviations in magnitude?

6.53 The display below summarizes the results of analyzing a 2^4 factorial design.

Term	Effect Estimate	Sum of Squares	% Contribution
Intercept			
A		6.25	3.25945
B	5.25	110.25	57.4967
C	3.5	49	25.5541
D	0.75		1.1734
AB	0.75	2.25	1.1734
AC	−0.5	1	0.521512
AD	0.75	2.25	1.1734
BC	1.5	9	
BD	0.25	0.25	0.130378
CD	0.5	1	0.521512
ABC	−1	4	2.08605
ABD		2.25	1.1734
ACD	−0.5		0.521512
BCD	0	0	0
ABCD	−0.5	1	

(a) Fill in the missing information in this table.

(b) Construct a normal probability plot of the effects. Which factors seem to be active?

Chapter 7 Problems

SS **7.1** Consider the experiment described in Problem 6.5. Analyze this experiment assuming that each replicate represents a block of a single production shift.

7.2 Consider the experiment described in Problem 6.9. Analyze this experiment assuming that each one of the four replicates represents a block.

SS **7.3** Consider the data from the first replicate of Problem 6.5. Suppose that these observations could not all be run using the same bar stock. Set up a design to run these observations in two blocks of four observations each with ABC confounded. Analyze the data.

7.4 Consider the data from the first replicate of Problem 6.5. Construct a design with two blocks of eight observations each with $ABCD$ confounded. Analyze the data.

7.5 Repeat Problem 7.4 assuming that four blocks are required. Confound ABD and ABC (and consequently CD) with blocks.

SS **7.6** Using the data from the 2^5 design in Problem 6.24, construct and analyze a design in two blocks with $ABCDE$ confounded with blocks.

SS **7.7** Repeat Problem 7.6 assuming that four blocks are necessary. Suggest a reasonable confounding scheme.

SS **7.8** Consider the data from the 2^5 design in Problem 6.24. Suppose that it was necessary to run this design in four blocks with $ACDE$ and BCD (and consequently ABE) confounded. Analyze the data from this design.

SS **7.9** Consider the fill height deviation experiment in Problem 6.20. Suppose that each replicate was run on a separate day. Analyze the data assuming that days are blocks.

7.10 Consider the fill height deviation experiment in Problem 6.20. Suppose that only four runs could be made on each shift. Set up a design with ABC confounded in replicate I and AC confounded in replicate II. Analyze the data and comment on your findings.

SS **7.11** Consider the putting experiment in Problem 6.21. Analyze the data considering each replicate as a block.

7.12 The experiment in Problem 6.35 is a 2^5 factorial. Suppose that this design had been run in four blocks of eight runs each.

(a) Recommend a blocking scheme and set up the design.

(b) Analyze the data from this blocked design. Is blocking important?

7.13 Repeat Problem 7.12 using a design in two blocks.

7.14 The design in Problem 6.38 is a 2^3 factorial replicated twice. Suppose that each replicate was a block. Analyze all of the responses from this blocked design. Are the results comparable to those from Problem 6.38? Is the block effect large?

7.15 Consider the 2^6 design in eight blocks of eight runs each with $ABCD$, ACE, and $ABEF$ as the independent effects chosen to be confounded with blocks. Generate the design. Find the other effects confounded with blocks.

7.16 Consider the 2^2 design in two blocks with AB confounded. Prove algebraically that $SS_{AB} = SS_{\text{Blocks}}$. SS

7.17 Consider the data in Example 7.2. Suppose that all the observations in block 2 are increased by 20. Analyze the data that would result. Estimate the block effect. Can you explain its magnitude? Do blocks now appear to be an important factor? Are any other effect estimates impacted by the change you made to the data?

7.18 Suppose that in Problem 6.5 we had confounded ABC in replicate I, AB in replicate II, and BC in replicate III. Calculate the factor effect estimates. Construct the analysis of variance table.

7.19 Repeat the analysis of Problem 6.5 assuming that ABC was confounded with blocks in each replicate.

7.20 Suppose that in Problem 6.11 $ABCD$ was confounded in replicate I and ABC was confounded in replicate II. Perform the statistical analysis of this design.

7.21 Construct a 2^3 design with ABC confounded in the first two replicates and BC confounded in the third. Outline the analysis of variance and comment on the information obtained.

7.22 The block effect in a two-level design with two blocks can be calculated directly as the difference in the average response between the two blocks.

(a) True

(b) False

7.23 When constructing the 2^7 design confounded in eight blocks, three independent effects are chosen to generate the blocks, and there are a total of eight interactions confounded with blocks.

(a) True

(b) False

7.24 Consider the 2^5 factorial design in two blocks. If *ABCDE* is confounded with blocks, then which of the following runs is in the same block as run *acde*?

(a) *a* **(b)** *acd* **(c)** *bcd*

(d) *be* **(e)** *abe* **(f)** None of the above

7.25 The information on the interaction confounded with the block can always be separated from the block effect.

(a) True

(b) False

7.26 Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run in two blocks with *ABCDE* confounded with the blocks. Set up the blocked design and perform the analysis. Compare your results with the results obtained for the completely randomized design in Problem 6.42.

7.27 Suppose that you are designing an experiment for four factors and that due to material properties it is necessary to conduct the experiment in blocks. Material availability restricts you to the use of two blocks; however, each batch of material is only sufficient for six runs. So the standard 2^4 factorial in two blocks of eight runs each with *ABCD* confounded will not work. Recommend a design. Suggestion: this is a reasonable application for a *D*-optimal design. What type of design do you find in each block?

Chapter 8 Problems

8.1 Suppose that in the chemical process development experiment described in Problem 6.11, it was only possible to run a one-half fraction of the 2^4 design. Construct the design and perform the statistical analysis, using the data from replicate I.

SS **8.2** Consider the plasma etch experiment described in Example 6.1. Suppose that only a one-half fraction of the design could be run. Set up the design and analyze the data.

SS **8.3** Problem 6.24 describes a process improvement study in the manufacturing process of an integrated circuit. Suppose that only eight runs could be made in this process. Set up an appropriate 2^{5-2} design and find the alias structure. Use the appropriate observations from Problem 6.5R as the observations in this design and estimate the factor effects. What conclusions can you draw?

SS **8.4** ***Continuation of Problem 8.3.*** Suppose you have made the eight runs in the 2^{5-2} design in Problem 8.3. What additional runs would be required to identify the factor effects that are of interest? What are the alias relationships in the combined design?

8.5 In Example 6.10, a 2^4 factorial design was used to improve the response rate to a credit card mail marketing offer. Suppose that the researchers had used the 2^{4-1} fractional factorial design with $I = ABCD$ instead. Set up the design and select the responses for the runs from the full factorial data in Example 6.6. Analyze the data and draw conclusions. Compare your findings with those from the full factorial in Example 6.6.

8.6 ***Continuation of Problem 8.5.*** In Problem 6.6, we found that all four main effects and the two-factor AB interaction were significant. Show that if the alternate fraction ($I = -ABCD$) is added to the 2^{4-1} design in Problem 8.5 that the analysis of the results from the combined design produce results identical to those found in Problem 6.6.

8.7 ***Continuation of Problem 8.5.*** Reconsider the 2^{4-1} fractional factorial design with $I = ABCD$ from Problem 8.5. Set a partial fold over of this fraction to isolate the AB interaction. Select the appropriate set of responses from the full factorial data in Example 6.6 and analyze the resulting data.

8.8 An article by J. J. Pignatiello Jr. and J. S. Ramberg in the *Journal of Quality Technology* (Vol. 17, 1985, pp. 198–206) describes the use of a replicated fractional factorial to investigate the effect of five factors on the free height of leaf springs used in an automotive application. The factors are A = furnace temperature, B = heating time, C = transfer time, D = hold down time, and E = quench oil temperature. The data are shown in Table P8.1.

(a) Write out the alias structure for this design. What is the resolution of this design?

(b) Analyze the data. What factors influence the mean free height?

■ **TABLE P8.1**
Leaf Spring Experiment

A	*B*	*C*	*D*	*E*	Free Height		
−	−	−	−	−	7.78	7.78	7.81
+	−	−	+	−	8.15	8.18	7.88
−	+	−	+	−	7.50	7.56	7.50
+	+	−	−	−	7.59	7.56	7.75
−	−	+	+	−	7.54	8.00	7.88
+	−	+	−	−	7.69	8.09	8.06
−	+	+	−	−	7.56	7.52	7.44
+	+	+	+	−	7.56	7.81	7.69
−	−	−	−	+	7.50	7.25	7.12
+	−	−	+	+	7.88	7.88	7.44
−	+	−	+	+	7.50	7.56	7.50
+	+	−	−	+	7.63	7.75	7.56
−	−	+	+	+	7.32	7.44	7.44
+	−	+	−	+	7.56	7.69	7.62
−	+	+	−	+	7.18	7.18	7.25
+	+	+	+	+	7.81	7.50	7.59

(c) Calculate the range and standard deviation of the free height for each run. Is there any indication that any of these factors affects variability in the free height?

(d) Analyze the residuals from this experiment, and comment on your findings.

(e) Is this the best possible design for five factors in 16 runs? Specifically, can you find a fractional design for five factors in 16 runs with a higher resolution than this one?

8.9 Consider the leaf spring experiment in Problem 8.8. Suppose that factor E (quench oil temperature) is very difficult to control during manufacturing. Where would you set factors A, B, C, and D to reduce variability in the free height as much as possible regardless of the quench oil temperature used?

8.10 Consider the 2^5 design in Problem 6.24. Suppose that only a one-half fraction could be run. Furthermore, two days were required to take the 16 observations, and it was necessary to confound the 2^{5-1} design in two blocks. Construct the design and analyze the data.

8.11 Analyze the data in Problem 6.26 as if it came from a 2^{4-1}_{IV} design with $I = ABCD$. Project the design into a full factorial in the subset of the original four factors that appear to be significant.

8.12 Repeat Problem 8.11 using $I = -ABCD$. Does the use of the alternate fraction change your interpretation of the data?

8.13 Project the 2^{4-1}_{IV} design in Example 8.1 into two replicates of a 2^2 design in the factors A and B. Analyze the data and draw conclusions.

8.14 Construct a 2^{5-2}_{III} design. Determine the effects that may be estimated if a full fold over of this design is performed.

8.15 Fold over the 2^{7-4}_{III} design in Table 8.19 to produce an eight-factor design. Verify that the resulting design is a 2^{8-4}_{IV} design. Is this a minimal design?

8.16 Fold over a 2^{5-2}_{III} design to produce a six-factor design. Verify that the resulting design is a 2^{6-2}_{IV} design. Compare this design to the 2^{6-2}_{IV} design in Table 8.10.

8.17 An industrial engineer is conducting an experiment using a Monte Carlo simulation model of an inventory system. The independent variables in her model are the order quantity (A), the reorder point (B), the setup cost (C), the backorder cost (D), and the carrying cost rate (E). The response variable is average annual cost. To conserve computer time, she decides to investigate these factors using a 2^{5-2}_{III} design with $I = ABD$ and $I = BCE$. The results she obtains are $de = 95$, $ae = 134$, $b = 158$, $abd = 190$, $cd = 92$, $ac = 187$, $bce = 155$, and $abcde = 185$.

(a) Verify that the treatment combinations given are correct. Estimate the effects, assuming three-factor and higher interactions are negligible.

(b) Suppose that a second fraction is added to the first, for example, $ade = 136$, $e = 93$, $ab = 187$, $bd = 153$, $acd = 139$, $c = 99$, $abce = 191$, and $bcde = 150$. How was this second fraction obtained? Add this data to the original fraction, and estimate the effects.

(c) Suppose that the fraction $abc = 189$, $ce = 96$, $bcd = 154$, $acde = 135$, $abe = 193$, $bde = 152$, $ad = 137$, and $(1) = 98$ was run. How was this fraction obtained? Add this data to the original fraction and estimate the effects.

8.18 Construct a 2^{5-1} design. Show how the design may be run in two blocks of eight observations each. Are any main effects or two-factor interactions confounded with blocks?

8.19 Construct a 2^{7-2} design. Show how the design may be run in four blocks of eight observations each. Are any main effects or two-factor interactions confounded with blocks?

8.20 ***Nonregular fractions of the 2^k [John (1971)].*** Consider a 2^4 design. We must estimate the four main effects and the six two-factor interactions, but the full 2^4 factorial cannot be run. The largest possible block size contains 12 runs. These 12 runs can be obtained from the four one-quarter replicates defined by $I = \pm AB = \pm ACD = \pm BCD$ by omitting the principal fraction. Show how the remaining three 2^{4-2} fractions can be combined to estimate the required effects, assuming three-factor and higher interactions are negligible. This design could be thought of as a three-quarter fraction.

8.21 Carbon anodes used in a smelting process are baked in a ring furnace. An experiment is run in the furnace to determine which factors influence the weight of packing material that is stuck to the anodes after baking. Six variables are of interest, each at two levels: A = pitch/fines ratio (0.45, 0.55), B = packing material type (1, 2), C = packing material temperature (ambient, 325°C), D = flue location (inside, outside), E = pit temperature (ambient, 195°C), and F = delay time before packing (zero, 24 hours). A 2^{6-3} design is run, and three replicates are obtained at each of the design points. The weight of packing material stuck to the anodes is measured in grams. The data in run order are as follows: $abd = (984, 826, 936)$; $abcdef = (1275, 976, 1457)$; $be = (1217, 1201, 890)$; $af = (1474, 1164, 1541)$; $def = (1320, 1156, 913)$; $cd = (765, 705, 821)$; $ace = (1338, 1254, 1294)$; and $bcf = (1325, 1299, 1253)$. We wish to minimize the amount of stuck packing material.

SS

(a) Verify that the eight runs correspond to a 2^{6-3}_{III} design. What is the alias structure?

(b) Use the average weight as a response. What factors appear to be influential?

(c) Use the range of the weights as a response. What factors appear to be influential?

(d) What recommendations would you make to the process engineers?

SS **8.22** A 16-run experiment was performed in a semiconductor manufacturing plant to study the effects of six factors on the curvature or camber of the substrate devices produced. The six variables and their levels are shown in Table P8.2.

(a) What type of design did the experimenters use?

(b) What are the alias relationships in this design?

(c) Do any of the process variables affect average camber?

(d) Do any of the process variables affect the variability in camber measurements?

(e) If it is important to reduce camber as much as possible, what recommendations would you make?

8.23 A spin coater is used to apply photoresist to a bare silicon wafer. This operation usually occurs early in the semiconductor manufacturing process, and the average coating thickness and the variability in the coating thickness have an important impact on downstream manufacturing steps. Six variables are used in the experiment. The variables and their high and low levels are as follows:

Factor	Low Level	High Level
Final spin speed	7350 rpm	6650 rpm
Acceleration rate	5	20
Volume of resist applied	3 cc	5 cc
Time of spin	14 sec	6 sec
Resist batch variation	Batch 1	Batch 2
Exhaust pressure	Cover off	Cover on

■ **TABLE P8.2**
Factor Levels for the Experiment in Problem 8.22

Run	Lamination Temperature (°C)	Lamination Time (sec)	Lamination Pressure (tn)	Firing Temperature (°C)	Firing Cycle Time (h)	Firing Dew Point (°C)
1	55	10	5	1580	17.5	20
2	75	10	5	1580	29	26
3	55	25	5	1580	29	20
4	75	25	5	1580	17.5	26
5	55	10	10	1580	29	26
6	75	10	10	1580	17.5	20
7	55	25	10	1580	17.5	26
8	75	25	10	1580	29	20
9	55	10	5	1620	17.5	26
10	75	10	5	1620	29	20
11	55	25	5	1620	29	26
12	75	25	5	1620	17.5	20
13	55	10	10	1620	29	20
14	75	10	10	1620	17.5	26
15	55	25	10	1620	17.5	20
16	75	25	10	1620	29	26

The experimenter decides to use a 2^{6-1} design and to make three readings on resist thickness on each test wafer. The data are shown in Table P8.4.

(a) Verify that this is a 2^{6-1} design. Discuss the alias relationships in this design.

(b) What factors appear to affect average resist thickness?

(c) Because the volume of resist applied has little effect on average thickness, does this have any important practical implications for the process engineers?

(d) Project this design into a smaller design involving only the significant factors. Graphically display the results. Does this aid in interpretation?

(e) Use the range of resist thickness as a response variable. Is there any indication that any of these factors affect the variability in resist thickness?

(f) Where would you recommend that the engineers run the process?

8.24 Harry Peterson-Nedry (a friend of the author) owns a vineyard and winery in Newberg, Oregon. He grows several varieties of grapes and produces wine. Harry has used factorial designs for process and product development in the winemaking segment of the business. This problem describes the experiment conducted for the 1985 Pinot Noir. Eight variables, shown in Table P8.5, were originally studied in this experiment: Harry decided to use a 2_{IV}^{8-4} design with 16 runs. The wine was taste-tested by a panel of experts on March 8, 1986. Each expert ranked the 16 samples of wine tasted, with rank 1 being the best. The design and the taste-test panel results are shown in Table P8.6.

(a) What are the alias relationships in the design selected by Harry?

(b) Use the average ranks ($\bar{y}$) as a response variable. Analyze the data and draw conclusions. You will find it helpful to examine a normal probability plot of the effect estimates.

(c) Use the standard deviation of the ranks (or some appropriate transformation such as log s) as a response variable. What conclusions can you draw about the effects of the eight variables on variability in wine quality?

(d) After looking at the results, Harry decides that one of the panel members (DCM) knows more about beer than he does about wine, so they decide to delete his ranking. What effect would this have on the results and conclusions from parts (b) and (c)?

(e) Suppose that just before the start of the experiment, Harry and Judy discovered that the eight new barrels they ordered from France for use in the experiment would not arrive in time, and all 16 runs would have to be made with old barrels. If Harry just drops column C from their design, what does this do to the alias relationships? Does he need to start over and construct a new design?

■ **TABLE P8.3**
Data from the Experiment in Problem 8.22

Run	Camber for Replicate (in./in.) 1	2	3	4	Total (10^{-4}in./in.)	Mean (10^{-4}in./in.)	Standard Deviation
1	0.0167	0.0128	0.0149	0.0185	629	157.25	24.418
2	0.0062	0.0066	0.0044	0.0020	192	48.00	20.976
3	0.0041	0.0043	0.0042	0.0050	176	44.00	4.083
4	0.0073	0.0081	0.0039	0.0030	223	55.75	25.025
5	0.0047	0.0047	0.0040	0.0089	223	55.75	22.410
6	0.0219	0.0258	0.0147	0.0296	920	230.00	63.639
7	0.0121	0.0090	0.0092	0.0086	389	97.25	16.029
8	0.0255	0.0250	0.0226	0.0169	900	225.00	39.42
9	0.0032	0.0023	0.0077	0.0069	201	50.25	26.725
10	0.0078	0.0158	0.0060	0.0045	341	85.25	50.341
11	0.0043	0.0027	0.0028	0.0028	126	31.50	7.681
12	0.0186	0.0137	0.0158	0.0159	640	160.00	20.083
13	0.0110	0.0086	0.0101	0.0158	455	113.75	31.12
14	0.0065	0.0109	0.0126	0.0071	371	92.75	29.51
15	0.0155	0.0158	0.0145	0.0145	603	150.75	6.75
16	0.0093	0.0124	0.0110	0.0133	460	115.00	17.45

Each run was replicated four times, and a camber measurement was taken on the substrate. The data are shown in Table P8.3.

■ TABLE P8.4
Data for Problem 8.23

	A	*B*	*C*	*D*	*E*	*F*	Resist Thickness				
Run	Volume	Batch	Time (sec)	Speed	Acc.	Cover	Left	Center	Right	Avg.	Range
1	5	Batch 2	14	7350	5	Off	4531	4531	4515	4525.7	16
2	5	Batch 1	6	7350	5	Off	4446	4464	4428	4446	36
3	3	Batch 1	6	6650	5	Off	4452	4490	4452	4464.7	38
4	3	Batch 2	14	7350	20	Off	4316	4328	4308	4317.3	20
5	3	Batch 1	14	7350	5	Off	4307	4295	4289	4297	18
6	5	Batch 1	6	6650	20	Off	4470	4492	4495	4485.7	25
7	3	Batch 1	6	7350	5	On	4496	4502	4482	4493.3	20
8	5	Batch 2	14	6650	20	Off	4542	4547	4538	4542.3	9
9	5	Batch 1	14	6650	5	Off	4621	4643	4613	4625.7	30
10	3	Batch 1	14	6650	5	On	4653	4670	4645	4656	25
11	3	Batch 2	14	6650	20	On	4480	4486	4470	4478.7	16
12	3	Batch 1	6	7350	20	Off	4221	4233	4217	4223.7	16
13	5	Batch 1	6	6650	5	On	4620	4641	4619	4626.7	22
14	3	Batch 1	6	6650	20	On	4455	4480	4466	4467	25
15	5	Batch 2	14	7350	20	On	4255	4288	4243	4262	45
16	5	Batch 2	6	7350	5	On	4490	4534	4523	4515.7	44
17	3	Batch 2	14	7350	5	On	4514	4551	4540	4535	37
18	3	Batch 1	14	6650	20	Off	4494	4503	4496	4497.7	9
19	5	Batch 2	6	7350	20	Off	4293	4306	4302	4300.3	13
20	3	Batch 2	6	7350	5	Off	4534	4545	4512	4530.3	33
21	5	Batch 1	14	6650	20	On	4460	4457	4436	4451	24
22	3	Batch 2	6	6650	5	On	4650	4688	4656	4664.7	38
23	5	Batch 1	14	7350	20	Off	4231	4244	4230	4235	14
24	3	Batch 2	6	7350	20	On	4225	4228	4208	4220.3	20
25	5	Batch 1	14	7350	5	On	4381	4391	4376	4382.7	15
26	3	Batch 2	6	6650	20	Off	4533	4521	4511	4521.7	22
27	3	Batch 1	14	7350	20	On	4194	4230	4172	4198.7	58
28	5	Batch 2	6	6650	5	Off	4666	4695	4672	4677.7	29
29	5	Batch 1	6	7350	20	On	4180	4213	4197	4196.7	33
30	5	Batch 2	6	6650	20	On	4465	4496	4463	4474.7	33
31	5	Batch 2	14	6650	5	On	4653	4685	4665	4667.7	32
32	3	Batch 2	14	6650	5	Off	4683	4712	4677	4690.7	35

(f) Harry knows from experience that some treatment combinations are unlikely to produce good results. For example, the run with all eight variables at the high level generally results in a poorly rated wine. This was confirmed in the March 8, 1986 taste test. He wants to set up a new design for their 1986 Pinot Noir using these same eight variables, but he does not want to make the run with all eight factors at the high level. What design would you suggest?

■ **TABLE P8.5**
Factors and Levels for the Winemaking Experiment for Problem 8.24

Variable	Low Level (−)	High Level (+)
A = Pinot Noir clone	Pommard	Wadenswil
B = Oak type	Allier	Troncais
C = Age of barrel	Old	New
D = Yeast/skin contact	Champagne	Montrachet
E = Stems	None	All
F = Barrel toast	Light	Medium
G = Whole cluster	None	10%
H = Fermentation temperature	Low (75°F max)	High (92°F max)

8.25 Consider the 2^5 factorial in Problem 6.35. Suppose that the experimenters could only afford 16 runs. Set up the 2^{5-1} fractional factorial design with $I = ABCDE$ and select the responses for the runs from the full factorial data in Problem 6.35.

(a) Analyze the data and draw conclusions.

(b) Compare your findings with those from the full factorial in Problem 6.35.

(c) Are there any potential interactions that need further study? What additional runs do you recommend? Select these runs from the full factorial design in Problem 6.35 and analyze the new design. Discuss your conclusions.

8.26 Consider the 2^4 factorial experiment for surfactin production in Problem 6.36. Suppose that the experimenters could only afford eight runs. Set up the 2^{4-1} fractional factorial design with $I = ABCD$ and select the responses for the runs from the full factorial data in Problem 6.36.

(a) Analyze the data and draw conclusions.

(b) Compare your findings with those from the full factorial in Problem 6.36.

8.27 Consider the 2^4 factorial experiment in Problem 6.38. Suppose that the experimenters could only afford eight runs. Set up the 2^{4-1} fractional factorial design with $I = ABCD$ and select the responses for the runs from the full factorial data in Problem 6.38.

(a) Analyze the data for all of the responses and draw conclusions.

(b) Compare your findings with those from the full factorial in Problem 6.38.

■ **TABLE P8.6**
Design and Results for Wine Tasting Experiment for Problem 8.24

	Variable								Panel Rankings					Summary	
Run	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	HPN	JPN	CAL	DCM	RGB	$\bar{y}$	*s*
1	−	−	−	−	−	−	−	−	12	6	13	10	7	9.6	3.05
2	+	−	−	−	−	+	+	+	10	7	14	14	9	10.8	3.11
3	−	+	−	−	+	−	+	+	14	13	10	11	15	12.6	2.07
4	+	+	−	−	+	+	−	−	9	9	7	9	12	9.2	1.79
5	−	−	+	−	+	+	+	−	8	8	11	8	10	9.0	1.41
6	+	−	+	−	+	−	−	+	16	12	15	16	16	15.0	1.73
7	−	+	+	−	−	+	−	+	6	5	6	5	3	5.0	1.22
8	+	+	+	−	−	−	+	−	15	16	16	15	14	15.2	0.84
9	−	−	−	+	+	+	−	+	1	2	3	3	2	2.2	0.84
10	+	−	−	+	+	−	+	−	7	11	4	7	6	7.0	2.55
11	−	+	−	+	−	+	+	−	13	3	8	12	8	8.8	3.96
12	+	+	−	+	−	−	−	+	3	1	5	1	4	2.8	1.79
13	−	−	+	+	−	−	+	+	2	10	2	4	5	4.6	3.29
14	+	−	+	+	−	+	−	−	4	4	1	2	1	2.4	1.52
15	−	+	+	+	+	−	−	−	5	15	9	6	11	9.2	4.02
16	+	+	+	+	+	+	+	+	11	14	12	13	13	12.6	1.14

■ TABLE P8.7
The 2^{5-1} Fractional Factorial Design for Problem 8.28

Experiment	*P* (MPa)	*T* (°C)	*s* (min)	*d* (min)	*c* (μl)	Resolution	Sensitivity
1	13.8	50	2	2	100	0.00025	0.057
2	55.1	50	2	2	10	0.33333	0.094
3	13.8	120	2	2	10	0.02857	0.017
4	55.1	120	2	2	100	0.20362	1.561
5	13.8	50	15	2	10	0.00027	0.010
6	55.1	50	15	2	100	0 52632	0.673
7	13.8	120	15	2	100	0.00026	0.028
8	55.1	120	15	2	10	0.52632	1.144
9	13.8	50	2	15	10	0 42568	0.142
10	55.1	50	2	15	100	0.60150	0.399
11	13.8	120	2	15	100	0.06098	0.767
12	55.1	120	2	15	10	0.74165	1.086
13	13.8	50	15	15	100	0.08780	0.252
14	55.1	50	15	15	10	0.40000	0.379
15	13.8	120	15	15	10	0.00026	0.028
16	55.1	120	15	15	100	0.28091	3.105
Central	34.5	85	8.5	8.5	55	0.75000	1.836

8.28 An article in the *Journal of Chromatography A* ("Simultaneous Supercritical Fluid Derivatization and Extraction of Formaldehyde by the Hantzsch Reaction," 2000, Vol. 896, pp. 51–59) describes an experiment where the Hantzsch reaction is used to produce the chemical derivatization of formaldehyde in a supercritical medium. Pressure, temperature, and other parameters such as static and dynamic extraction time must be optimized to increase the yield of this kinetically controlled reaction. A 2^{5-1} fractional factorial design with one center run was used to study the significant parameters affecting the supercritical process in terms of resolution and sensitivity. Ultraviolet–visible spectrophotometry was used as the detection technique. The experimental design and the responses are shown in Table P8.7.

(a) Analyze the data from this experiment and draw conclusions.

(b) Analyze the residuals. Are there any concerns about model adequacy or violations of assumptions?

(c) Does the single center point cause any concerns about curvature or the possible need for second-order terms?

(d) Do you think that running one center point was a good choice in this design?

8.29 An article in *Soldering & Surface Mount Technology* ("Characterization of a Solder Paste Printing Process and Its Optimization," 1999, Vol. 11, No. 3, pp. 23–26) describes the use of a 2^{8-3} fractional factorial experiment to study the effect of eight factors on two responses; percentage volume matching (PVM)—the ratio of the actual printed solder paste volume to the designed volume; and nonconformities per unit (NPU)—the number of solder paste printing defects determined by visual inspection (20′ magnification) after printing according to an industry workmanship standard. The factor levels are shown below and the test matrix and response data are shown in Table P8.8.

Parameters	Levels	
	Low (−)	High (+)
A. Squeegee pressure, MPa	0.1	0.3
B. Printing speed, mm/s	24	32
C. Squeegee angle, deg	45	65
D. Temperature, °C	20	28
E. Viscosity, kCps	1100–1150	1250–1300
F. Cleaning interval, stroke	8	15
G. Separation speed, mm/s	0.4	0.8
H. Relative humidity, %	30	70

■ TABLE P8.8
The Solder Paste Experiment for Problem 8.29

Run Order	Parameters								PVM	NPU (%)
	A	*B*	*C*	*D*	*E*	*F*	*G*	*H*		
4	−	−	−	−	−	−	−	+	1.00	5
13	+	−	−	−	−	+	+	+	1.04	13
6	−	+	−	−	−	+	+	−	1.02	16
3	+	+	−	−	−	−	−	−	0.99	12
19	−	−	+	−	−	+	−	−	1.02	15
25	+	−	+	−	−	−	+	−	1.01	9
21	−	+	+	−	−	−	+	+	1.01	12
14	+	+	+	−	−	+	−	+	1.03	17
10	−	−	−	+	−	−	+	−	1.04	21
22	+	−	−	+	−	+	−	−	1.14	20
1	−	+	−	+	−	+	−	+	1.20	25
2	+	+	−	+	−	−	+	+	1.13	21
30	−	−	+	+	−	+	+	+	1.14	25
8	+	−	+	+	−	−	−	+	1.07	13
9	−	+	+	+	−	−	−	−	1.06	20
20	+	+	+	+	−	+	+	−	1.13	26
17	−	−	−	−	+	−	−	−	1.02	10
18	+	−	−	−	+	+	+	−	1.10	13
5	−	+	−	−	+	+	+	+	1.09	17
26	+	+	−	−	+	−	−	+	0.96	13
31	−	−	+	−	+	+	−	+	1.02	14
11	+	−	+	−	+	−	+	+	1.07	11
29	−	+	+	−	+	−	+	−	0.98	10
23	+	+	+	−	+	+	−	−	0.95	14
32	−	−	−	+	+	−	+	+	1.10	28
7	+	−	−	+	+	+	−	+	1.12	24
15	−	+	−	+	+	+	−	−	1.19	22
27	+	+	−	+	+	−	+	−	1.13	15
12	−	−	+	+	+	+	+	−	1.20	21
28	+	−	+	+	+	−	−	−	1.07	19
24	−	+	+	+	+	−	−	+	1.12	21
16	+	+	+	+	+	+	+	+	1.21	27

(a) Verify that the generators are $I = ABCF$, $I = ABDG$, and I = $BCDEH$ for this design.

(b) What are the aliases for the main effects and two-factor interactions? You can ignore all interactions of order three and higher.

(c) Analyze both PVM and NPU responses.

(d) Analyze the residual for both responses. Are there any problems with model adequacy?

(e) The ideal value of PVM is unity and the NPU response should be as small as possible. Recommend suitable operating conditions for the process based on the experimental results.

8.30 An article in the *International Journal of Research in Marketing* ("Experimental design on the front lines of marketing: Testing new ideas to increase direct mail sales," 2006, Vol. 23, pp. 309–319) describes the use of a 20-run Plackett–Burman design to investigate the effects of 19 factors to improve the response rate to a direct mail sales campaign to attract new customers to a credit card. The 19 factors are as follows:

Factor	(−) Control	(+) New Idea
A: Envelope teaser	General offer	Product-specific offer
B: Return address	Blind	Add company name
C: "Official" ink-stamp on envelope	Yes	No
D: Postage	Pre-printed	Stamp
E: Additional graphic on envelope	Yes	No
F: Price graphic on letter	Small	Large
G: Sticker	Yes	No
H: Personalize letter copy	No	Yes
I: Copy message	Targeted	Generic
J: Letter headline	Headline 1	Headline 2
K: List of benefits	Standard layout	Creative layout
L: Postscript on letter	Control version	New P.S.
M: Signature	Manager	Senior executive
N: Product selection	Many	Few
O: Value of free gift	High	Low
P: Reply envelope	Control	New style
Q: Information on buckslip	Product info	Free gift info
R: 2nd buckslip	No	Yes
S: Interest rate	Low	High

The 20-run Plackett–Burman design is shown in Table P8.9. Each test combination in Table P8.9 was mailed to 5000 potential customers, and the response rate is the percentage of customers who responded positively to the offer.

(a) Verify that in this design each main effect is aliased with all two-factor interactions except those that involve that main effect. That is, in the 19-factor design, the main effect for each factor is aliased with all two-factor interactions involving the other 18 factors, or 153 two-factor interactions (18!/2!16!).

(b) Show that for the 20-run Plackett–Burman design in Table P8.9, the weights (or correlations) that multiple the two-factor interactions in each alias chain are either −0.2, +0.2, or −0.6. Of the 153 interactions that are aliased with each main effect, 144 have weights of −0.2 or +0.2, while 9 interactions have weights of −0.6.

(c) Verify that the five largest main effects are *S*, *G*, *R*, *I*, and *J*.

(d) Factors *S* (interest rate) and *G* (presence of a sticker) are by far the largest main effects. The correlation between the main effect of *R* (2nd buckslip) and the *SG* interaction is −0.6. This means that a significant *SG* interaction would bias the estimate of the main effect of *R* by −0.6 times the value of the interaction. This suggests that it may not be the main effect of factor *R* that is important, but the two-factor interaction between *S* and *G*.

(e) Since this design projects into a full factorial in any three factors, obtain the projection in factors *S*, *G*, and *R* and verify that it is a full factorial with some runs replicated. Fit a full factorial model involving all three of these factors and the interactions (you will need to use a regression program to do this). Show that *S*, *G*, and the *SG* interaction are significant.

■ TABLE P8.9

The Plackett-Burman Design for the Direct Mail Experiment in Problem 8.30

Test Cell	Envelope Teaser *A*	Return Address *B*	"Official" Ink-stamp on Envelope *C*	Postage *D*	Additional Graphic on Envelope *E*	Price Graphic on Letter *F*	Sticker *G*	Personalize Letter Copy *H*	Copy Message *I*	Letter Headline *J*
1	+	+	−	−	+	+	+	+	−	+
2	−	+	+	−	−	+	+	+	+	−
3	+	−	+	+	−	−	+	+	+	+
4	+	+	−	+	+	−	−	+	+	+
5	−	+	+	−	+	+	−	−	+	+
6	−	−	+	+	−	+	+	−	−	+
7	−	−	−	+	+	−	+	+	−	−
8	−	−	−	−	+	+	−	+	+	−
9	+	−	−	−	−	+	+	−	+	+
10	−	+	−	−	−	−	+	+	−	+
11	+	−	+	−	−	−	−	+	+	−
12	−	+	−	+	−	−	−	−	+	+
13	+	−	+	−	+	−	−	−	−	+
14	+	+	−	+	−	+	−	−	−	−
15	+	+	+	−	+	−	+	−	−	−
16	+	+	+	+	−	+	−	+	−	−
17	−	+	+	+	+	−	+	−	+	−
18	−	−	+	+	+	+	−	+	−	+
19	+	−	−	+	+	+	+	−	+	−
20	−	−	−	−	−	−	−	−	−	−

List of Benefits *K*	Postscript on Letter *L*	Signature *M*	Product Selection *N*	Value of Free gift *O*	Reply Envelope *P*	Information on Buckslip *Q*	2nd Buckslip *R*	Interest Rate *S*	Orders	Response Rate
−	+	−	−	−	−	+	+	−	52	1.04%
+	−	+	−	−	−	−	+	+	38	0.76%
−	+	−	+	−	−	−	−	+	42	0.84%
+	−	+	−	+	−	−	−	−	134	2.68%
+	+	−	+	−	+	−	−	−	104	2.08%
+	+	+	−	+	−	+	−	−	60	1.20%
+	+	+	+	−	+	−	+	−	61	1.22%
−	+	+	+	+	−	+	−	+	68	1.36%
−	−	+	+	+	+	−	+	−	57	1.14%
+	−	−	+	+	+	+	−	+	30	0.60%
+	+	−	−	+	+	+	+	−	108	2.16%
−	+	+	−	−	+	+	+	+	39	0.78%
+	−	+	+	−	−	+	+	+	40	0.80%
+	+	−	+	+	−	−	+	+	49	0.98%
−	+	+	−	+	+	−	−	+	37	0.74%
−	−	+	+	−	+	+	−	−	99	1.98%
−	−	−	+	+	−	+	+	−	86	1.72%
−	−	−	−	+	+	−	+	+	43	0.86%
+	−	−	−	−	+	+	−	+	47	0.94%
−	−	−	−	−	−	−	−	−	104	2.08%

8.31 Consider the following experiment:

Run	Treatment Combination
1	*d*
2	*ae*
3	*b*
4	*abde*
5	*cde*
6	*ac*
7	*bce*
8	abcd

Answer the following questions about this experiment:

(a) How many factors did this experiment investigate?

(b) How many factors are in the basic design?

(c) Assume that the factors in the experiment are represented by the initial letters of the alphabet (i.e., *A*, *B*, etc.), what are the design generators for the factors beyond the basic design?

(d) Is this design a principal fraction?

(e) What is the complete defining relation?

(f) What is the resolution of this design?

8.32 Consider the following experiment:

Run	Treatment Combination	y
1	(1)	8
2	*ad*	10
3	*bd*	12
4	*ab*	7
5	*cd*	13
6	*ac*	6
7	*bc*	5
8	*abcd*	11

Answer the following questions about this experiment:

(a) How many factors did this experiment investigate?

(b) What is the resolution of this design?

(c) Calculate the estimates of the main effects.

(d) What is the complete defining relation for this design?

8.33 An unreplicated 2^{5-1} fractional factorial experiment with four center points has been run in a chemical process. The response variable is molecular weight. The experimenter has used the following factors:

Factor	Natural Levels	Coded Levels (x's)
***A*—time**	20, 40 (minutes)	−1, 1
***B*—temperature**	160, 180 (°C)	−1, 1
***C*—concentration**	30, 60 (percent)	−1, 1
***D*—stirring rate**	100, 150 (rpm)	−1, 1
***E*—catalyst type**	1, 2 (Type)	−1. 1

Suppose that the prediction equation that results from this experiment is $\hat{y} = 10 + 3x_1 + 2x_2 - 1x_1x_2$. What is the predicted response at $A = 30$, $B = 165$, $C = 50$, $D = 135$, and $E = 1$?

8.34 An unreplicated 2^{4-1} fractional factorial experiment has been run. The experimenter has used the following factors:

Factor	Natural Levels	Coded Levels (x's)
A	20, 50	−1, 1
B	200, 280	−1, 1
C	50, 100	−1, 1
D	150, 200	−1, 1

(a) Suppose that this design has four center runs that average 100. The average of the 16 factorial design points is 95. What is the sum of squares for pure quadratic curvature?

(b) Suppose that the prediction equation that results from this experiment is $\hat{y} = 100 + -2x_1 + 10x_2 - 4x_1x_2$. What is the predicted response at $A = 41$, $B = 280$, $C = 60$, and $D = 185$?

8.35 A 2^{6-2} factorial experiment with three replicates has been run in a pharmaceutical drug manufacturing process. The experimenter has used the following factors:

Factor	Natural Levels	Coded Levels (x's)
A	50, 100	−1, 1
B	20, 60	−1, 1
C	10, 30	−1, 1
D	12, 18	−1, 1
E	15, 30	−1, 1
F	60, 100	−1, 1

(a) If two main effects and one two-factor interaction are included in the final model, how many degrees of freedom for error will be available?

(b) Suppose that the significant factors are A, C, AB, and AC. What other effects need to be included to obtain a hierarchical model?

8.36 Consider the following design:

Run	A	B	C	D	E	y
1	−1	−1	−1	−1	−1	63
2	1	−1	−1	−1	1	21
3	−1	1	−1	−1	1	36
4	1	1	−1	−1	−1	99
5	−1	−1	1	−1	1	24
6	1	−1	1	−1	−1	66
7	−1	1	1	−1	−1	71
8	1	1	1	−1	1	54
9	−1	−1	−1	1	−1	23
10	1	−1	−1	1	1	74
11	−1	1	−1	1	1	80
12	1	1	−1	1	−1	33
13	−1	−1	1	1	1	63
14	1	−1	1	1	−1	21
15	−1	1	1	1	−1	44
16	1	1	1	1	1	96

(a) What is the generator for column E?

(b) If ABC is confounded with blocks, run 1 above goes in the _____ block. Answer either + or −.

(c) What is the resolution of this design?

(d) (8 pts) Find the estimates of the main effects and their aliases.

8.37 Consider the following design:

Run	A	B	C	D	E	y
1	−1	−1	−1	−1	−1	65
2	1	−1	−1	−1	1	25
3	−1	1	−1	−1	1	30
4	1	1	−1	−1	−1	89
5	−1	−1	1	−1	1	25
6	1	−1	1	−1	−1	60
7	−1	1	1	−1	−1	70
8	1	1	1	−1	1	50
9	−1	−1	−1	1	1	20
10	1	−1	−1	1	−1	70
11	−1	1	−1	1	−1	80
12	1	1	−1	1	1	30
13	−1	−1	1	1	−1	60
14	1	−1	1	1	1	20
15	−1	1	1	1	1	40
16	1	1	1	1	−1	90

(a) What is the generator for column E?

(b) If ABE is confounded with blocks, run 16 goes in the _____ block. Answer either − or +.

(c) The resolution of this design is _____.

(d) Find the estimates of the main effects and their aliases.

8.38 Consider the following design:

Run	A	B	C	D	E	y
1	−1	−1	−1	1	−1	50
2	1	−1	−1	−1	−1	20
3	−1	1	−1	−1	1	40
4	1	1	−1	1	1	25
5	−1	−1	1	−1	1	45
6	1	−1	1	1	1	30
7	−1	1	1	1	−1	40
8	1	1	1	−1	−1	30

(a) What is the generator for column D?

(b) What is the generator for column E?

(c) If this design were run in two blocks with the AB interaction confounded with blocks, the run d would be in the block where the sign on AB is _____. Answer either − or +.

8.39 Consider the following design:

Std	A	B	C	D	E	y
1	−1	−1	−1	1	1	40
2	1	−1	−1	−1	1	10
3	−1	1	−1	−1	−1	30
4	1	1	−1	1	−1	20
5	−1	−1	1	−1	−1	40
6	1	−1	1	1	−1	30
7	−1	1	1	1	1	20
8	1	1	1	−1	1	30

(a) What is the generator for column D?

(b) What is the generator for column E?

(c) If this design were folded over, what is the resolution of the combined design?

8.40 In an article in *Quality Engineering* ("An Application of Fractional Factorial Experimental Designs," 1988, Vol. 1, pp. 19–23), M. B. Kilgo describes an experiment to determine the effect of CO_2 pressure (A), CO_2 temperature (B), peanut

SS

moisture (C), CO_2 flow rate (D), and peanut particle size (E) on the total yield of oil per batch of peanuts (y). The levels that she used for these factors are shown in Table P8.10. She conducted the 16-run fractional factorial experiment shown in Table P8.11.

■ **TABLE P8.10**
Factor Levels for the Experiment in Problem 8.40

Coded Level	*A*, Pressure (bar)	*B*, Temp, (°C)	*C*, Moisture (% by weight)	*D*, Flow (liters/ min)	*E*, Part. Size (mm)
−1	415	25	5	40	1.28
1	550	95	15	60	4.05

■ **TABLE P8.11**
The Peanut Oil Experiment for Problem 8.40

A	*B*	*C*	*D*	*E*	*y*
415	25	5	40	1.28	63
550	25	5	40	4.05	21
415	95	5	40	4.05	36
550	95	5	40	1.28	99
415	25	15	40	4.05	24
550	25	15	40	1.28	66
415	95	15	40	1.28	71
550	95	15	40	4.05	54
415	25	5	60	4.05	23
550	25	5	60	1.28	74
415	95	5	60	1.28	80
550	95	5	60	4.05	33
415	25	15	60	1.28	63
550	25	15	60	4.05	21
415	95	15	60	4.05	44
550	95	15	60	1.28	96

(a) What type of design has been used? Identify the defining relation and the alias relationships.

(b) Estimate the factor effects and use a normal probability plot to tentatively identify the important factors.

(c) Perform an appropriate statistical analysis to test the hypotheses that the factors identified in part (b) have a significant effect on the yield of peanut oil.

(d) Fit a model that could be used to predict peanut oil yield in terms of the factors that you have identified as important.

(e) Analyze the residuals from this experiment and comment on model adequacy.

8.41 A 16-run fractional factorial experiment in nine factors was conducted by Chrysler Motors Engineering and described in the article "Sheet Molded Compound Process Improvement," by P. I. Hsieh and D. E. Goodwin (*Fourth Symposium on Taguchi Methods*, American Supplier Institute, Dearborn, MI, 1986, pp. 13–21). The purpose was to reduce the number of defects in the finish of sheet-molded grill opening panels. The design, and the resulting number of defects, c, observed on each run, is shown in Table P8.12. This is a resolution III fraction with generators $E = BD$, $F = BCD$, $G = AC$, $H = ACD$, and $J = AB$.

(a) Find the defining relation and the alias relationships in this design.

(b) Estimate the factor effects and use a normal probability plot to tentatively identify the important factors.

(c) Fit an appropriate model using the factors identified in part (b).

(d) Plot the residuals from this model versus the predicted number of defects. Also, prepare a normal probability plot of the residuals. Comment on the adequacy of these plots.

(e) In part (d) you should have noticed an indication that the variance of the response is not constant. (Considering that the response is a count, you should have expected this.) The previous table also shows a transformation on c, the square root, that is a widely used *variance stabilizing transformation* for count data. (Refer to the discussion of variance stabilizing transformations in Chapter 3.) Repeat parts (a) through (d) using the transformed response and comment on your results. Specifically, are the residual plots improved?

(f) There is a modification to the square root transformation, proposed by Freeman and Tukey ("Transformations Related to the Angular and the Square Root," *Annals of Mathematical Statistics*, Vol. 21, 1950, pp. 607–611) that improves its performance. F&T's modification to the square root transformation is

$$[\sqrt{c} + \sqrt{(c+1)}]/2$$

Rework parts (a) through (d) using this transformation and comment on the results. (For an interesting discussion and analysis of this experiment, refer to "Analysis of Factorial Experiments with Defects or Defectives as the Response," by S. Bisgaard and H. T. Fuller, *Quality Engineering*, Vol. 7, 1994–95, pp. 429–443.)

8.42 An experiment is run in a semiconductor factory to investigate the effect of six factors on transistor gain. The design selected is the 2_{IV}^{6-2} shown in Table P8.13.

■ **TABLE P8.12**
The Grill Defects Experiment for Problem 8.41

Run	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*c*	$\sqrt{c}$	F&T's Modification
1	−	−	−	−	+	−	+	−	+	56	7.48	7.52
2	+	−	−	−	+	−	−	+	−	17	4.12	4.18
3	−	+	−	−	−	+	+	−	−	2	1.41	1.57
4	+	+	−	−	−	+	−	+	+	4	2.00	2.12
5	−	−	+	−	+	+	−	+	+	3	1.73	1.87
6	+	−	+	−	+	+	+	−	−	4	2.00	2.12
7	−	+	+	−	−	−	−	+	−	50	7.07	7.12
8	+	+	+	−	−	−	+	−	+	2	1.41	1.57
9	−	−	−	+	−	+	+	+	+	1	1.00	1.21
10	+	−	−	+	−	+	−	−	−	0	0.00	0.50
11	−	+	−	+	+	−	+	+	−	3	1.73	1.87
12	+	+	−	+	+	−	−	−	+	12	3.46	3.54
13	−	−	+	+	−	−	−	−	+	3	1.73	1.87
14	+	−	+	+	−	−	+	+	−	4	2.00	2.12
15	−	+	+	+	+	+	−	−	−	0	0.00	0.50
16	+	+	+	+	+	+	+	+	+	0	0.00	0.50

■ **TABLE P8.13**
The Transistor Gain Experiment for Problem 8.42

Standard Order	Run Order	*A*	*B*	*C*	*D*	*E*	*F*	Gain
1	2	−	−	−	−	−	−	1455
2	8	+	−	−	−	+	−	1511
3	5	−	+	−	−	+	+	1487
4	9	+	+	−	−	−	+	1596
5	3	−	−	+	−	+	+	1430
6	14	+	−	+	−	−	+	1481
7	11	−	+	+	−	−	−	1458
8	10	+	+	+	−	+	−	1549
9	15	−	−	−	+	−	+	1454
10	13	+	−	−	+	+	+	1517
11	1	−	+	−	+	+	−	1487
12	6	+	+	−	+	−	−	1596
13	12	−	−	+	+	+	−	1446
14	4	+	−	+	+	−	−	1473
15	7	−	+	+	+	−	+	1461
16	16	+	+	+	+	+	+	1563

(a) Use a normal plot of the effects to identify the significant factors.

(b) Conduct appropriate statistical tests for the model identified in part (a).

(c) Analyze the residuals and comment on your findings.

(d) Can you find a set of operating conditions that produce gain of 1500 ± 25?

8.43 Heat treating is often used to carbonize metal parts, such as gears. The thickness of the carbonized layer is a critical output variable from this process, and it is usually measured by performing a carbon analysis on the gear pitch (the top of the gear tooth). Six factors were studied in a 2_{IV}^{6-2} design: A = furnace temperature, B = cycle time, C = carbon concentration, D = duration of the carbonizing cycle, E = carbon concentration of the diffuse cycle, and F = duration of the diffuse cycle. The experiment is shown in Table P8.14.

(a) Estimate the factor effects and plot them on a normal probability plot. Select a tentative model.

(b) Perform appropriate statistical tests on the model.

(c) Analyze the residuals and comment on model adequacy.

(d) Interpret the results of this experiment. Assume that a layer thickness of between 140 and 160 is desirable.

■ **TABLE P8.14**
The Heat Treating Experiment for Problem 8.43

Standard Order	Run Order	A	B	C	D	E	F	Pitch
1	5	−	−	−	−	−	−	74
2	7	+	−	−	−	+	−	190
3	8	−	+	−	−	+	+	133
4	2	+	+	−	−	−	+	127
5	10	−	−	+	−	+	+	115
6	12	+	−	+	−	−	+	101
7	16	−	+	+	−	−	−	54
8	1	+	+	+	−	+	−	144
9	6	−	−	−	+	−	+	121
10	9	+	−	−	+	+	+	188
11	14	−	+	−	+	+	−	135
12	13	+	+	−	+	−	−	170
13	11	−	−	+	+	+	−	126
14	3	+	−	+	+	−	−	175
15	15	−	+	+	+	−	+	126
16	4	+	+	+	+	+	+	193

8.44 An article by L. B. Hare ("In the Soup: A Case Study to Identify Contributors to Filling Variability," *Journal of Quality Technology*, Vol. 20, pp. 36–43) describes a factorial experiment used to study the filling variability of dry soup mix packages. The factors are A = number of mixing ports through which the vegetable oil was added (1, 2), B = temperature surrounding the mixer (cooled, ambient), C = mixing time (60, 80 sec), D = batch weight (1500, 2000 lb), and E = number of days of delay between mixing and packaging (1, 7). Between 125 and 150 packages of soup were sampled over an 8-hour period for each run in the design, and the standard deviation of package weight was used as the response variable. The design and resulting data are shown in Table P8.15.

(a) What is the generator for this design?

(b) What is the resolution of this design?

(c) Estimate the factor effects. Which effects are large?

(d) Does a residual analysis indicate any problems with the underlying assumptions?

(e) Draw conclusions about this filling process.

■ **TABLE P8.15**
The Soup Experiment for Problem 8.43

Std. Order	A Mixer Ports	B Temp.	C Time	D Batch Weight	E Delay	y Std. Dev
1	−	−	−	−	−	1.13
2	+	−	−	−	+	1.25
3	−	+	−	−	+	0.97
4	+	+	−	−	−	1.7
5	−	−	+	−	+	1.47
6	+	−	+	−	−	1.28
7	−	+	+	−	−	1.18
8	+	+	+	−	+	0.98
9	−	−	−	+	+	0.78
10	+	−	−	+	−	1.36
11	−	+	−	+	−	1.85
12	+	+	−	+	+	0.62
13	−	−	+	+	−	1.09
14	+	−	+	+	+	1.1
15	−	+	+	+	+	0.76
16	+	+	+	+	−	2.1

8.45 Consider the 2_{IV}^{6-2} design.

(a) Suppose that the design had been folded over by changing the signs in column B instead of column A. What changes would have resulted in the effects that can be estimated from the combined design?

(b) Suppose that the design had been folded over by changing the signs in column E instead of column A. What changes would have resulted in the effects that can be estimated from the combined design?

8.46 Consider the 2_{IV}^{7-3} design. Suppose that a fold over of this design is run by changing the signs in column A. Determine the alias relationships in the combined design.

8.47 Reconsider the 2_{IV}^{7-3} design in Problem 8.46.

(a) Suppose that a fold over of this design is run by changing the signs in column B. Determine the alias relationships in the combined design.

(b) Compare the aliases from this combined design to those from the combined design from Problem 8.28. What differences resulted by changing the signs in a different column?

8.48 Consider the 2_{IV}^{7-3} design.

(a) Suppose that a partial fold over of this design is run using column A (+ signs only). Determine the alias relationships in the combined design.

(b) Rework part (a) using the negative signs to define the partial fold over. Does it make any difference which signs are used to define the partial fold over?

8.49 Consider a partial fold over for the 2_{IV}^{6-2} design. Suppose that the signs are reversed in column A, but the eight runs that are retained are the runs that have positive signs in column C. Determine the alias relationships in the combined design.

8.50 Consider a partial fold over for the 2_{III}^{7-4} design. Suppose that the partial fold over of this design is constructed using column A (+ signs only). Determine the alias relationships in the combined design.

8.51 Consider a partial fold over for the 2_{III}^{5-2} design. Suppose that the partial fold over of this design is constructed using column A (+ signs only). Determine the alias relationships in the combined design.

8.52 Reconsider the 2^{4-1} design in Example 8.1. The significant factors are A, C, D, $AC + BD$, and $AD + BC$. Find a partial fold-over design that will allow the AC, BD, AD, and BC interactions to be estimated.

8.53 Construct a supersaturated design for $k = 8$ factors in $P = 6$ runs.

8.54 Consider the 2^{8-3} design in Problem 8.29. Suppose that the alias chain involving the AB interaction was large. Recommend a partial fold-over design to resolve the ambiguity about this interaction.

8.55 Construct a supersaturated design for $h = 12$ factors in $N = 10$ runs.

8.56 Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-1} fractional factorial. Set up the fractional design using the principal fraction. Using the 16 runs associated with this design from the original experiment, analyze the data and compare your results with those obtained from the analysis of the original full factorial.

8.57 Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-1} fractional factorial. Set up the fractional design using the alternate fraction. Using the 16 runs associated with this design from the original experiment, analyze the data and compare your results with those obtained from the analysis of the principal fraction in Problem 8.56.

8.58 Consider the full 2^5 factorial design in Problem 6.42. Suppose that this experiment had been run as a 2^{5-2} fractional factorial. Set up the fractional design using the principal fraction. Using the eight runs associated with this design from the original experiment, analyze the data and compare your results with those obtained from the analysis of the original full factorial. Then construct the fold-over design and analyze the data from the combined design.

8.59 Consider the fold over of the 2^{5-2} fractional factorial constructed in Problem 8.58. Compare this design with the two one-half fractions of the 2^{5-1}. Is the fold-over design the same as either of the one-half fractions?

8.60 The resolution of a two-level fractional factorial design is the number of words in the defining relation.

(a) True

(b) False

8.61 For a half fraction of a two-level factorial design the maximum resolution possible is equal to the number of factors.

(a) True

(b) False

8.62 The 12-run Plackett–Burman for up to 11 factors design is a regular fraction.

(a) True

(b) False

8.63 The design points of the 2^{k-p} family are at the corners of a cube in a k-dimensional space and they project into a full factorial in any subset of the original k factors.

(a) True

(b) False

8.64 It is good practice to keep the number of factor levels low and region of interest small in a screening experiment.

(a) True

(b) False

8.65 Consider a 2^{4-1} fractional factorial design. If the principal fraction is run first—first block (I = ABCD)—and then later augmented with the alternate fraction—second block—the four-factor interaction effect is confounded with blocks.

(a) True

(b) False

8.66 In a 2^{k-2} design, every effect has four aliases.

(a) True

(b) False

8.67 In a 2^{k-3} design, the complete defining relation has 15 words.

(a) True

(b) False

8.68 The aberration of a fractional factorial design is related to the length of the longest word in the defining relation.

(a) True

(b) False

Chapter 9 Problems

S **9.1** The effects of developer strength (A) and development time (B) on the density of photographic plate film are being studied. Three strengths and three times are used, and four replicates of a 3^2 factorial experiment are run. The data from this experiment follow. Analyze the data using the standard methods for factorial experiments.

Developer Strength	Development Time (minutes)					
	10		14		18	
1	0	2	1	3	2	5
	5	4	4	2	4	6
2	4	6	6	8	9	10
	7	5	7	7	8	5
3	7	10	10	10	12	10
	8	7	8	7	9	8

9.2 Compute the I and J components of the two-factor interaction in Problem 9.1.

S **9.3** An experiment was performed to study the effect of three different types of 32-ounce bottles (A) and three different shelf types (B)—smooth permanent shelves, end-aisle displays with grilled shelves, and beverage coolers—on the time it takes to stock ten 12-bottle cases on the shelves. Three workers (factor C) were employed in the experiment, and two replicates of a 3^3 factorial design were run. The observed time data are shown in the following table. Analyze the data and draw conclusions.

		Replicate I		
Worker	Bottle Type	Permanent	End Aisle	Cooler
	Plastic	3.45	4.14	5.80
1	28-mm glass	4.07	4.38	5.48
	38-mm glass	4.20	4.26	5.67
	Plastic	4.80	5.22	6.21
2	28-mm glass	4.52	5.15	6.25
	38-mm glass	4.96	5.17	6.03
	Plastic	4.08	3.94	5.14
3	28-mm glass	4.30	4.53	4.99
	38-mm glass	4.17	4.86	4.85

		Replicate II		
		Permanent	End Aisle	Cooler
	Plastic	3.36	4.19	5.23
1	28-mm glass	3.52	4.26	4.85
	38-mm glass	3.68	4.37	5.58
	Plastic	4.40	4.70	5.88
2	28-mm glass	4.44	4.65	6.20
	38-mm glass	4.39	4.75	6.38
	Plastic	3.65	4.08	4.49
3	28-mm glass	4.04	4.08	4.59
	38-mm glass	3.88	4.48	4.90

9.4 An experiment is run in a chemical process using a 3^2 factorial design. The design factors are temperature and pressure, and the response variable is yield. The data that result from this experiment are as follows.

Temperature, °C	Pressure, psig		
	100	120	140
80	47.58, 48.77	64.97, 69.22	80.92, 72.60
90	51.86, 82.43	88.47, 84.23	93.95, 88.54
100	71.18, 92.77	96.57, 88.72	76.58, 83.04

(a) Analyze the data from this experiment by conducting an analysis of variance. What conclusions can you draw?

(b) Graphically analyze the residuals. Are there any concerns about underlying assumptions or model adequacy?

(c) Verify that if we let the low, medium, and high levels of both factors in this design take on the levels $-1, 0$, and $+1$, then a least squares fit to a second-order model for yield is

$$\hat{y} = 86.81 + 10.4x_1 + 8.42x_2 - 7.17x_1^2 - 7.84x_2^2 - 7.69x_1x_2$$

(d) Confirm that the model in part (c) can be written in terms of the natural variables temperature (T) and pressure (P) as

$$\hat{y} = -1335.63 + 18.56T + 8.59P - 0.072T^2 - 0.0196P^2 - 0.0384TP$$

(e) Construct a contour plot for yield as a function of pressure and temperature. Based on examination of this plot, where would you recommend running this process?

9.5 (a) Confound a 3^3 design in three blocks using the ABC^2 component of the three-factor interaction. Compare your results with the design in Figure 9.7.

(b) Confound a 3^3 design in three blocks using the AB^2C component of the three-factor interaction. Compare your results with the design in Figure 9.7.

(c) Confound a 3^3 design in three blocks using the ABC component of the three-factor interaction. Compare your results with the design in Figure 9.7.

(d) After looking at the designs in parts (a), (b), and (c) and Figure 9.7, what conclusions can you draw?

9.6 Confound a 3^4 design in three blocks using the AB^2CD component of the four-factor interaction.

9.7 Consider the data from the first replicate of Problem 9.3. Assuming that not all 27 observations could be run on the same day, set up a design for conducting the experiment over three days with AB^2C confounded with blocks. Analyze the data. SS

9.8 Outline the analysis of variance table for the 3^4 design in nine blocks. Is this a practical design?

9.9 Consider the data in Problem 9.3. If ABC is confounded in replicate I and ABC^2 is confounded in replicate II, perform the analysis of variance. SS

9.10 Consider the data from replicate I of Problem 9.3. Suppose that only a one-third fraction of this design with $I = ABC$ is run. Construct the design, determine the alias structure, and analyze the data.

9.11 From examining Figure 9.9, what type of design would remain if after completing the first nine runs, one of the three factors could be dropped?

9.12 Construct a 3^{4-1}_{IV} design with $I = ABCD$. Write out the alias structure for this design. SS

9.13 Verify that the design in Problem 9.12 is a resolution IV design.

9.14 Construct a 3^{5-2} design with $I = ABC$ and $I = CDE$. Write out the alias structure for this design. What is the resolution of this design? SS

9.15 Construct a 3^{9-6} design and verify that it is a resolution III design.

9.16 Construct a 4×2^3 design confounded in two blocks of 16 observations each. Outline the analysis of variance for this design.

9.17 Outline the analysis of variance table for a $2^2 3^2$ factorial design. Discuss how this design may be confounded in blocks.

9.18 Starting with a 16-run 2^4 design, show how two three-level factors can be incorporated in this experiment. How many two-level factors can be included if we want some information on two-factor interactions?

9.19 Starting with a 16-run 2^4 design, show how one three-level factor and three two-level factors can be accommodated and still allow the estimation of two-factor interactions.

9.20 In Problem 8.24, you met Harry Peterson-Nedry, a friend of the author who has a winery and vineyard in Newberg, Oregon. That problem described the application of two-level fractional factorial designs to their 1985 Pinot Noir

product. In 1987, he wanted to conduct another Pinot Noir experiment. The variables for this experiment were

Variable	Levels
Clone of Pinot Noir	Wadenswil, Pommard
Berry size	Small, large
Fermentation temperature	80°, 85°, 90/80°, and 90°F
Whole berry	None, 10%
Maceration time	10 and 21 days
Yeast type	Assmanhau, Champagne
Oak type	Tronçais, Allier

Harry decided to use a 16-run two-level fractional factorial design, treating the four levels of fermentation temperature as two two-level variables. As he has done previously, Harry used the rankings from a taste-test panel as the response variable. The design and the resulting average ranks are as follows.

Run	Clone	Berry Size	Ferm. Temp.	Whole Berry	Macer. Time	Yeast Type	Oak Type	Average Rank
1	−	−	− −	−	−	−	−	4
2	+	−	− −	−	+	+	+	10
3	−	+	− −	+	−	+	+	6
4	+	+	− −	+	+	−	−	9
5	−	−	+ −	+	+	+	−	11
6	+	−	+ −	+	−	−	+	1
7	−	+	+ −	−	+	−	+	15
8	+	+	+ −	−	−	+	−	5
9	−	−	− +	+	+	−	+	12
10	+	−	− +	+	−	+	−	2
11	−	+	− +	−	+	+	−	16
12	+	+	− +	−	−	−	+	3
13	−	−	+ +	−	−	+	+	8
14	+	−	+ +	−	+	−	−	14
15	−	+	+ +	+	−	−	−	7
16	+	+	+ +	+	+	+	+	13

(a) Describe the aliasing in this design.

(b) Analyze the data and draw conclusions.

9.21 Suppose that there are four three-level categorical factors and a single two-level continuous factor. What is the minimum number of runs required to estimate all main effects and two-factor interactions? Construct this design.

S **9.22** Reconsider the experiment in Problem 9.21. Construct a design with $N = 48$ runs and compare it to the design you constructed in Problem 9.21.

9.23 Reconsider the experiment in Problem 9.21. Suppose that you are only interested in main effects. Construct a design with $N = 12$ runs for this experiment.

9.24 An article in the *Journal of Chemical Technology and Biotechnology* ("A Study of Antifungal Antibiotic Production by *Thermomonospora* sp MTCC 3340 Using Full Factorial Design," 2003, Vol. 78, pp. 605–610) investigated three independent variables—concentration of carbon source (glucose), concentration of nitrogen source (soybean meal), and temperature of incubation for their effects on the production of antifungal antibiotic by the isolate *Thermomonospora* sp MTCC 3340. A 3^3 factorial design was conducted and the results are shown in Table P9.1.

(a) Analyze the data from this experiment.

(b) Fit a second-order model to the activity response. Construct contour plots and response surface plots to assist in interpreting the results of this experiment.

(c) What operating conditions would you recommend to optimize this process?

9.25 Construct a minimum-run D-optimal resolution IV design for 10 factors. Find the alias relationships. What approach would you recommend for analyzing the data from this experiment?

9.26 Construct a minimum-run D-optimal resolution IV design for 12 factors. Find the alias relationships. What approach would you recommend for analyzing the data from this experiment?

9.27 Suppose that you must design an experiment to investigate nine continuous factors. It is thought that running all factors at two levels is adequate but that all two-factor interactions are of interest.

(a) How many runs are required to estimate all main effects and two-factor interactions?

(b) Find a minimum-run D-optimal design that is suitable for this problem.

9.28 Suppose that you must design an experiment to investigate seven continuous factors. Running all factors at two levels is thought to be appropriate but that only the two-factor interactions involving factor A are of interest.

(a) How many runs are required to estimate all of the relevant effects?

(b) Find a minimum-run D-optimal design that is suitable for this problem.

9.29 Suppose that you must design an experiment to investigate six continuous factors. It is thought that running all factors at two levels is adequate but that only the AB, AC, and AD two-factor interactions are of interest. SS

(a) How many runs are required to estimate all of the effects that are of interest?

(b) Find a minimum-run D-optimal design that is suitable for this problem.

■ **TABLE P9.1**
Data for Problem 9.24

Run No	Percent Carbon (Glucose)	Percent Nitrogen (Soybean Meal)	Temperature (°C)	Activity Against *C. albicans*
1	2	0.5		25.84
2	2	1.0		51.86
3	2	3.0		32.59
4	5	0.5		20.48
5	5	1.0	25	25.84
6	5	3.0		12.87
7	7.5	0.5		20.48
8	7.5	1.0		25.84
9	7.5	3.0		10.20
10	2	0.5		51.86
11	2	1.0		131.33
12	2	3.0		41.11
13	5	0.5		41.11
14	5	1.0	30	104.11
15	5	3.0		32.59
16	7.5	0.5		65.42
17	7.5	1.0		82.53
18	7.5	3.0		51.86
19	2	0.5		41.11
20	2	1.0		104.11
21	2	3.0		32.59
22	5	0.5		32.59
23	5	1.0	35	82.53
24	5	3.0		25.84
25	7.5	0.5		51.86
26	7.5	1.0		65.42
27	7.5	3.0		41.11

9.30 Suppose that you must design an experiment with six categorical factors. Factor A has six levels, factor B has five levels, factor C has five levels, factor D has three levels, and factors E and F have two levels. You are interested in main effects and two-factor interactions.

(a) How many runs are required to estimate all of the effects that are of interest?

(b) Find a D-optimal design that is suitable for this problem.

(c) Suppose that the experimenter decides that this design requires too many runs. What strategy would you recommend?

9.31 Suppose that you need to design an experiment for nine factors. Three of these factors are categorical with three levels and the remaining six are continuous. You are interested in the main effects and the two-factor interactions.

(a) How many runs are required for this experiment?

(b) Construct a D-optimal design with the minimum number of runs. What are the relative variances of the model parameters?

(c) Construct an I-optimal design with the minimum number of runs. What are the relative variances of the model parameters? How does this design compare to the D-optimal design you constructed in part (b)?

9.32 Suppose that you need to design an experiment for five factors. You are interested in the main effects of all factors, but only the AB and AC factor interactions.

(a) How many runs are required for this experiment?

(b) Construct a D-optimal design with the minimum number of runs. What are the relative variances of the model parameters?

(c) Construct an I-optimal design with the minimum number of runs. What are the relative variances of the model parameters? How does this design compare to the D-optimal design you constructed in part (b)?

(d) Construct an I-optimal design with 12 runs. How does this design compare to the minimum-run design from part (c)?

9.33 Construct a minimum-run D-optimal design for eight two-level factors. Design this experiment to estimate all main effects and as many two-factor interactions as possible. Compare this design to the 16-run no-confounding design for eight factors. Which design would you prefer? Why?

9.34 Suppose that you need to conduct an experiment for four categorical factors. Factor A has five levels, factor B has four levels, and the other two factors each have three levels. You are interested in all main effects and the two-factor interactions.

(a) How many runs are required?

(b) How many runs would you actually recommend?

(c) Find an appropriate D-optimal design.

Chapter 10 Problems

10.1 The tensile strength of a paper product is related to the amount of hardwood in the pulp. Ten samples are produced in the pilot plant, and the data obtained are shown in the following table.

Strength	Percent Hardwood	Strength	Percent Hardwood
160	10	181	20
171	15	188	25
175	15	193	25
182	20	195	28
184	20	200	30

(a) Fit a linear regression model relating strength to percent hardwood.

(b) Test the model in part (a) for significance of regression.

(c) Find a 95 percent confidence interval on the parameter β_1.

10.2 Plot the residuals from Problem 10.1 and comment on model adequacy.

10.3 Using the results of Problem 10.1, test the regression model for lack of fit.

10.4 A study was performed on wear of a bearing y and its relationship to x_1 = oil viscosity and x_2 = load. The following data were obtained:

y	x_1	x_2
193	1.6	851
230	15.5	816
172	22.0	1058
91	43.0	1201
113	33.0	1357
125	40.0	1115

(a) Fit a multiple linear regression model to the data.

(b) Test for significance of regression.

(c) Compute t statistics for each model parameter. What conclusions can you draw?

10.5 The brake horsepower developed by an automobile engine on a dynamometer is thought to be a function of the engine speed in revolutions per minute (rpm), the road octane number of the fuel, and the engine compression. An experiment is run in the laboratory and the data that follow are collected:

Brake Horsepower	rpm	Road Octane Number	Compression
225	2000	90	100
212	1800	94	95
229	2400	88	110
222	1900	91	96
219	1600	86	100
278	2500	96	110
246	3000	94	98
237	3200	90	100
233	2800	88	105
224	3400	86	97
223	1800	90	100
230	2500	89	104

(a) Fit a multiple regression model to these data.

(b) Test for significance of regression. What conclusions can you draw?

(c) Based on t-tests, do you need all three regressor variables in the model?

10.6 Analyze the residuals from the regression model in Problem 10.5. Comment on model adequacy.

10.7 The yield of a chemical process is related to the concentration of the reactant and the operating temperature. An experiment has been conducted with the following results.

Yield	Concentration	Temperature
81	1.00	150
89	1.00	180
83	2.00	150
91	2.00	180
79	1.00	150
87	1.00	180
84	2.00	150
90	2.00	180

(a) Suppose we wish to fit a main effects model to this data. Set up the $\mathbf{X}'\mathbf{X}$ matrix using the data exactly as it appears in the table.

(b) Is the matrix you obtained in part (a) diagonal? Discuss your response.

(c) Suppose we write our model in terms of the "usual" coded variables

$$x_1 = \frac{\text{Conc} - 1.5}{0.5} \qquad x_2 = \frac{\text{Temp} - 165}{15}$$

Set up the $\mathbf{X}'\mathbf{X}$ matrix for the model in terms of these coded variables. Is this matrix diagonal? Discuss your response.

(d) Define a new set of coded variables

$$x_1 = \frac{\text{Conc} - 1.0}{1.0} \qquad x_2 = \frac{\text{Temp} - 150}{30}$$

Set up the $\mathbf{X}'\mathbf{X}$ matrix for the model in terms of this set of coded variables. Is this matrix diagonal? Discuss your response.

(e) Summarize what you have learned from this problem about coding the variables.

10.8 Consider the 2^4 factorial experiment in Example 6.2. Suppose that the last two observations are missing. Reanalyze the data and draw conclusions. How do these conclusions compare with those from the original example?

10.9 Given the following data, fit the second-order polynomial regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \varepsilon$$

y	x_1	x_2
26	1.0	1.0
24	1.0	1.0
175	1.5	4.0
160	1.5	4.0
163	1.5	4.0
55	0.5	2.0
62	1.5	2.0
100	0.5	3.0
26	1.0	1.5
30	0.5	1.5
70	1.0	2.5
71	0.5	2.5

After you have fit the model, test for significance of regression.

10.10 **(a)** Consider the quadratic regression model from Problem 10.9. Compute t statistics for each model parameter and comment on the conclusions that follow from these quantities.

(b) Use the extra sum of squares method to evaluate the value of the quadratic terms x_1^2, x_2^2, and $x_1 x_2$ to the model.

10.11 ***Relationship between analysis of variance and regression.*** Any analysis of variance model can be expressed in terms of the general linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where the $\mathbf{X}$ matrix consists of 0s and 1s. Show that the single-factor model $y_{ij} = \mu + \tau_i + \varepsilon_{ij}, i = 1, 2, 3, j = 1, 2, 3, 4$ can be written in general linear model form. Then,

(a) Write the normal equations $(\mathbf{X}'\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$ and compare them with the normal equations found for this model in Chapter 3.

(b) Find the rank of $\mathbf{X}'\mathbf{X}$. Can $(\mathbf{X}'\mathbf{X})^{-1}$ be obtained?

(c) Suppose the first normal equation is deleted and the restriction $\sum_{i=1}^{3} n\hat{\tau}_i = 0$ is added. Can the resulting system of equations be solved? If so, find the solution. Find the regression sum of squares $\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$, and compare it to the treatment sum of squares in the single-factor model.

10.12 Suppose that we are fitting a straight line and we desire to make the variance of $\hat{\beta}_1$ as small as possible. Restricting ourselves to an even number of experimental points, where should we place these points so as to minimize $V(\hat{\beta}_1)$? [*Note:* Use the design called for in this exercise with *great* caution because, even though it minimizes $V(\hat{\beta}_1)$, it has some undesirable properties; for example, see Myers, Montgomery, and Anderson-Cook (2009). Only if you are *very sure* the true

functional relationship is linear, should you consider using this design.]

10.13 ***Weighted least squares.*** Suppose that we are fitting the straight line $y = \beta_0 + \beta_1 x + \varepsilon$, but the variance of the y's now depends on the level of x; that is,

$$V(y|x_i) = \sigma_i^2 = \frac{\sigma^2}{w_i} i = 1, 2, \ldots, n$$

where the w_i are known constants, often called weights. Show that if we choose estimates of the regression coefficients to minimize the weighted sum of squared errors given by $\sum_{i=1}^{n} w_i(y_i - \beta_0 - \beta_1 x_i)^2$, the resulting least squares normal equations are

$$\hat{\beta}_0 \sum_{i=1}^{n} w_i + \hat{\beta}_1 \sum_{i=1}^{n} w_i x_i = \sum_{i=1}^{n} w_i y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} w_i x_i + \hat{\beta}_1 \sum_{i=1}^{n} w_i x_i^2 = \sum_{i=1}^{n} w_i x_i y_i$$

10.14 Consider a 2_{III}^{7-4} design. Suppose after running the experiment, the largest observed effects are $A + BD$, $B + AD$, and $D + AB$. You wish to augment the original design with a group of four runs to de-alias these effects.

(a) Which four runs would you make?

(b) Find the variances and covariances of the regression coefficients in the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_4 x_4 + \beta_{12} x_1 x_2 + \beta_{14} x_1 x_4 + \beta_{24} x_2 x_4 + \varepsilon.$$

(c) Is it possible to de-alias these effects with fewer than four additional runs?

10.15 Consider the computer output from a regression model shown below.

Source	Sum of Squares	Degrees of Freedom	Mean Square	F
Model	532, 567.14	2		
Error				
Total	539, 534.33	14		

(a) Fill in the blanks in the table.

(b) How many regressor or predictor variables are in this model?

(c) Find bounds on the P-value for the test on significance of regression.

(d) Suppose that the t-statistic for one of the predictor variables in this model is 4.22. Find bounds on the P-value.

10.16 Consider the computer output from a regression model shown below.

Source	Sum of Squares	Degrees of Freedom	Mean Square	F
Model	534.66		133.67	
Error				
Total	590.25	19		

(a) Fill in the blanks in the table.

(b) How many regressor or predictor variables are in this model?

(c) Compute the F-statistic for significance of regression and find bounds on the P-value.

(d) Suppose that the t-statistic for one of the predictor variables in this model is 6.11. Find bounds on the P-value.

(e) What is the value of the R^2 statistic for this model?

10.17 A regression model with 5 regressors has been built to 75 observations. The individual t-statistics for each predictor are as follows: 4.21, 2.12, 6.98, 3.55, and 2.45.

(a) Find bounds on the P-values for each predictor.

(b) Based on your answers to part (a), are all of the regressor variables currently in the model necessary?

10.18 A regression model with 3 regressors has been fit to a sample of 45 observations. The total sum of squares for the model is 275.60 and the model sum of squares is 245.86.

(a) Find the value of the mean square error.

(b) Find the model mean square.

(c) Compute the value of the F-statistic for significance of regression and find bounds on the P-value. What are your conclusions?

(d) What is the value of R^2 for this model?

(e) What is the value of the adjusted R^2 statistic for this model?

10.19 A regression model with 4 regressors has been fit to a sample of 65 observations. The total sum of squares for the model is 300 and the model sum of squares is 280.

(a) What is the value of R^2 for this model?

(b) What is the value of the adjusted R^2 statistic for this model?

(c) Suppose that one of the regressors is removed from the model and the new model sum of squares is 250. What is the value of the adjusted R^2 for this reduced model? What is the impact on model fit of removing this regressor?

10.20 A regression model with 3 regressors has been built to a sample of 30 observations. The individual t-statistics for each predictor are as follows: 2.12, 5.45, and 17.50.

(a) Find bounds on the P-values for each predictor.

(b) Based on your answers to part (a), are all of the regressor variables currently in the model necessary?

10.21 The value of the adjusted R^2 statistic always increases when a new regressor variable is added to the model.

(a) True

(b) False

10.22 The value of the ordinary R^2 statistic can decrease when a new regressor variable is added to the model.

(a) True

(b) False

10.23 The ordinary R^2 statistic is a good indicator of the prediction capability of a regression model.

(a) True

(b) False

10.24 If the P-value for the test for significance of regression is <0.01, this is an indication that all of the regressor variables in the model are necessary.

(a) True

(b) False

10.25 If the P-value for the test for significance of regression is <0.01, this is an indication that the regression model will be a good predictor of new observations.

(a) True

(b) False

10.26 A small value of the PRESS statistic is a good indication that the regression model will be a good predictor of new observations.

(a) True

(b) False

10.27 The t-test on individual regression coefficients is a test on the utility of the regressor as if it were the last regressor added to a model that already contains the other regressors.

Chapter 11 Problems

SS **11.1** A chemical plant produces oxygen by liquefying air and separating it into its component gases by fractional distillation. The purity of the oxygen is a function of the main condenser temperature and the pressure ratio between the upper and lower columns. Current operating conditions are temperature $(\xi_1) = -200°C$ and pressure ratio $(\xi_2) = 1.2$. Using the following data, find the path of steepest ascent:

Temperature (ξ_1)	Pressure Ratio (ξ_2)	Purity
−225	1.1	82.8
−225	1.3	83.5
−215	1.1	84.7
−215	1.3	85.0
−220	1.2	84.1
−220	1.2	84.5
−220	1.2	83.9
−220	1.2	84.3

11.2 Verify that the following design is a simplex. Fit the first-order model and find the path of steepest ascent.

x_1	x_2	x_3	y
0	$\sqrt{2}$	−1	18.5
$-\sqrt{2}$	0	1	19.8
0	$-\sqrt{2}$	−1	17.4
$\sqrt{2}$	0	1	22.5

11.3 For the first-order model

$$\hat{y} = 60 + 1.5x_1 - 0.8x_2 + 2.0x_3$$

find the path of steepest ascent. The variables are coded as $-1 \le x_i \le 1$.

11.4 The region of experimentation for two factors are temperature ($100 \le T \le 300°F$) and catalyst feed rate ($10 \le C \le 30$ lb/in). A first-order model in the usual ± 1 coded variables has been fit to a molecular weight response, yielding the following model:

$$\hat{y} = 2000 + 125x_1 + 40x_2$$

(a) Find the path of steepest ascent.

(b) It is desired to move to a region where molecular weights are above 2500. On the basis of the information you have from experimentation in this region, about how many steps along the path of steepest ascent might be required to move to the region of interest?

11.5 The path of steepest ascent is usually computed assuming that the model is truly first order; that is, there is no interaction. However, even if there is interaction, steepest ascent ignoring the interaction still usually produces good results. To illustrate, suppose that we have fit the model

$$\hat{y} = 20 + 5x_1 - 8x_2 + 3x_1x_2$$

using coded variables $(-1 \le x_i \le +1)$.

(a) Draw the path of steepest ascent that you would obtain if the interaction were ignored.

(b) Draw the path of steepest ascent that you would obtain with the interaction included in the model. Compare this with the path found in part (a).

11.6 The data shown in the Table P11.1 were collected in an experiment to optimize crystal growth as a function of three variables x_1, x_2, and x_3. Large values of y (yield in grams) are desirable. Fit a second-order model and analyze the fitted surface. Under what set of conditions is maximum growth achieved?

■ **TABLE P11.1**
The Crystal Growth Experiment, Problem 11.6

x_1	x_2	x_3	y
−1	−1	−1	66
−1	−1	1	70
−1	1	−1	78
−1	1	1	60
1	−1	−1	80
1	−1	1	70
1	1	−1	100
1	1	1	75
−1.682	0	0	100
1.682	0	0	80
0	−1.682	0	68
0	1.682	0	63
0	0	−1.682	65
0	0	1.682	82
0	0	0	113
0	0	0	100
0	0	0	118
0	0	0	88
0	0	0	100
0	0	0	85

11.7 The data in Table P11.2 were collected by a chemical engineer. The response y is filtration time, x_1 is temperature, and x_2 is pressure. Fit a second-order model.

■ **TABLE P11.2**
The Experiment for Problem 11.7

x_1	x_2	y
−1	−1	54
−1	1	45
1	−1	32
1	1	47
−1.414	0	50
1.414	0	53
0	−1.414	47
0	1.414	51
0	0	41
0	0	39
0	0	44
0	0	42
0	0	40

(a) What operating conditions would you recommend if the objective is to minimize the filtration time?

(b) What operating conditions would you recommend if the objective is to operate the process at a mean filtration rate very close to 46?

11.8 An experimenter has run a Box–Behnken design and obtained the results as shown in Table P11.3, where the response variable is the viscosity of a polymer:

■ **TABLE P11.3**
The Box–Behnken Design for Problem 11.8

Level	Temp.	Agitation Rate	Pressure	x_1	x_2	x_3
High	200	10.0	25	+1	+1	+1
Middle	175	7.5	20	0	0	0
Low	150	5.0	15	−1	−1	−1

Run	x_1	x_2	x_3	y_1
1	−1	−1	0	535
2	+1	−1	0	580
3	−1	+1	0	596
4	+1	+1	0	563
5	−1	0	−1	645
6	+1	0	−1	458
7	−1	0	+1	350
8	+1	0	+1	600
9	0	−1	−1	595
10	0	+1	−1	648
11	0	−1	+1	532
12	0	+1	+1	656
13	0	0	0	653
14	0	0	0	599
15	0	0	0	620

(a) Fit the second-order model.

(b) Perform the canonical analysis. What type of surface has been found?

(c) What operating conditions on x_1, x_2, and x_3 lead to the stationary point?

(d) What operating conditions would you recommend if it is important to obtain a viscosity that is as close to 600 as possible?

11.9 Consider the three-variable central composite design shown in Table P11.4. Analyze the data and draw conclusions, assuming that we wish to maximize conversion (y_1) with activity (y_2) between 55 and 60.

■ **TABLE P11.4**
A Three-Variable CCD

Run	Time (min)	Temperature (°C)	Catalyst (%)	Conversion (%) y_1	Activity y_2
1	−1.000	−1.000	−1.000	74.00	53.20
2	1.000	−1.000	−1.000	51.00	62.90
3	−1.000	1.000	−1.000	88.00	53.40
4	1.000	1.000	−1.000	70.00	62.60
5	−1.000	−1.000	1.000	71.00	57.30
6	1.000	−1.000	1.000	90.00	67.90
7	−1.000	1.000	1.000	66.00	59.80
8	1.000	1.000	1.000	97.00	67.80
9	0.000	0.000	0.000	81.00	59.20
10	0.000	0.000	0.000	75.00	60.40
11	0.000	0.000	0.000	76.00	59.10
12	0.000	0.000	0.000	83.00	60.60
13	−1.682	0.000	0.000	76.00	59.10
14	1.682	0.000	0.000	79.00	65.90
15	0.000	−1.682	0.000	85.00	60.00
16	0.000	1.682	0.000	97.00	60.70
17	0.000	0.000	−1.682	55.00	57.40
18	0.000	0.000	1.682	81.00	63.20
19	0.000	0.000	0.000	80.00	60.80
20	0.000	0.000	0.000	91.00	58.90

11.10 A manufacturer of cutting tools has developed two empirical equations for tool life in hours (y_1) and for tool cost in dollars (y_2). Both models are linear functions of steel hardness (x_1) and manufacturing time (x_2). The two equations are

$$\hat{y} = 10 + 5x_1 + 2x_2$$
$$\hat{y}_2 = 23 + 3x_1 + 4x_2$$

and both equations are valid over the range $-1.5 \le x_i \le 1.5$. Unit tool cost must be below \$27.50 and life must exceed 12 hours for the product to be competitive. Is there a feasible set of operating conditions for this process? Where would you recommend that the process be run?

11.11 A central composite design is run in a chemical vapor deposition process, resulting in the experimental data shown in Table P11.5. Four experimental units were processed simultaneously on each run of the design, and the responses are the mean and the variance of thickness, computed across the four units.

■ **TABLE P11.5**
The CCD for Problem 11.11

x_1	x_2	$\bar{y}$	s^2
−1	−1	360.6	6.689
1	−1	445.2	14.230
−1	1	412.1	7.088
1	1	601.7	8.586
1.414	0	518.0	13.130
−1.414	0	411.4	6.644
0	1.414	497.6	7.649
0	−1.414	397.6	11.740
0	0	530.6	7.836
0	0	495.4	9.306
0	0	510.2	7.956
0	0	487.3	9.127

(a) Fit a model to the mean response. Analyze the residuals.

(b) Fit a model to the variance response. Analyze the residuals.

(c) Fit a model to $\ln(s^2)$. Is this model superior to the one you found in part (b)?

(d) Suppose you want the mean thickness to be in the interval 450 ± 25. Find a set of operating conditions that achieves this objective and simultaneously minimizes the variance.

(e) Discuss the variance minimization aspects of part (d). Have you minimized the *total* process variance?

11.12 Verify that an orthogonal first-order design is also first-order rotatable.

11.13 Show that augmenting a 2^k design with n_C center points does not affect the estimates of the $\beta_i (i = 1, 2, \ldots, k)$ but that the estimate of the intercept β_0 is the average of all $2^k + n_C$ observations.

11.14 Verify that the central composite design shown in Table P11.6 blocks orthogonally:

■ **TABLE P11.6**
A CCD in Three Blocks

Block 1			Block 2		
x_1	x_2	x_3	x_1	x_2	x_3
0	0	0	0	0	0
0	0	0	0	0	0
1	1	1	1	1	−1
1	−1	−1	1	−1	1
−1	−1	1	−1	1	1
−1	1	−1	−1	−1	−1

Block 3		
x_1	x_2	x_3
−1.633	0	0
1.633	0	0
0	−1.633	0
0	1.633	0
0	0	−1.633
0	0	1.633
0	0	0
0	0	0

11.15 ***Blocking in the central composite design.*** Consider a central composite design for $k = 4$ variables in two blocks. Can a rotatable design always be found that blocks orthogonally?

11.16 How could a hexagon design be run in two orthogonal blocks?

11.17 Yield during the first four cycles of a chemical process is shown in the following table. The variables are percentage of concentration (x_1) at levels 30, 31, and 32 and temperature (x_2) at 140, 142, and 144°F. Analyze by EVOP methods.

	Conditions				
Cycle	(1)	(2)	(3)	(4)	(5)
1	60.7	59.8	60.2	64.2	57.5
2	59.1	62.8	62.5	64.6	58.3
3	56.6	59.1	59.0	62.3	61.1
4	60.5	59.8	64.5	61.0	60.1

11.18 Suppose that we approximate a response surface with a model of order d_1, such as $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$, when the true surface is described by a model of order $d_2 > d_1$; that is, $E(y) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_1\boldsymbol{\beta}_2$.

(a) Show that the regression coefficients are biased, that is, $E(\hat{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 + \mathbf{A}\boldsymbol{\beta}_2$, where $\mathbf{A} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2$. $\mathbf{A}$ is usually called the alias matrix.

(b) If $d_1 = 1$ and $d_2 = 2$, and a full 2^k is used to fit the model, use the result in part (a) to determine the alias structure.

(c) If $d_1 = 1$, $d_2 = 2$, and $k = 3$, find the alias structure assuming that a 2^{3-1} design is used to fit the model.

(d) If $d_1 = 1$, $d_2 = 2$, and $k = 3$, and the simplex design in Problem 11.2 is used to fit the model, determine the alias structure and compare the results with part (c).

11.19 Suppose that you need to design an experiment to fit a quadratic model over the region $-1 \le x_i \le +1, i = 1,2$ subject to the constraint $x_1 + x_2 \le 1$. If the constraint is violated, the process will not work properly. You can afford to make no more than $n = 12$ runs. Set up the following designs:

(a) An "inscribed" CCD with center point at $x_1 = x_2 = 0$.

(b) An "inscribed" 3^2 factorial with center point at $x_1 = x_2 = -0.25$.

(c) A D-optimal design.

(d) A modified D-optimal design that is identical to the one in part (c), but with all replicate runs at the design center.

(e) Evaluate the $|(\mathbf{X}'\mathbf{X})^{-1}|$ criterion for each design.

(f) Evaluate the D-efficiency for each design relative to the D-optimal design in part (c).

(g) Which design would you prefer? Why?

11.20 Consider a 2^3 design for fitting a first-order model.

(a) Evaluate the D-criterion $|(\mathbf{X}'\mathbf{X})^{-1}|$ for this design.

(b) Evaluate the A-criterion $\text{tr}(\mathbf{X}'\mathbf{X})^{-1}$ for this design.

(c) Find the maximum scaled prediction variance for this design. Is this design G-optimal?

11.21 Repeat problem 11.20 using a first-order model with the two-factor interactions.

11.22 Suppose that you want to fit a second-order model in $k = 5$ factors. You cannot afford more than 25 runs. Construct both a D-optimal and on I-optimal design for this situation. Compare the prediction variance properties of the designs. Which design would you prefer?

11.23 Suppose that you want to fit a second-order response surface model in a situation where there are $k = 4$ factors; however, one of the factors is categorical with two levels. What model should you consider for this experiment? Suggest an appropriate design for this situation.

SS **11.24** An experimenter wishes to run a three-component mixture experiment. The constraints in the component proportions are as follows:

$$0.2 \le x_1 \le 0.4$$
$$0.1 \le x_2 \le 0.3$$
$$0.4 \le x_2 \le 0.7$$

(a) Set up an experiment to fit a quadratic mixture model. Use $n = 14$ runs, with four replicates. Use the D-criterion.

(b) Draw the experimental region.

(c) Set up an experiment to fit a quadratic mixture model with $n = 12$ runs, assuming that three of these runs are replicates. Use the D-criterion.

(d) Comment on the two designs you have found.

11.25 Myers, Montgomery, and Anderson-Cook (2009) describe a gasoline blending experiment involving three mixture components. There are no constraints on the mixture proportions, and the following 10-run design is used:

Design Point	x_1	x_2	x_3	y (mi/gal)
1	1	0	0	24.5, 25.1
2	0	1	0	24.8, 23.9
3	0	0	1	22.7, 23.6
4	$\frac{1}{2}$	$\frac{1}{2}$	0	25.1
5	$\frac{1}{2}$	0	$\frac{1}{2}$	24.3
6	0	$\frac{1}{2}$	$\frac{1}{2}$	23.5
7	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	24.8, 24.1
8	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$	24.2
9	$\frac{1}{6}$	$\frac{2}{3}$	$\frac{1}{6}$	23.9
10	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{3}$	23.7

(a) What type of design did the experimenters use?

(b) Fit a quadratic mixture model to the data. Is this model adequate?

(c) Plot the response surface contours. What blend would you recommend to maximize the miles per gallon?

11.26 Table P11.7 shows a six-variable RSM design from Jones and Nachtsheim (2011). Analyze the response data from this experiment.

■ **TABLE P11.7**
The Design for Problem 11.25

Run (i)	$x_{i,1}$	$x_{i,2}$	$x_{i,3}$	$x_{i,4}$	$x_{i,5}$	$x_{i,6}$	y_i
1	0	1	−1	−1	−1	−1	21.04
2	0	−1	1	1	1	1	10.48
3	1	0	−1	1	1	−1	17.89
4	−1	0	1	−1	−1	1	10.07
5	−1	−1	0	1	−1	−1	7.74
6	1	1	0	−1	1	1	21.01
7	−1	1	1	0	1	−1	16.53
8	1	−1	−1	0	−1	1	20.38
9	1	−1	1	−1	0	−1	8.62
10	−1	1	−1	1	0	1	7.80
11	1	1	1	1	−1	0	23.56
12	−1	−1	−1	−1	1	0	15.24
13	0	0	0	0	0	0	19.91

11.27 An article in *Quality Progress* ("For Starbucks, It's in the Bag," March 2011, pp. 18–23) describes using a central composite design to improve the packaging of one-pound coffee. The objective is to produce an airtight seal that is easy to open without damaging the top of the coffee bag. The experimenters studied three factors—x_1 = plastic viscosity (300–400 centipoise), x_2 = clamp pressure (170–190 psi), and x_3 = plate gap (−3, +3 mm) and two responses—y_1 = tear and y_2 = leakage. The design is shown in Table P11.8. The tear response was measure on a scale from 0 to 9 (good to bad) and leakage was proportion failing. Each run used a sample of 20 bags for response measurement.

(a) Build a second-order model for the tear response.

(b) Build a second-order model for the leakage response.

(c) Analyze the residuals for both models. Do transformations seem necessary for either response? If so, refit the models in the transformed metric.

(d) Construct response surface plots and contour plots for both responses. Provide interpretations for the fitted surfaces.

(e) What conditions would you recommend for process operation to minimize leakage and keep tear below 0.75?

■ **TABLE P11.8**
The Coffee Bag Experiment in Problem 11.27

Run	Viscosity	Pressure	Plate Gap	Tear	Leakage
Center	350	180	0	0	0.15
Axial	350	170	0	0	0.5
Factorial	319	186	1.8	0.45	0.15
Factorial	380	174	1.8	0.85	0.05
Center	350	180	0	0.35	0.15
Axial	300	180	0	0.3	0.45
Axial	400	180	0	0.7	0.25
Axial	350	190	0	1.9	0
Center	350	180	0	0.25	0.05
Factorial	319	186	−1.8	0.1	0.35
Factorial	380	186	−1.8	0.15	0.4
Axial	350	180	3	3.9	0
Factorial	380	174	−1.8	0	0.45
Center	350	180	0	0.55	0.2
Axial	350	180	−3	0	1
Factorial	319	174	−1.8	0.05	0.2
Factorial	319	174	1.8	0.4	0.25
Factorial	380	186	1.8	4.3	0.05
Center	350	180	0	0	0

11.28 Box and Liu (1999) describe an experiment flying paper helicopters where the objective is to maximize flight time. They used the central composite design shown in Table P11.9. Each run involved a single helicopter made to the following specifications: x_1 = wing area (in^2), $-1 = 11.80$ and $+1 = 13.00$; x_2 = wing-length to width ratio, $-1 = 2.25$ and $+1 = 2.78$; x_3 = base width (in), $-1 = 1.00$ and $+1 = 1.50$; and x_4 = base length (in), $-1 = 1.50$ and $+1 = 2.50$. Each helicopter was flown four times and the average flight time and the standard deviation of flight time were recorded.

(a) Fit a second-order model to the average flight time response.

(b) Fit a second-order model to the standard deviation of flight time response.

(c) Analyze the residuals for both models from parts (a) and (b). Are transformations on the response(s) necessary? If so, fit the appropriate models.

(d) What design would you recommend to maximize the flight time?

(e) What design would you recommend to maximize the flight time while simultaneously minimizing the standard deviation of flight time?

11.29 The Paper Helicopter Experiment Revisited. Reconsider the paper helicopter experiment in Problem 11.28. This experiment was actually run in two blocks. Block 1 consisted of the first 16 runs in Table P11.9 (standard order runs 1–16) and two center points (standard order runs 25 and 26).

(a) Fit main effects plus two-factor interaction models to the block 1 data, using both responses.

(b) For the models in part (a) use the two center points to test for lack of fit. Is there an indication that second-order terms are needed?

(c) Now use the data from block 2 (standard order runs 17–24 and the remaining center points, standard order runs 27–30) to augment block 1 and fit second-order models to both responses. Check the adequacy of the fit for both models. Does blocking seem to have been important in this experiment?

(d) What design would you recommend to maximize the flight time while simultaneously minimizing the standard deviation of flight time?

11.30 An article in the *Journal of Chromatography A* ("Optimization of the Capillary Electrophoresis Separation of Ranitidine and Related Compounds," Vol. 766, pp. 245–254) describes an experiment to optimize the production of ranitidine, a compound that is the primary active ingredient of Zantac, a pharmaceutical product used to treat ulcers, gastroesophageal reflux disease (a condition in which backward flow of acid from the stomach causes heartburn and injury of the esophagus), and other conditions where the stomach produces too much acid, such as Zollinger–Ellison syndrome. The authors used three factors (x_1 = pH of the buffer solution, x_2 = the electrophoresis voltage, and the concentration of one component of the buffer solution) in a central composite design. The response is chromatographic exponential function (CEF), which should be minimized. Table P11.10 shows the design.

(a) Fit a second-order model to the CEF response. Analyze the residuals from this model. Does it seem that all model terms are necessary?

(b) Reduce the model from part (a) as necessary. Did model reduction improve the fit?

(c) Does transformation of the CEF response seem like a useful idea? What aspect of either the data or the residual analysis suggests that transformation would be helpful?

(d) Fit a second-order model to the transformed CEF response. Analyze the residuals from this model. Does it seem that all model terms are necessary? What would you choose as the final model?

(e) Suppose that you had some information that suggests that the separation process malfunctioned during run 7. Delete this run and analyze the data from this experiment again.

(f) What conditions would you recommend to minimize CEF?

■ **TABLE P11.9**
The Paper Helicopter Experiment

Std. Order	Run Order	Wing Area	Wing Ratio	Base Width	Base Length	Avg. Flight Time	Std. Dev Flight Time
1	9	−1	−1	−1	−1	3.67	0.052
2	21	1	−1	−1	−1	3.69	0.052
3	14	−1	1	−1	−1	3.74	0.055
4	4	1	1	−1	−1	3.7	0.062
5	2	−1	−1	1	−1	3.72	0.052
6	19	1	−1	1	−1	3.55	0.065
7	22	−1	1	1	−1	3.97	0.052
8	25	1	1	1	−1	3.77	0.098
9	27	−1	−1	−1	−1	3.5	0.079
10	13	1	−1	−1	1	3.73	0.072
11	20	−1	1	−1	1	3.58	0.083
12	6	1	1	−1	1	3.63	0.132
13	12	−1	−1	1	1	3.44	0.058
14	17	1	−1	1	1	3.55	0.049
15	26	−1	1	1	1	3.7	0.081
16	1	1	1	1	1	3.62	0.051
17	8	−2	0	0	0	3.61	0.129
18	15	2	0	0	0	3.64	0.085
19	7	0	−2	0	0	3.55	0.1
20	5	0	2	0	0	3.73	0.063
21	29	0	0	−2	0	3.61	0.051
22	28	0	0	2	0	3.6	0.095
23	16	0	0	0	−2	3.8	0.049
24	18	0	0	0	2	3.6	0.055
25	24	0	0	0	0	3.77	0.032
26	10	0	0	0	0	3.75	0.055
27	23	0	0	0	0	3.7	0.072
28	11	0	0	0	0	3.68	0.055
29	3	0	0	0	0	3.69	0.078
30	30	0	0	0	0	3.66	0.058

11.31 An article in the *Electronic Journal of Biotechnology* ("Optimization of Medium Composition for Transglutaminase Production by a Brazilian Soil *Streptomyces* sp," available at The correct URL is http://www.ejbiotechnology.info/index.php/ejbiotechnology/article/viewFile/v10n4-10/110) describes the use of designed experiments to improve the medium for cells used in a new microbial source of transglutaminase (MTGase), an enzyme that catalyzes an acyl transfer reaction using peptide-bond glutamine residues as acyl donors and some primary amines as acceptors. Reactions catalyzed by MTGase can be used in food processing. The article describes two phases of experimentation—screening with a fractional factorial and optimization. We will use only the optimization experiment. The design was a central composite design in four factors—$x_1 = KH_2PO_4$, $x_2 = MgSO4{\cdot}7H_2O$, x_3 = soybean flower, and x_4 = peptone. MTGase activity is the response, which should be maximized. Table P11.11 contains the design and the response data.

(a) Fit a second-order model to the MTGase activity response.

(b) Analyze the residuals from this model.

(c) Recommend operating conditions that maximize MTGase activity.

11.32 Given the fitted response function

$$\hat{y} = 72.0 + 3.6x_1 - 2.5x_2$$

which is found to fit well in a localized region of (x_1, x_2).

(a) Plot contours of constant response, y, in the (x_1, x_2) plane.

(b) Calculate and plot the path of steepest ascent generated from this response function. Use x_1 to determine the path, with $\Delta x_1 = 1$.

11.33 It is stated in the text that the development of the path of steepest ascent makes use of the assumption that the model is truly first-order in nature. However, even if there is a modest amount of curvature or interaction in the system, the use of steepest ascent can be extremely useful in determining a future experimental region. Suppose that in a system involving x_1 and x_2 the actual model is given by

$$E(y) = 14 + 5x_1 - 10x_2 + 3x_1x_2$$

Assume that x_1 and x_2 are in coded form.

(a) Show a plot of the path of steepest ascent (based on actual parameters) if the interaction is ignored.

(b) Show a plot of the path of steepest ascent for the model with interaction. Note that this path is not linear.

(c) Comment on the difference in the two paths.

11.34 Suppose that you have fit the following model SS

$$\hat{y} = 20 + 10x_1 - 4x_2 + 12x_3$$

(a) The experimenter decides to choose x_1 as the variable to define the step size for steepest ascent. Do you think that this is a good choice for defining step size? Explain why or why not.

(b) The experimenter decides to make the step size exactly one coded unit in the x_1 direction. Find the coordinates of this point from the design center on the path of steepest ascent.

■ **TABLE P11.10**
The Ranitidine Separation Experiment

Standard Order	X1	X2	X3	CEF
1	−1	−1	−1	17.3
2	1	−1	−1	45.5
3	−1	1	−1	10.3
4	1	1	−1	11757.1
5	−1	−1	1	16.942
6	1	−1	1	25.4
7	−1	1	1	31697.2
8	1	1	1	12039.2
9	−1.68	0	0	7.5
10	1.68	0	0	6.3
11	0	−1.68	0	11.1
12	0	1.68	0	6.664
13	0	0	−1.68	16548.7
14	0	0	1.68	26351.8
15	0	0	0	9.9
16	0	0	0	9.6
17	0	0	0	8.9
18	0	0	0	8.8
19	0	0	0	8.013
20	0	0	0	8.059

■ **TABLE P11.11**
The MTGase Optimization Experiment

Standard Order	X1	X2	X3	X4	MTGase Activity
1	−1	−1	−1	−1	0.87
2	1	−1	−1	−1	0.74
3	−1	1	−1	−1	0.51
4	1	1	−1	−1	0.99
5	−1	−1	1	−1	0.67
6	1	−1	1	−1	0.72
7	−1	1	1	−1	0.81
8	1	1	1	−1	1.01
9	−1	−1	−1	1	1.33
10	1	−1	−1	1	0.7
11	−1	1	−1	1	0.82
12	1	1	−1	1	0.78
13	−1	−1	1	1	0.36
14	1	−1	1	1	0.23
15	−1	1	1	1	0.21
16	1	1	1	1	0.44
17	−2	0	0	0	0.56
18	2	0	0	0	0.49
19	0	−2	0	0	0.57
20	0	2	0	0	0.81
21	0	0	−2	0	0.9
22	0	0	2	0	0.65
23	0	0	0	−2	0.91
24	0	0	0	2	0.49
25	0	0	0	0	1.43
26	0	0	0	0	1.17
27	0	0	0	0	1.5

(c) Suppose that you had used x_3 as the variable to define the direction of steepest ascent. Assuming a step size of one coded unit, find the coordinates of the first step in this direction away from the design center.

(d) Find the distance between the two points found in parts (b) and (c), respectively.

11.35 Any point on the path of steepest ascent is proportional to the unit vector of the first-order regression coefficients.

(a) True

(b) False

11.36 Consider the first-order model $\hat{y} = 150 + 20x_1 + 12x_2$. If the experimenter used one coded unit in the x_1 direction as the step size, the point $x_1 = 1.5, x_2 = 1.0$ is a point on the path of steepest ascent.

(a) True

(b) False

11.37 A 2^2 factorial design with four center points has been used to fit the first-order model $\hat{y} = 175 + 25x_1 + 10x_2 - 8x_3$. The variance of the predicted response at any design point other than the center point is _____

11.38 When conducting steepest ascent experiments, it may be a good idea to replicate runs at some or all of the points along the path.

(a) True

(b) False

11.39 If a first-order model with interaction is used, the path of steepest ascent is a straight line.

(a) True

(b) False

11.40 The coordinates of the points on the path of steepest ascent are proportional to the effect estimates from the first-order model.

(a) True

(b) False

11.41 For a first-order model with three variables, the variance of the predicted response at the design center (in coded units) is σ^2, if an unreplicated 2^3 design is used to collect the experimental data.

(a) True

(b) False

11.42 The method of steepest ascent is a gradient procedure.

(a) True

(b) False

11.43 Using a second-order model to estimate the underlying relationship is a common approach to approximating the surface.

(a) True

(b) False

11.44 The location of the stationary point corresponds to a global maximum or minimum of the surface.

(a) True

(b) False

11.45 The existence of a ridge system in the canonical variables can restrict the choice of optimal location.

(a) True

(b) False

11.46 If the sign of all of the eigenvalues is the same, the response surface will have a global maximum.

(a) True

(b) False

11.47 In designing mixture experiments, a good rule of thumb is to assume that at least a second-order mixture model is likely to be adequate.

(a) True

(b) False

11.48 The simplex-lattice and simplex-centroid designs have most of the design points on the boundary of the experimental region.

(a) True

(b) False

11.49 When axial check blends are added to a simplex-lattice, these blends are as far as possible (Euclidean distance) from the other design points.

(a) True

(b) False

11.50 The (3,2) simplex-lattice design is a rotatable design for fitting a second-order mixture model.

(a) True

(b) False

Chapter 12 Problems

12.1 Reconsider the leaf spring experiment in Table 12.1. Suppose that the objective is to find a set of conditions where the mean free height is as close as possible to 7.6 inches, with the variance of free height as small as possible. What conditions would you recommend to achieve these objectives?

SS **12.2** Consider the bottle-filling experiment in Problem 6.21. Suppose that the percentage of carbonation (A) is a noise variable ($\sigma_z^2 = 1$ in coded units).

(a) Fit the response model to these data. Is there a robust design problem?

(b) Find the mean model and either the variance model or the POE.

(c) Find a set of conditions that result in mean fill deviation as close to zero as possible with minimum transmitted variance.

12.3 Reconsider the leaf spring experiment from Table 12.1. Suppose that factors A, B, and C are controllable variables and that factors D and E are noise factors. Set up a crossed array design to investigate this problem, assuming that all of the two-factor interactions involving the controllable variables are thought to be important. What type of design have you obtained?

12.4 **Continuation of Problem 12.3.** Reconsider the leaf spring experiment from Table 12.1. Suppose that factors A, B and C are controllable variables and that factors D and E are noise factors. Show how a combined array design can be employed to investigate this problem that allows all two-factor interaction to be estimated and only requires 16 runs. Compare this with the crossed array design from Problem 12.3. Can you see how in general combined array designs have fewer runs than crossed array designs?

12.5 Consider the connector pull-off force experiment shown in Table 12.2. Show how an experiment can be designed for this problem that will allow a full quadratic model to be fit in the controllable variables along all main effects of the noise variables and their interactions with the controllable variables. How many runs will be required in this design? How does this compare with the design in Table 12.2?

12.6 Consider the experiment in Problem 11.8. Suppose that pressure is a noise variable ($\sigma_z^2 = 1$ in coded units). Fit the response model for the viscosity response. Find a set of conditions that result in viscosity as close as possible to 600 and that minimize the variability transmitted from the noise variable pressure.

12.7 ***A variation of Example 12.1.*** In Example 12.1 (which utilized data from Example 6.2), we found that one of the process variables (B = pressure) was not important. Dropping this variable produces two replicates of a 2^3 design. The data are as follows:

C	D	A (+)	A (−)	$\bar{y}$	s^2
−	−	45, 48	71, 65	57.75	121.19
+	−	68, 80	60, 65	68.25	72.25
−	+	43, 45	100, 104	73.00	1124.67
+	+	75, 70	86, 96	81.75	134.92

Assume that C and D are controllable factors and that A is a noise variable.

(a) Fit a model to the mean response.

(b) Fit a model to the $\ln(s^2)$ response.

(c) Find operating conditions that result in the mean filtration rate response exceeding 75 with minimum variance.

(d) Compare your results with those from Example 12.1, which used the transmission of error approach. How similar are the two answers?

12.8 In an article ("Let's All Beware the Latin Square," *Quality Engineering*, Vol. 1, 1989, pp. 453–465), J. S. Hunter illustrates some of the problems associated with 3^{k-p} fractional factorial designs. Factor A is the amount of ethanol added to a standard fuel and factor B represents the air/fuel ratio. The response variable is carbon monoxide (CO) emission in g/m^3. The design is as follows:

SS

Design				Observations	
A	B	x_1	x_2	y	
0	0	−1	−1	66	62
1	0	0	−1	78	81
2	0	+1	−1	90	94
0	1	−1	0	72	67
1	1	0	0	80	81
2	1	+1	0	75	78
0	2	−1	+1	68	66
1	2	0	+1	66	69
2	2	+1	+1	60	58

Notice that we have used the notation system of 0, 1, and 2 to represent the low, medium, and high levels for the factors. We have also used a "geometric notation" of −1, 0, and +1. Each run in the design is replicated twice.

(a) Verify that the second-order model

$$\hat{y} = 78.5 + 4.5x_1 - 7.0x_2 - 4.5x_1^2 - 4.0x_2^2 - 9.0x_1x_2$$

is a reasonable model for this experiment. Sketch the CO concentration contours in the x_1, x_2 space.

(b) Now suppose that instead of only two factors, we had used *four* factors in a 3^{4-2} fractional factorial design and obtained *exactly* the same data as in part (a). The design would be as follows:

Design								Observations y	
A	B	C	D	x_1	x_2	x_3	x_4		
0	0	0	0	−1	−1	−1	−1	66	62
1	0	1	1	0	−1	0	0	78	81
2	0	2	2	+1	−1	+1	+1	90	94
0	1	2	1	−1	0	+1	0	72	67
1	1	0	2	0	0	−1	+1	80	81
2	1	1	0	+1	0	0	−1	75	78
0	2	1	2	−1	+1	0	+1	68	66
1	2	2	0	0	+1	+1	−1	66	69
2	2	0	1	+1	+1	−1	0	60	58

Calculate the marginal averages of the CO response at each level of the four factors A, B, C, and D. Construct plots of these marginal averages and interpret the results. Do factors C and D appear to have strong effects? Do these factors *really* have any effect on CO emission? Why is their apparent effect strong?

(c) The design in part (b) allows the model

$$y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_i x_i^2 + \epsilon$$

to be fitted. Suppose that the *true* model is

$$y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_{ij} x_i^2 + \sum\sum_{i<j} \beta_{ij} x_i x_j + \epsilon$$

Show that if $\hat{\beta}_i$ represents the least squares estimate of the coefficients in the fitted model, then

$$\begin{aligned}
E(\hat{\beta}_0) &= \beta_0 - \beta_{13} - \beta_{14} - \beta_{34} \\
E(\hat{\beta}_1) &= \beta_1 - (\beta_{23} + \beta_{24})/2 \\
E(\hat{\beta}_2) &= \beta_2 - (\beta_{13} + \beta_{14} + \beta_{34})/2 \\
E(\hat{\beta}_3) &= \beta_3 - (\beta_{12} + \beta_{24})/2 \\
E(\hat{\beta}_4) &= \beta_4 - (\beta_{12} + \beta_{23})/2 \\
E(\hat{\beta}_{11}) &= \beta_{11} - (\beta_{23} - \beta_{24})/2 \\
E(\hat{\beta}_{22}) &= \beta_{22} + (\beta_{13} + \beta_{14} + \beta_{34})/2 \\
E(\hat{\beta}_{33}) &= \beta_{33} - (\beta_{24} - \beta_{12})/2 + \beta_{14} \\
E(\hat{\beta}_{44}) &= \beta_{44} - (\beta_{12} - \beta_{23})/2 + \beta_{13}
\end{aligned}$$

Does this help explain the strong effects for factors C and D observed graphically in part (b)?

12.9 An experiment has been run in a process that applies a coating material to a wafer. Each run in the experiment produced a wafer, and the coating thickness was measured several times at different locations on the wafer. Then the mean y_1 and the standard deviation y_2 of the thickness measurement were obtained. The data [adapted from Box and Draper (2007)] are shown in Table P12.1:

■ **TABLE P12.1**
The Coating Experiment in Problem 12.9

Run	Speed	Pressure	Distance	Mean y_1	Std. Dev. y_2
1	−1	−1	−1	24.0	12.5
2	0	−1	−1	120.3	8.4
3	+1	−1	−1	213.7	42.8
4	−1	0	−1	86.0	3.5
5	0	0	−1	136.6	80.4
6	+1	0	−1	340.7	16.2
7	−1	+1	−1	112.3	27.6
8	0	+1	−1	256.3	4.6
9	+1	+1	−1	271.7	23.6
10	−1	−1	0	81.0	0.0
11	0	−1	0	101.7	17.7
12	+1	−1	0	357.0	32.9
13	−1	0	0	171.3	15.0
14	0	0	0	372.0	0.0
15	+1	0	0	501.7	92.5
16	−1	+1	0	264.0	63.5
17	0	+1	0	427.0	88.6
18	+1	+1	0	730.7	21.1
19	−1	−1	+1	220.7	133.8
20	0	−1	+1	239.7	23.5
21	+1	−1	+1	422.0	18.5
22	−1	0	+1	199.0	29.4
23	0	0	+1	485.3	44.7
24	+1	0	+1	673.7	158.2
25	−1	+1	+1	176.7	55.5
26	0	+1	+1	501.0	138.9
27	+1	+1	+1	1010.0	142.4

(a) What type of design did the experimenters use? Is this a good choice of design for fitting a quadratic model?

(b) Build models of both responses.

(c) Find a set of optimum conditions that result in the mean as large as possible with the standard deviation less than 60.

SS **12.10** Suppose that there are four controllable variables and two noise variables. It is necessary to estimate the main effects and two-factor interactions of all of the controllable variables, the main effects of the noise variables, and the two-factor interactions between all controllable and noise factors. If all factors are at two levels, what is the minimum number of runs that can be used to estimate all of the model parameters using a combined array design? Use a *D*-optimal algorithm to find a design.

SS **12.11** Suppose that there are four controllable variables and two noise variables. It is necessary to fit a complete quadratic model in the controllable variables, the main effects of the noise variables, and the two-factor interactions between all controllable and noise factors. Set up a combined array design for this by modifying a central composite design.

12.12 Reconsider the situation in Problem 12.11. What is the minimum number of runs that can be used to estimate all of the model parameters using a combined array design? Use a *D*-optimal algorithm to find a reasonable design for this problem.

12.13 Rework Problem 12.12 using the *I*-criterion to construct the design. Compare this design to the *D*-optimal design in Problem 12.12. Which design would you prefer?

12.14 Rework Problem 12.10 using the *I*-criterion. Compare this design to the *D*-optimal design in Problem 12.10. Which design would you prefer?

12.15 An experiment was run in a wave soldering process. There are five controllable variables and three noise variables. The response variable is the number of solder defects per million opportunities. The experimental design employed was the following crossed array.

						Outer Array			
					F	−1	1	1	−1
Inner Array					*G*	−1	1	−1	1
A	*B*	*C*	*D*	*E*	*H*	−1	−1	1	1
1	1	1	−1	−1		194	197	193	275
1	1	−1	1	1		136	136	132	136
1	−1	1	−1	1		185	261	264	264
1	−1	−1	1	−1		47	125	127	42
−1	1	1	1	−1		295	216	204	293
−1	1	−1	−1	1		234	159	231	157
−1	−1	1	1	1		328	326	247	322
−1	−1	−1	−1	−1		186	187	105	104

(a) What types of designs were used for the inner and outer arrays? What are the alias relationships in these designs?

(b) Develop models for the mean and variance of solder defects. What set of operating conditions would you recommend?

12.16 Reconsider the wave soldering experiment in Problem 12.15. Find a combined array design for this experiment that requires fewer runs.

12.17 Reconsider the wave soldering experiment in Problem 12.15. Suppose that it was necessary to fit a complete quadratic model in the controllable variables, all main effects of the noise variables, and all controllable variable–noise variable interactions. What design would you recommend?

12.18 Consider the alloy cracking experiment in Problem 6.2R. Suppose that temperature (A) is a noise variable. Find the response model, the model for the mean response, and the model for the transmitted variability. Can you find settings for the controllable factors that minimize crack length and make the transmitted variability small?

12.19 Signal-to-noise factors generally confound location and dispersion effects.

(a) True

(b) False

12.20 A crossed array design allows estimation of the control-by-noise factor interactions.

(a) True

(b) False

12.21 If a 3^{3-1} design is used as the inner array and a 2^3 is used as the outer array, all interactions can be estimated.

(a) True

(b) False

12.22 Suppose that the response model is $\hat{y} = 6 + 4x_1 + 3x_2 + 3z + 2x_1z + 2x_2z$. The value of the controllable factor x_1 that minimizes the transmitted variance is

(a) -1

(b) $+1$

(c) -0.5

(d) none of the above

12.23 Consider the response model in Problem 12.22. The value of the controllable factor x_2 that minimizes the transmitted variance is

(a) -1

(b) $+1$

(c) -0.5

(d) none of the above

12.24 A log transformation could be useful in directly modeling the variance as a response.

(a) True

(b) False

12.25 Computer-generated optimal designs can be a useful approach for designing combined array experiments with both control and noise variables.

(a) True

(b) False

Chapter 13 Problems

SS **13.1** An experiment was performed to investigate the capability of a measurement system. Ten parts were randomly selected, and two randomly selected operators measured each part three times. The tests were made in random order, and the data are shown in Table P13.1.

■ **TABLE P13.1**
Measurement Systems Data for Problem 13.1

	Operator 1 Measurements			Operator 2 Measurements		
Part No.	**1**	**2**	**3**	**1**	**2**	**3**
1	50	49	50	50	48	51
2	52	52	51	51	51	51
3	53	50	50	54	52	51
4	49	51	50	48	50	51
5	48	49	48	48	49	48
6	52	50	50	52	50	50
7	51	51	51	51	50	50
8	52	50	49	53	48	50
9	50	51	50	51	48	49
10	47	46	49	46	47	48

(a) Analyze the data from this experiment.

(b) Estimate the variance components using the ANOVA method.

SS **13.2** An article by Hoof and Berman ("Statistical Analysis of Power Module Thermal Test Equipment Performance," *IEEE Transactions on Components, Hybrids, and Manufacturing Technology* Vol. 11, pp. 516–520, 1988) describes an experiment conducted to investigate the capability of measurements in thermal impedance ($C^\circ/w \times 100$) on a power module for an induction motor starter. There are 10 parts, 3 operators, and 3 replicates. The data are shown in Table P13.2.

■ **TABLE P13.2**
Power Module Thermal Test Equipment Data for Problem 13.2

	Inspector 1			Inspector 2			Inspector 3		
	Test			Test			Test		
Part No.	**1**	**2**	**3**	**1**	**2**	**3**	**1**	**2**	**3**
1	37	38	37	41	41	40	41	42	41
2	42	41	43	42	42	42	43	42	43
3	30	31	31	31	31	31	29	30	28
4	42	43	42	43	43	43	42	42	42
5	28	30	29	29	30	29	31	29	29
6	42	42	43	45	45	45	44	46	45
7	25	26	27	28	28	30	29	27	27
8	40	40	40	43	42	42	43	43	41
9	25	25	25	27	29	28	26	26	26
10	35	34	34	35	35	34	35	34	35

(a) Analyze the data from this experiment, assuming that both parts and operators are random effects.

(b) Estimate the variance components using the analysis of variance method.

(c) Estimate the variance components using the REML method. Use the confidence intervals on the variance components to assist in drawing conclusions.

13.3 Reconsider the data in Problem 5.11. Suppose that both factors, machines and operators are chosen at random.

(a) Analyze the data from this experiment.

(b) Find point estimates of the variance components using the analysis of variance method.

13.4 Suppose that in Problem 5.15 the furnace positions were randomly selected, resulting in a mixed model experiment. Reanalyze the data from this experiment under this new assumption. Estimate the appropriate model components using the ANOVA method.

SS **13.5** Reanalyze the measurement systems experiment in Problem 13.1, assuming that operators are a fixed factor. Estimate the appropriate model components using the ANOVA method.

13.6 Reanalyze the measurement system experiment in Problem 13.2, assuming that operators are a fixed factor. Estimate the appropriate model components using the ANOVA method.

SS **13.7** In Problem 5.11, suppose that there are only four machines of interest, but the operators were selected at random.

(a) What type of model is appropriate?

(b) Perform the analysis and estimate the model components using the ANOVA method.

13.8 Rework Problem 13.4 using the REML method.

13.9 Rework Problem 13.5 using the REML method.

13.10 Rework Problem 13.6 using the REML method.

13.11 Rework Problem 13.7 using the REML method.

13.12 Consider the three-factor factorial design in Example 13.4. Propose appropriate test statistics for all main effects and interactions. Repeat for the case where A and B are fixed and C is random.

13.13 Consider the experiment in Example 13.5. Analyze the data for the case where A, B, and C are random.

13.14 Derive the expected mean squares shown in Table 13.10.

SS **13.15** In Problem 5.21, assume that the three operators were selected at random. Analyze the data under these conditions and draw conclusions. Estimate the variance components.

13.16 Consider the three-factor factorial model

$$y_{ijk} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\beta\gamma)_{jk} + \varepsilon_{ijk} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, b \\ k = 1, 2, \ldots, c \end{cases}$$

Assuming that all the factors are random, develop the analysis of variance table, including the expected mean squares. Propose appropriate test statistics for all effects.

13.17 In the two-factor mixed model analysis of variance, show that $\text{Cov}[(\tau\beta)_{ij}, (\tau\beta)_{i'j}] = -(1/a)\sigma^2_{\tau\beta}$ for $i \neq i'$.

13.18 Show that the method of analysis of variance always produces unbiased point estimates of the variance components in any random or mixed model.

13.19 Analyze the data in Problem 13.1, assuming that operators are fixed, using both the unrestricted and the restricted forms of the mixed models. Compare the results obtained from the two models.

13.20 Consider the two-factor mixed model. Show that the standard error of the fixed factor mean (e.g., A) is $[MS_{AB}/bn]^{1/2}$.

13.21 Consider the variance components in the random model from Problem 13.1.

(a) Find an exact 95 percent confidence interval on σ^2.

(b) Find approximate 95 percent confidence intervals on the other variance components using the Satterthwaite method.

13.22 Use the experiment described in Problem 5.11 and assume that both factors are random. Find an exact 95 percent confidence interval on σ^2. Construct approximate 95 percent confidence intervals on the other variance components using the Satterthwaite method.

13.23 Consider the three-factor experiment in Problem 5.21 and assume that operators were selected at random. Find an approximate 95 percent confidence interval on the operator variance component.

13.24 Rework Problem 13.19 using the modified large-sample approach described in Section 13.6.2. Compare the two sets of confidence intervals obtained and discuss. SS

13.25 Rework Problem 13.21 using the modified large-sample method described in Section 13.6.2. Compare this confidence interval with the one obtained previously and discuss.

13.26 Consider the experiment described in Problem 5.11. Estimate the variance components using the REML method. Compare the confidence intervals to the approximate CIs found in Problem 13.22.

13.27 Consider the experiment in Problem 13.1. Analyze the data using REML. Compare the CIs to those obtained in Problem 13.21. SS

13.28 Rework Problem 13.24 using REML. Compare all sets of CIs for the variance components.

13.29 The levels of a random factor are sampled from a large population of possible levels.

(a) True

(b) False

13.30 The random factor is assumed to have a normal distribution with a mean of zero and a variance component that may be nonzero.

(a) True

(b) False

13.31 The REML method is preferred as a technique for estimating variance components because it can also provide the standard error of the variance component leading to a confidence interval.

(a) True

(b) False

13.32 Both the ANOVA method and REML will produce the same point estimate of a variance component if the design is balanced.

(a) True

(b) False

13.33 Random factors are always categorical.

(a) True

(b) False

13.34 In a mixed model, the statistical tests on fixed factors are always exactly the same as they are in the fixed effects model.

(a) True

(b) False

13.35 A negative point estimate of a variance component implies that the variance of that fixed effect is zero.

(a) True

(b) False

13.36 In a two-factor factorial experiment with at least one random factor

(a) the statistical tests on main effects are identical to the corresponding main effects tests in the fixed effects case.

(b) the statistical tests on main effects are different from the corresponding main effects tests in the fixed effects case.

(c) no test on the two-factor interaction may be made.

(d) none of the above (**a–c**) are true.

(e) all of the above (**a–c**) are true.

13.37 In a two-factor factorial experiment with at least one random factor

(a) the statistical test on the interaction effect is identical to the interaction effect test in the fixed effects case.

(b) the statistical test on the interaction effect is different from the interaction effect test in the fixed effects case.

(c) no test on the two-factor interaction may be made.

(d) none of the above (**a–c**) are true.

(e) all of the above (**a–c**) are true.

Chapter 14 Problems

SS **14.1** A rocket propellant manufacturer is studying the burning rate of propellant from three production processes. Four batches of propellant are randomly selected from the output of each process, and three determinations of burning rate are made on each batch. The results follow. Analyze the data and draw conclusions.

	Process 1				Process 2				Process 3			
Batch	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**
	25	19	15	15	19	23	18	35	14	35	38	25
	30	28	17	16	17	24	21	27	15	21	54	29
	26	20	14	13	14	21	17	25	20	24	50	33

SS **14.2** A manufacturing engineer is studying the dimensional variability of a particular component that is produced on three machines. Each machine has two spindles, and four components are randomly selected from each spindle. The results follow. Analyze the data, assuming that machines and spindles are fixed factors.

	Machine 1		Machine 2		Machine 3	
Spindle	**1**	**2**	**1**	**2**	**1**	**2**
	12	8	14	12	14	16
	9	9	15	10	10	15
	11	10	13	11	12	15
	12	8	14	13	11	14

14.3 To simplify production scheduling, an industrial engineer is studying the possibility of assigning one time standard to a particular class of jobs, believing that differences between jobs are negligible. To see if this simplification is possible, six jobs are randomly selected. Each job is given to a different group of three operators. Each operator completes the job twice at different times during the week, and the following results are obtained. What are your conclusions about the use of a common time standard for all jobs in this class? What value would you use for the standard?

Job	**Operator 1**		**Operator 2**		**Operator 3**	
1	158.3	159.4	159.2	159.6	158.9	157.8
2	154.6	154.9	157.7	156.8	154.8	156.3
3	162.5	162.6	161.0	158.9	160.5	159.5
4	160.0	158.7	157.5	158.9	161.1	158.5
5	156.3	158.1	158.3	156.9	157.7	156.9
6	163.7	161.0	162.3	160.3	162.6	161.8

14.4 Consider the three-stage nested design shown in Figure 14.5 to investigate alloy hardness. Using the data that follow, analyze the design, assuming that alloy chemistry and heats are fixed factors and ingots are random. Use the restricted form of the mixed model. SS

	Alloy Chemistry 1					
Heats	1		2		3	
Ingots	1	2	1	2	1	2
	40	27	95	69	65	78
	63	30	67	47	54	45

	Alloy Chemistry 2					
Heats	1		2		3	
Ingots	1	2	1	2	1	2
	22	23	83	75	61	35
	10	39	62	64	77	42

14.5 Reanalyze the experiment in Problem 14.4 using the unrestricted form of the mixed model. Comment on any differences you observe between the restricted and the unrestricted model results. You may use a computer software package.

14.6 Derive the expected mean squares for a balanced three-stage nested design, assuming that A is fixed and that B and C are random. Obtain formulas for estimating the variance components. Assume the restricted form of the mixed model.

14.7 Derive the expected mean squares for a balanced three-stage nested design if all three factors are random. Obtain formulas for estimating the variance components.

14.8 Verify the expected mean squares given in Table 14.1.

14.9 ***Unbalanced nested designs.*** Consider an unbalanced two-stage nested design with b_j levels of B under the ith level of A and n_{ij} replicates in the ijth cell.

Factor *A*	**1**		**2**		
Factor *B*	**1**	**2**	**1**	**2**	**3**
	6	−3	5	2	1
	4	1	7	4	0
	8		9	3	−3
			6		

(a) Write down the least squares normal equations for this situation. Solve the normal equations.

(b) Construct the analysis of variance table for the unbalanced two-stage nested design.

(c) Analyze the following data, using the results in part (b).

14.10 A process engineer is testing the yield of a product manufactured on three machines. Each machine can be operated at two power settings. Furthermore, a machine has three stations on which the product is formed. An experiment is conducted in which each machine is tested at both power settings, and three observations on yield are taken from each station. The runs are made in random order, and the results are shown in Table P14.1. Analyze this experiment, assuming that all three factors are fixed.

■ **TABLE P14.1**
Yield Experiment in Problem 14.10

	Machine 1			**Machine 2**		
Station	**1**	**2**	**3**	**1**	**2**	**3**
Power	34.1	33.7	36.2	31.1	33.1	32.8
setting 1	30.3	34.9	36.8	33.5	34.7	35.1
	31.6	35.0	37.1	34.0	33.9	34.3
Power	24.3	28.1	25.7	24.1	24.1	26.0
setting 2	26.3	29.3	26.1	25.0	25.1	27.1
	27.1	28.6	24.9	26.3	27.9	23.9

	Machine 3		
Station	**1**	**2**	**3**
Power	32.9	33.8	33.6
setting 1	33.0	33.4	32.8
	33.1	32.8	31.7
Power	24.2	23.2	24.7
setting 2	26.1	27.4	22.0
	25.3	28.0	24.8

14.11 Suppose that in Problem 14.10 a large number of power settings could have been used and that the two selected for the experiment were chosen randomly. Obtain the expected mean squares for this situation assuming the restricted form of the mixed model and modify the previous analysis appropriately.

14.12 Reanalyze the experiment in Problem 14.11 assuming the unrestricted form of the mixed model. You may use a computer software package to do this. Comment on any differences between the restricted and unrestricted model analysis and conclusions.

14.13 A structural engineer is studying the strength of aluminum alloy purchased from three vendors. Each vendor submits the alloy in standard-sized bars of 1.0, 1.5, or 2.0 inches. The processing of different sizes of bar stock from a common ingot involves different forging techniques, and so this factor may be important. Furthermore, the bar stock is forged from ingots made in different heats. Each vendor submits two test specimens of each size bar stock from three heats.

The resulting strength data are shown in Table P14.2. Analyze the data, assuming that vendors and bar size are fixed and heats are random. Use the restricted form of the mixed model.

■ **TABLE P14.2**
Strength Data in Problem 14.13

	Vendor 1			Vendor 2		
Heat	**1**	**2**	**3**	**1**	**2**	**3**
Bar size						
1 in.	1.230	1.346	1.235	1.301	1.346	1.315
	1.259	1.400	1.206	1.263	1.392	1.320
$1\frac{1}{2}$ in.	1.316	1.329	1.250	1.274	1.384	1.346
	1.300	1.362	1.239	1.268	1.375	1.357
2 in.	1.287	1.346	1.273	1.247	1.362	1.336
	1.292	1.382	1.215	1.215	1.328	1.342

	Vendor 3		
Heat	**1**	**2**	**3**
Bar size			
1 in.	1.247	1.275	1.324
	1.296	1.268	1.315
$1\frac{1}{2}$ in.	1.273	1.260	1.392
	1.264	1.265	1.364
2 in.	1.301	1.280	1.319
	1.262	1.271	1.323

14.14 Rework Problem 14.13 using the unrestricted form of the mixed model. You may use a computer software package to do this. Comment on any differences between the restricted and unrestricted model analysis and conclusions.

SS **14.15** Steel is normalized by heating above the critical temperature, soaking, and then air cooling. This process increases the strength of the steel, refines the grain, and homogenizes the structure. An experiment is performed to determine the effect of temperature and heat treatment time on the strength of normalized steel. Two temperatures and three times are selected. The experiment is performed by heating the oven to a randomly selected temperature and inserting three specimens. After 10 minutes one specimen is removed, after 20 minutes the second is removed, and after 30 minutes the final specimen is removed. Then the temperature is changed to the other level and the process is repeated. Four shifts are required to collect the data, which are shown below. Analyze the data and draw conclusions, assuming both factors are fixed.

		Temperature (°F)	
Shift	**Time (min)**	**1500**	**1600**
	10	63	89
1	20	54	91
	30	61	62
	10	50	80
2	20	52	72
	30	59	69
	10	48	73
3	20	74	81
	30	71	69
	10	54	88
4	20	48	92
	30	59	64

14.16 An experiment is designed to study pigment dispersion in paint. Four different mixes of a particular pigment are studied. The procedure consists of preparing a particular mix and then applying that mix to a panel by three application methods (brushing, spraying, and rolling). The response measured is the percentage reflectance of pigment. Three days are required to run the experiment, and the data obtained follow. Analyze the data and draw conclusions, assuming that mixes and application methods are fixed. SS

	Application	Mix			
Day	**Method**	**1**	**2**	**3**	**4**
	1	64.5	66.3	74.1	66.5
1	2	68.3	69.5	73.8	70.0
	3	70.3	73.1	78.0	72.3
	1	65.2	65.0	73.8	64.8
2	2	69.2	70.3	74.5	68.3
	3	71.2	72.8	79.1	71.5
	1	66.2	66.5	72.3	67.7
3	2	69.0	69.0	75.4	68.6
	3	70.8	74.2	80.1	72.4

14.17 Repeat Problem 14.16, assuming that the mixes are random and the application methods are fixed.

14.18 Consider the split-split-plot design described in Example 14.4. Suppose that this experiment is conducted as described and that the data shown in Table P14.3 are obtained. Analyze the data and draw conclusions.

■ TABLE P14.3
The Absorption Time Experiment

Replicates (or Blocks)	Dosage Strengths	Technician 1			Technician 2			Technician 3		
		1	2	3	1	2	3	1	2	3
	Wall Thickness									
	1	95	71	108	96	70	108	95	70	100
1	2	104	82	115	99	84	100	102	81	106
	3	101	85	117	95	83	105	105	84	113
	4	108	85	116	97	85	109	107	87	115
	1	95	78	110	100	72	104	92	69	101
2	2	106	84	109	101	79	102	100	76	104
	3	103	86	116	99	80	108	101	80	109
	4	109	84	110	112	86	109	108	86	113
	1	96	70	107	94	66	100	90	73	98
3	2	105	81	106	100	84	101	97	75	100
	3	106	88	112	104	87	109	100	82	104
	4	113	90	117	121	90	117	110	91	112
	1	90	68	109	98	68	106	98	72	101
4	2	100	84	112	102	81	103	102	78	105
	3	102	85	115	100	85	110	105	80	110
	4	114	88	118	118	85	116	110	95	120

14.19 Rework Problem 14.18, assuming that the technicians are chosen at random. Use the restricted form of the mixed model.

14.20 Suppose that in Problem 14.18 four technicians had been used. Assuming that all the factors are fixed, how many blocks should be run to obtain an adequate number of degrees of freedom on the test for differences among technicians?

14.21 Consider the experiment described in Example 14.4. Demonstrate how the order in which the treatment combinations are run would be determined if this experiment were run as (a) a split-split-plot, (b) a split-plot, (c) a factorial design in a randomized block, and (d) a completely randomized factorial design.

14.22 An article in *Quality Engineering* ("Quality Quandaries: Two-Level Factorials Run as Split-Plot Experiments," Bisgaard et al., Vol. 8, No. 4, pp. 705–708, 1996) describes a 2^5 factorial experiment in a plasma process focused on making paper more susceptible to ink. Four of the factors (A–D) are difficult to change from run to run, so the experimenters set up the reactor at the eight sets of conditions specified by the low and high levels of those factors, and then processed the two paper types (factor E) together. The placement of the paper specimens in the reactor (right versus left) was randomized. This produces a split-plot design with A–D as the whole-plot factors and factor E as the subplot factor. The data from this experiment are shown in Table P14.4. Analyze the data from this experiment and draw conclusions.

14.23 There is always an interaction term in a nested design.

(a) True

(b) False

14.24 The nested factor is always random.

(a) True

(b) False

14.25 Both the ANOVA method and REML can be applied to a nested design.

(a) True

(b) False

14.26 If there are three levels of the main factor and five levels of the nested factor there will be four degrees of freedom for the nested factor.

(a) True

(b) False

■ **TABLE P14.4**
The Paper-Making Experiment in Problem 14.22

Standard Order	Run Number	A = Pressure	B = Power	C = Gas Flow	D = Gas Type	E = Paper Type	y Contact Angle
5	1	−1	−1	+1	Oxygen	*E*1	37.6
21	2	−1	−1	+1	Oxygen	*E*2	43.5
2	3	+1	−1	−1	Oxygen	*E*1	41.2
18	4	+1	−1	−1	Oxygen	*E*2	38.2
10	5	+1	−1	−1	SiCl4	*E*1	56.8
26	6	+1	−1	−1	SiCl4	*E*2	56.2
14	7	+1	−1	+1	SiCl4	*E*1	47.5
30	8	+1	−1	+1	SiCl4	*E*2	43.2
11	9	−1	+1	−1	SiCl4	*E*1	25.6
27	10	−1	+1	−1	SiCl4	*E*2	33.0
3	11	−1	+1	−1	Oxygen	*E*1	55.8
19	12	−1	+1	−1	Oxygen	*E*2	62.9
13	13	−1	−1	+1	SiCl4	*E*1	13.3
29	14	−1	−1	+1	SiCl4	*E*2	23.7
6	15	+1	−1	+1	Oxygen	*E*1	47.2
22	16	+1	−1	+1	Oxygen	*E*2	44.8
16	17	+1	+1	+1	SiCl4	*E*1	49.5
32	18	+1	+1	+1	SiCl4	*E*2	48.2
9	19	−1	−1	−1	SiCl4	*E*1	5.0
25	20	−1	−1	−1	SiCl4	*E*2	18.1
15	21	−1	+1	+1	SiCl4	*E*1	11.3
31	22	−1	+1	+1	SiCl4	*E*2	23.9
1	23	−1	−1	−1	Oxygen	*E*1	48.6
17	24	−1	−1	−1	Oxygen	*E*2	57.0
8	25	+1	+1	+1	Oxygen	*E*1	48.7
24	26	+1	+1	+1	Oxygen	*E*2	44.4
7	27	−1	+1	+1	Oxygen	*E*1	47.2
23	28	−1	+1	+1	Oxygen	*E*2	54.6
4	29	+1	+1	−1	Oxygen	*E*1	53.5
20	30	+1	+1	−1	Oxygen	*E*2	51.3
12	31	+1	+1	−1	SiCl4	*E*1	41.8
28	32	+1	+1	−1	SiCl4	*E*2	37.8

14.27 In a split-plot design the interaction between the whole-plot factor and the subplot factor is always tested against the whole-plot error.

(a) True

(b) False

14.28 A good reason to consider a split-plot design is a situation where some experimental units are larger than others.

(a) True

(b) False

14.29 In a split-plot design the subplot error is usually smaller than the whole-plot error.

(a) True

(b) False

14.30 If an experiment is conducted as a split plot but analyzed as a factorial the resulting error estimate will usually be

too large for properly testing the subplot factor and could lead to misleading conclusions.

(a) True

(b) False

14.31 It is usually easier or cheaper to change the levels of the whole-plot factor in a split-plot design.

(a) True

(b) False

14.32 In an experiment that has at least one nested factor

(a) the nested factor will always be a random effect.

(b) the primary factor will always be a fixed effect.

(c) one may identify the key factor by arbitrarily renumbering its levels.

(d) none of the above (a–c) are true.

(e) all of the above (a–c) are true.